Bias-Reduced and Parallel Heavy-Tailed Stochastic Convex Optimization in ℓ_p Geometries

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Abstract

We consider the fundamental problem of parallel stochastic convex optimization (SCO), whose aim is to approximately minimize a convex objective when given the ability to query a moment-bounded stochastic gradient oracle multiple times in parallel per round. Whereas most previous results have focused on the variance-bounded Euclidean (ℓ_2-norm) setting, meaning the objective is Lipschitz with respect to the ℓ_2-norm over an ℓ_2-bounded feasible set in $\mathbb{R}^d$, in this work we study parallel SCO in ℓ_p settings for $2 \le p < \infty$ under a heavy-tailed noise model.

To this end, we begin by introducing a new class of ℓ_p-optimal-distance-convergence (ODC_p) algorithms, drawing on prior stochastic bias-reduced techniques [ACJ+21], which combine with multilevel Monte Carlo methods to efficiently estimate the optimum of uniformly convex objectives. As a consequence of our ODC_p algorithms, we first establish rates for projection-efficient ℓ_p-Lipschitz optimization, composite convex minimization, and efficient ℓ_p-proximal point estimation.

Further combining our ODC_p algorithms with a new ℓ_p-generalized Gaussian-based stochastic gradient estimator (p-GREG), along with advances in ℓ_p ball-oracle acceleration [ABJS26, CGMR25], we provide two efficient ℓ_p parallel algorithms, EpochDA and ℓ_p-AC-SA, which achieve optimization error $\varepsilon > 0$ with at most $\tilde{O}(d^{\frac{p}{2p+2}}\varepsilon^{-\frac{p}{p+1}})$ query depth, improving upon the sequential heavy-tailed stochastic query depth of $O(\varepsilon^{-p})$ for $\varepsilon \le \tilde{O}(d^{-1/(2p)})$. In addition, we establish dimension-dependent lower bounds of $\Omega\big(\min\{\varepsilon^{-p}, d^{p/(3p-2)}\}\big)$ adaptive rounds for nonsmooth convex optimization over a subset of the unit ℓ_p ball, with polynomially many parallel local-oracle queries per round. Our results improve upon the $\Omega(\min\{\varepsilon^{-p}, \varepsilon^{2/3}d^{1/3}\})$ bounds of [DG19], when adapted to the same feasible set.

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1 Introduction

Stochastic methods are an essential part of modern optimization, with applications spanning machine learning, statistics, and theoretical computer science. The foundational work of Nemirovski and Yudin [NY83] introduced the *oracle model* for convex optimization, whereby the goal is to understand, for a particular oracle class, how many queries are needed to minimize a function to sufficient accuracy. Often considered in the case of stochastic convex optimization (SCO) is the stochastic first-order oracle, which for a convex function $f : \mathbb{R}^d \to \mathbb{R}$ queried at a point x, provides an unbiased estimate of the subgradient with bounded q-th moment ($q \in (1, 2]$), that is, $\widehat{\nabla} f(x)$ such that $\mathbb{E}[\widehat{\nabla} f(x)] \in \partial f(x)$ and $\mathbb{E}[\|\widehat{\nabla} f(x)\|_q^q] \le \sigma^q$ for $\sigma > 0$.

An additional line of work, initiated by Nemirovski [Nem94], has further considered the setting of *parallel* convex optimization, in which we may query an oracle multiple times in parallel for each round of interaction, and where we denote the *query depth* as the total number of such rounds. While several works have studied first-order methods for parallel settings in the Euclidean (ℓ_2-norm) case [BJL+19, CJJ+23, DBW12], meaning f is Lipschitz continuous with respect to the ℓ_2-norm over an ℓ_2-bounded feasible set, parallel SCO has been less explored in non-Euclidean settings that deal with ℓ_p geometries for $2 \le p < \infty$. For the $p = \infty$ case, Nemirovski [Nem94] established $\Omega((\frac{d}{\ln(2Kd)})^{1/3} \ln(1/\varepsilon))$ lower bounds on the number of rounds required to achieve an ε-accurate solution, where K denotes the number of parallel queries in each round, while Diakonikolas and Guzmán [DG19] provided lower bounds of the form $\Omega(\min\{\varepsilon^{-p}, \varepsilon^{2/3} d^{1/3}\})$ for $p \ge 2$.[1] Meanwhile, work by Chakrabarty, Graur, Jiang, and Sidford [CGJS23] presented algorithms that achieve $O(d^{1/3}\varepsilon^{-2/3})$ parallel stochastic query depth for ℓ_∞-Lipschitz convex problems, based on leveraging additional Gaussian smoothing for this setting, thereby matching the lower bounds for constant accuracy, though their results rely on stronger bounded second-moment (i.e., variance) assumptions that are standard when analyzing stochastic first-order methods. Moreover, ℓ_p-based optimization methods have been shown to be broadly useful across a variety of contexts including training deep neural networks [LBL+25, MFS26, Pet26], p-norm diffusion and flow problems [AS20, FWY20, KLOS14, LS20], and stochastic optimization under heavy-tailed noise assumptions [FHL25, SBZ+26, VYB+22].

In this work, we study parallel stochastic ℓ_p-Lipschitz convex optimization under bounded q-th moment assumptions (for $q = \frac{p}{p-1}$), for which we provide new query depth complexities that improve upon previous sequential results for certain ranges of solution accuracy. Underlying our results are two key technical contributions:

1. First, we introduce a class of ℓ_p-optimal-distance-convergence (ODC_p) algorithms, which we then show may be used to efficiently estimate the optimum of uniformly convex objectives. Although initially inspired by ideas on stochastic bias-reduced gradient methods [ACJ+21], we show how extending this type of approach to generalized ℓ_p-based Moreau envelopes leads to several technical complications that need to be addressed, including the fact that the gradient function for the generalized Moreau envelope is a nonlinear mapping.

2. Second, we develop and analyze a new ℓ_p-generalized Gaussian reweighting scheme for gradient estimation, which broadly extends previous approaches based on standard Gaussian convolution. As a result of these tools, we are able to handle several basic challenges that arise when attempting to estimate the gradients of these ℓ_p-based Moreau envelopes that are central to our techniques.

By combining these techniques with recent algorithms for ℓ_p ball-oracle acceleration [ABJS26,

[1] Diakonikolas and Guzmán [DG19] also established $\Omega(\varepsilon^{-2})$ and $\Omega(\varepsilon^{-2/3})$ lower bounds for the $1 < p < 2$ and $p = 1$ cases, respectively.

CGMR25], we arrive at our improved stochastic query depth complexities for parallel ℓ_p-Lipschitz SCO. In addition, we establish new dimension-dependent lower bounds that improve upon the previous results of [DG19], under certain ℓ_p domain restrictions.

1.1 ℓ_p Stochastic Bias-Reduced Gradient Methods

We begin by discussing the setting of bias-reduced gradient methods. When considering the problem of minimizing a function $f : \mathcal{X} \to \mathbb{R}$ given access to a stochastic subgradient estimator $\hat{\nabla} f$ that is unbiased and has bounded variance, a natural question is to ask whether this estimator can be transformed into a nearly unbiased estimator of the true minimizer $x_* = \text{argmin}_{x \in \mathcal{X}} f(x)$. For strongly convex f, [ACJ+21] showed how to do so by proposing an estimator $\hat{x}_*$ such that for any $\delta > 0$, it holds that

$$||\mathbb{E}\hat{x}_* - x_*||_2 \leq \delta \quad \text{and} \quad \mathbb{E}||\hat{x}_* - \mathbb{E}\hat{x}_*||_2^2 = O\left(\frac{G^2}{\mu^2} \log\left(\frac{G}{\mu\delta}\right)\right),$$

where $\mathbb{E}||\nabla \hat{F}(x)||_2^2 \leq G^2$. Given the importance of these estimators and the extensive number of applications that these are used for such as estimating proximal points [PB14, MS13], minimizing the maximal loss [CJJS21], and gradient-efficient composite optimization [Lan16] and the prevalence of uniformly convex optimization contexts such as ℓ_p regression [Bul20, ABJS26, APS19, JLS22] and online learning [LZ20], it is natural to ask whether such estimators can be developed for the ℓ_p setting.

In this work, we first answer this in the affirmative, and provide a class of ℓ_p-optimal-distance-convergence (ODC$_p$) algorithms that, when combined with multilevel Monte Carlo methods [Gil15, BG15], efficiently estimate the optimum of a uniformly convex function with low bias and bounded q-th moment in the *dual* ℓ_q norm, where $q = \frac{p}{p-1}$. Further, we use these estimators to yield improved rates for projection-efficient ℓ_p Lipschitz optimization, derive complementary rates to [ACJ+21] for composite minimization, and develop efficient methods for estimating proximal points, where the proximal point of y is $P_{f,\lambda}(y) = \text{argmin}_{x \in \mathbb{R}^d}\{f(x) + \frac{\lambda}{p}||x - y||_p^p\}$.

In Section 3, we describe our ODC$_p$ algorithms, which are part of a larger set of ODC algorithms that take as input an unbiased subgradient estimator which satisfies a given moment bound, and provides a (hopefully) efficient convergence guarantee for convergence to x_*. In the Euclidean setting, [AFKT21] defines an ODC algorithm to minimize a strongly convex function $F = f + \psi$, where f is convex and ψ is simple such that the algorithm's output $x = \text{ODC}(\hat{\nabla} f, \psi, T)$ satisfies $\mathbb{E}||x - x_*||_2^2 \leq c\frac{G^2}{\mu^2 T}$ for some constant c. Moreover, such algorithms are widely found in the literature [HK11, RSS11, AFKT21]. Given the advantages that these algorithms have provided, we develop analogous ODC$_p$ algorithms for $p \geq 2$, for minimizing a μ-uniformly convex function $F = f + \psi$, where f is convex. As in [ACJ+21], we leverage the multilevel Monte Carlo (MLMC) technique of [BG15] to construct a low-bias, bounded-dual-moment estimator of the ℓ_p-Moreau gradient. However, unlike in the Euclidean-case, when considering $p > 2$, the map from the proximal point to the Moreau gradient is nonlinear. Therefore, we depart from previous analyses, by applying the MLMC construction directly to approximate the Moreau gradient, rather than first constructing a low-bias estimator of the proximal point and then using this to compute the Moreau gradient estimation. We prove that for any λ-uniformly convex function $F = f + \psi$ where f is convex and iteration budget $T_{\max}$, our ODC$_p$ algorithms provide good estimators g' of the true gradient g_* in a reasonably fast manner (see Lemma 3 for more details).

EpochDA (Section 3.1). Next, we provide a concrete ODC$_p$ algorithm, EpochDA (Algorithm 3), which extends Algorithm 3 of [Nes09] to a composite setting using an ℓ_p-based uniformly convex prox

function, and will be useful in achieving efficient ℓ_p parallel rates (see Section 4 for more details). Within each epoch, it applies a dual-averaging solver to the regularized objective, incorporating stochastic gradient estimates into successive updates while controlling the iterates through the global prox function centered at $\overline{x}$. It uses geometrically increasing epoch lengths and decreasing step sizes, averaging the iterates at the end of each epoch to initialize the next.

Projection-Efficient ℓ_p-Lipschitz Optimization (Section 3.2). In stochastic nonsmooth optimization, constrained proximal or projection computations may be substantially more expensive than stochastic subgradient evaluations [ACJ+21]. In Section 3.2, we study this tradeoff in the ℓ_p setting for $p > 2$, where f is convex and accessed through an unbiased stochastic subgradient oracle $\widehat{\nabla} f$. In the Euclidean setting, [ACJ+21] combine Moreau envelope smoothing with accelerated gradient descent to obtain $O(GR/\epsilon)$ expensive projections while retaining near-optimal stochastic subgradient complexity. For $p > 2$, we combine our ℓ_p-Moreau envelope gradient estimator with accelerated updates using p-power regularization, inspired by [ABJS26]. Our analysis uses the smoothness of the envelope with respect to ℓ_p, while all constrained updates use p-power regularization. The resulting algorithm requires $O_p((GR/\epsilon)^{2p/(p+2)})$ expensive constrained updates and $\widetilde{O}_p((GR/\epsilon)^{p(p-1)})$ stochastic subgradient evaluations. Thus, compared with direct stochastic mirror descent, our method reduces the number of expensive constrained updates at the cost of a larger stochastic subgradient bound.

Convex Composite Minimization (Section 3.4). We consider the convex composite optimization problem

$$\min_{x \in \mathcal{X}} \Psi(x) := \Lambda(x) + f(x),$$

where Λ is (L, κ)-weakly smooth with respect to $\|\cdot\|_p$, f satisfies Assumption 1, and there exists a minimizer $x^\star$ such that $\|x_0 - x^\star\|_p \leq R$. A closely related setting was studied in [DG24], who considered composite minimization with a weakly smooth component and a uniformly convex component. Their generalized AGD+ framework yields an iteration complexity of $O\left((LR^\kappa/\epsilon)^{\frac{p}{\kappa p - p + \kappa}}\right)$ when specialized to ℓ_p geometry without exploiting uniform convexity. Building on the stochastic composite AGD framework of [ACJ+21], we develop an ℓ_p extension using Bregman proximal steps and our reduced-bias gradient estimators. Our algorithm (Algorithm 5) achieves the same dependence on ϵ in the number of $\nabla\Lambda$ evaluations, up to logarithmic factors, while allowing f to be nonsmooth and accessed through a stochastic subgradient oracle.

ℓ_p Accelerated Proximal Point Methods (Section 3.5). Here, we develop Monteiro-Svaiter-type ℓ_p accelerated proximal point methods which leverage our reduced-bias Moreau envelope gradient estimators. Inspired by [ACJ+21], we present a stochastic accelerated ℓ_p proximal point method (Algorithm 6), that combines adaptive proximal regularization with an accelerated first-order update. At each iteration, the algorithm selects a regularization parameter (λ_{k+1}), from a designed NextLambda algorithm (Algorithm 7), and computes an acceleration weight (a_{k+1}), which determines the cumulative weight (A_{k+1}) and the interpolation point (y_k) between the current iterate and the auxiliary sequence. We obtain the next iterate (x_{k+1}) by approximately solving a regularized problem centered at y_k, with the algorithm terminating when the cumulative accelerated weight reaches a prescribed threshold or the iteration budget is exhausted.

1.2 Parallel Optimization in ℓ_p Geometries

Parallel Upper Bounds. In Section 4, we consider the following problem, simplified for the purposes of the introduction. We assume that there is a convex function $f : \mathbb{R}^d \to \mathbb{R}$ whose minimizer lies in an ℓ_p ball of radius R, which can be queried through a stochastic gradient oracle g, where $\mathbb{E}g \in \partial f$ and $\mathbb{E}||g||_q^q \leq L^q$, for some $L > 0$. Moreover, throughout the text, unless otherwise specified, $q = \frac{p}{p-1}$. Our goal is to minimize f up to an expected additive error ε_{opt}. Furthermore, we assume a setting where multiple queries can be submitted simultaneously. Along with the query depth, we also compute the computational depth and total number of queries to the stochastic gradient oracle to characterize the efficiency of our algorithms. As in [CJJ+23], the *computational depth* measures the parallel depth of the algorithm, assuming that each oracle query has $O(1)$ depth, and the *total number of queries* refers to the total number of queries made to the stochastic gradient oracle over all rounds.

Generalized Gaussian. A key technical tool that we introduce to achieve the parallel rates for our EpochDA and ℓ_p-AC-SA solvers is an ℓ_p-Gaussian Reweighting for Gradient Estimation (p-GREG) estimator. Beginning with a generalized ℓ_p-norm Gaussian as studied by [BGN21], our estimator computes an unbiased estimate of a convolution of our objective f with this ℓ_p Gaussian, which furthermore offers the potential for parallelism.

EpochDA Based Solver. Our first algorithm (Theorem 14) is based on an epoch-based stochastic solver using our p-GREG estimators within the ball acceleration framework. By implementing the stochastic gradient queries in parallel, our algorithm achieves an oracle query depth of $\widetilde{O}(d^{\frac{p}{2(p+1)}}\epsilon_{\text{opt}}^{-\frac{p}{p+1}})$ when $L_p = R_p = 1$. This improves upon the $O(\epsilon_{\text{opt}}^{-p})$ oracle query depth of stochastic ℓ_p mirror descent in the regime $\epsilon_{\text{opt}} \leq d^{-\frac{1}{2p}}$, up to logarithmic factors.

ℓ_p-AC-SA Based Solver. Our second algorithm (Theorem 17) replaces EpochDA with an accelerated counterpart based on ℓ_p-AC-SA, achieving a computational depth of $\widetilde{O}(d^{\frac{p}{2(p+1)}}\epsilon_{\text{opt}}^{-\frac{p}{p+1}} + d^{\frac{p}{2(p+2)}}\epsilon_{\text{opt}}^{-\frac{2p}{p+2}})$ when $L_p = R_p = 1$. This improves upon both the oracle query depth and computational depth of stochastic ℓ_p mirror descent in the regime $\epsilon_{\text{opt}} < d^{-\frac{1}{2p}}$, up to logarithmic factors. Compared with our EpochDA-based solver, the AC-SA variant further reduces computational depth while maintaining the same oracle query depth and total query complexity up to logarithmic factors.

Parallel Lower Bounds. Several prior works have established fundamental limitations on parallelizing nonsmooth convex optimization. [BS18] developed an adaptivity-based lower-bound framework, showing that polynomially many parallel queries cannot improve upon sequential convergence rates for up to $\widetilde{\Omega}(d^{1/3})$ rounds in Euclidean geometry. Subsequently, [BJL+19] strengthened the Euclidean lower bound to $\widetilde{\Omega}(d^{1/2})$ by introducing a wall construction. For general ℓ_p geometries with $p \geq 2$, [DG19] established a lower bound of $\widetilde{\Omega}(\min\{\varepsilon^{-p}, \varepsilon^{2/3}d^{1/3}\})$. Building on the proof framework of [BS18] and incorporating the wall function adapted to ℓ_p geometry, we establish an improved lower bound of

$$\widetilde{\Omega}_p\left(\min\left\{\varepsilon^{-p}, d^{\frac{p}{3p-2}}\right\}\right)$$

for nonsmooth convex optimization over a subset of the unit ℓ_p ball, with polynomially many parallel local-oracle queries per round. Our result improves the dimension-dependent rate obtained by adapting the construction of [DG19] to the same feasible set, while recovering the known $d^{1/2}$

Method	Query depth	Computational depth
Stochastic Mirror Descent ([VYB+22], Theorem 6)	$\left(\frac{L_pR_p}{\varepsilon_{\text{opt}}}\right)^p$	$\left(\frac{L_pR_p}{\varepsilon_{\text{opt}}}\right)^p$
BallAccel + EpochSGD ([CJJ+23], Theorem 1)	$d^{1/3}\left(\frac{L_2R_2}{\varepsilon_{\text{opt}}}\right)^{2/3}$	$d^{1/3}\left(\frac{L_2R_2}{\varepsilon_{\text{opt}}}\right)^{2/3} + \left(\frac{L_2R_2}{\varepsilon_{\text{opt}}}\right)^2$
BallAccel + AC-SA ([CJJ+23], Theorem 2)	$d^{1/3}\left(\frac{L_2R_2}{\varepsilon_{\text{opt}}}\right)^{2/3}$	$d^{1/3}\left(\frac{L_2R_2}{\varepsilon_{\text{opt}}}\right)^{2/3} + d^{1/4}\left(\frac{L_2R_2}{\varepsilon_{\text{opt}}}\right)$
[JST24] (Theorem 2)	$d^{1/3}\left(\frac{L_2R_2}{\varepsilon_{\text{opt}}}\right)^{2/3}$	$d^{1/3}\left(\frac{L_2R_2}{\varepsilon_{\text{opt}}}\right)^{2/3}$
ℓ_p BallAccel + EpochDA (Theorem 14)	$d^{\frac{p}{2p+2}}\left(\frac{L_pR_p}{\varepsilon_{\text{opt}}}\right)^{\frac{p}{p+1}}$	$d^{\frac{p}{2p+2}}\left(\frac{L_pR_p}{\varepsilon_{\text{opt}}}\right)^{\frac{p}{p+1}} + d^{\frac{p(p-2)}{2(p+1)}}\left(\frac{L_pR_p}{\varepsilon_{\text{opt}}}\right)^{\frac{p(2p-1)}{p+1}}$
ℓ_p BallAccel + ℓ_p-AC-SA (Theorem 17)	$d^{\frac{p}{2p+2}}\left(\frac{L_pR_p}{\varepsilon_{\text{opt}}}\right)^{\frac{p}{p+1}}$	$d^{\frac{p}{2p+2}}\left(\frac{L_pR_p}{\varepsilon_{\text{opt}}}\right)^{\frac{p}{p+1}} + d^{\frac{p}{2p+4}}\left(\frac{L_pR_p}{\varepsilon_{\text{opt}}}\right)^{\frac{2p}{p+2}}$

Table 1: Query and computational depths to achieve ε_{opt}-accurate solution, for general $L_2, R_2, L_p, R_p > 0$, where L_2 and L_p represent ℓ_2 and ℓ_p Lipschitz constants respectively and R_2 and R_p represent the ℓ_2 and ℓ_p norm bounds on the feasible set (omitting additional polylogarithmic factors).

exponent when $p = 2$. However, our current upper bound retains a dimension exponent of $1/2$, whereas our lower bound has exponent $p/(3p-2) < 1/2$. Closing this gap and determining the optimal parallel complexity in general ℓ_p geometry remains an open question.

1.3 Paper Organization and Notation

In Section 2, we introduce an ℓ_p-generalized Gaussian distribution and present the p-GREG estimator, which estimates the gradient of a convolution of f with this ℓ_p-generalized Gaussian. In Section 3, we introduce our ODC$_p$ algorithms, and discuss their applications to projection-efficient (Section 3.2) and convex composite (Section 3.4) optimization, in addition to efficient ℓ_p-proximal point estimation (Section 3.5).

In Section 4, we present two algorithms, EpochDA and ℓ_p-AC-SA and present upper bounds on their query depths, computational depths, and total number of gradient queries. In Section 5, we present our lower bounds for the number of adaptive rounds needed to acheive at most ε error.

Throughout the paper, we use $\tilde{O}(\cdot)$ to hide polylogarithmic factors in the parameters and will refer to p and q as Hölder conjugate exponents (i.e., $\frac{1}{p} + \frac{1}{q} = 1$). Moreover, unless otherwise specified, we consider $p \in [2, \infty)$. We denote $\mathbb{B}_p(\overline{x}, r)$ to denote an ℓ_p ball around $\overline{x}$ of radius r, i.e., the set

Method	Total query complexity
Stochastic Mirror Descent ([VYB+22], Theorem 6)	$\left(\frac{L_p R_p}{\varepsilon_{\text{opt}}}\right)^p$
BallAccel + EpochSGD ([CJJ+23], Theorem 1)	$d^{1/3}\left(\frac{L_2 R_2}{\varepsilon_{\text{opt}}}\right)^{2/3} + \left(\frac{L_2 R_2}{\varepsilon_{\text{opt}}}\right)^2$
BallAccel + AC-SA ([CJJ+23], Theorem 2)	$d^{1/3}\left(\frac{L_2 R_2}{\varepsilon_{\text{opt}}}\right)^{2/3} + \left(\frac{L_2 R_2}{\varepsilon_{\text{opt}}}\right)^2$
[JST24] (Theorem 2)	$d^{1/3}\left(\frac{L_2 R_2}{\varepsilon_{\text{opt}}}\right)^{2/3} + \left(\frac{L_2 R_2}{\varepsilon_{\text{opt}}}\right)^2$
ℓ_p BallAccel + EpochDA (Theorem 14)	$d^{\frac{p}{2p+2}}\left(\frac{L_p R_p}{\varepsilon_{\text{opt}}}\right)^{\frac{p}{p+1}} + d^{\frac{p(p-2)}{2(p+1)}}\left(\frac{L_p R_p}{\varepsilon_{\text{opt}}}\right)^{\frac{p(2p-1)}{p+1}}$
ℓ_p BallAccel + ℓ_p-AC-SA (Theorem 17)	$d^{\frac{p}{2p+2}}\left(\frac{L_p R_p}{\varepsilon_{\text{opt}}}\right)^{\frac{p}{p+1}} + d^{\frac{p}{2p+4}}\left(\frac{L_p R_p}{\varepsilon_{\text{opt}}}\right)^{\frac{2p}{p+2}} + d^{\frac{p(p-2)}{2(p+1)}}\left(\frac{L_p R_p}{\varepsilon_{\text{opt}}}\right)^{\frac{p(2p-1)}{p+1}}$

Table 2: Total query complexities to achieve ε_{opt}-accurate solution, for general $L_2, R_2, L_p, R_p > 0$, where L_2 and L_p represent ℓ_2 and ℓ_p Lipschitz constants respectively and R_2 and R_p represent the ℓ_2 and ℓ_p norm bounds on the feasible set (omitting additional polylogarithmic factors).

$\{x' \in \mathbb{R}^d : ||x' - \overline{x}||_p \leq r\}$. Furthermore, for $a \in (0, 1]$, let $\text{Geom}(a)$ denote the geometric distribution with parameter a. We also refer to $\mathcal{N}(\mu, \sigma^2 I)$ as a multivariate Gaussian distribution with mean $\mu \in \mathbb{R}^d$ and covariance $\sigma^2 I$, where I denotes the identity matrix in $\mathbb{R}^{d \times d}$.

We say that a function $f : \mathbb{R}^d \to \mathbb{R}$ is L-Lipschitz with respect to the ℓ_p-norm if, for all $x, y \in \mathbb{R}^d$, it holds that $|f(x) - f(y)| \leq L||x - y||_p$. We say that f is (L, κ)-weakly smooth with respect to $||\cdot||_p$ for $L > 0$ and $\kappa \in (1, 2]$, if its gradient is $(L, \kappa - 1)$-Hölder continuous, i.e.,

$$\forall x, y \in \mathbb{R}^d, ||\nabla f(x) - \nabla f(y)||_q \leq L||x - y||_p^{\kappa - 1}.$$

We say that f is convex if for all $x, y \in \mathbb{R}^d$ and $\lambda \in [0, 1]$, $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda) f(y)$. Moreover, we say that f is uniformly convex with convexity parameter μ and order p if for all $x, y \in \mathbb{R}^d$, it holds that

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\mu}{p}||y - x||_p^p$$

or equivalently, $\langle \nabla f(x) - \nabla f(y), x - y \rangle \geq \mu ||x - y||_p^p$.

2 Framework

2.1 Generalized ℓ_p Gaussian

First, we state the density of our Generalized ℓ_p Gaussian ($\mathcal{GG}_p$) [BGN21]. For any $X \sim \mathcal{GG}_p(\mu, \sigma^2)$, its density is represented as

$$\left(\frac{(2\Gamma(1+1/p))^d}{\Gamma(1+d/p)}(2\sigma^2)^{d/2}\Gamma(1+d/2)\right)^{-1}\exp\left(-\frac{\|x-\mu\|_p^2}{2\sigma^2}\right).$$

We will use $\gamma_{\rho,p}$ to denote the probability density function of $\mathcal{GG}_p(0,\rho^2)$. Now, we define the convolution operation for our distribution.

Definition 1. *($\mathcal{GG}_p$-Convolution) For a function $f:\mathbb{R}^d\to\mathbb{R}$, we denote its convolution with a $\mathcal{GG}_p$ random variable of mean 0 and variance ρ^2 by*

$$\hat{f}_{\rho,p}(x) := \mathbb{E}_{y\sim\mathcal{GG}_p(0,\rho^2)}f(x+y) = \int_{y\in\mathbb{R}^d} f(x-y)\gamma_{\rho,p}(y)dy.$$

2.2 p-GREG Estimator

We now present our p-GREG (ℓ_p-Generalized Gaussian Reweighting for Gradient Estimation) estimator for the ℓ_p norm, which is related to the ReSQue estimator [CJJ+23], as well as the estimator in [BJL+19].

Definition 2. *(p-GREG Estimator) Let $\overline{x}\in\mathbb{R}^d$ and let $f:\mathbb{R}^d\to\mathbb{R}$ be convex. Suppose we have a gradient estimator $g:\mathbb{R}^d\to\mathbb{R}^d$ such that $\mathbb{E}[g]\in\partial f$. We define the p-GREG estimator of radius ρ as the following:*

$$\tilde{\nabla}^g_{\overline{x}}\hat{f}_{\rho,p}(x) := \frac{\gamma_{\rho,p}(x-\overline{x}-\xi)}{\gamma_{\rho,p}(\xi)}g(\overline{x}+\xi), \xi\sim\mathcal{GG}_p(0,\rho^2)$$

where ξ is sampled first and g is independently queried at $\overline{x}+\xi$.

We next prove that if g is an unbiased estimate of ∂f and has a certain moment bound, then our estimator is also unbiased for the convolved function and enjoys a similar moment bound.

Lemma 1. *The following statements hold about our estimator presented in Definition 2:*

- $\mathbb{E}[\tilde{\nabla}^g_{\overline{x}}\hat{f}_{\rho,p}(x)] = \nabla\hat{f}_{\rho,p}(x)$
- *If $\mathbb{E}||g(z)||_q^q \le G^q$ for all $z\in\mathbb{R}^d$ and $x\in\mathbb{B}_p(\overline{x},\rho)$, then $\mathbb{E}||\tilde{\nabla}^g_{\overline{x}}\hat{f}_{\rho,p}(x)||_q^q \le 3G^q$*

Proof. To prove the first statement, we note that

$$\begin{aligned}\mathbb{E}[\tilde{\nabla}^g_{\overline{x}}\hat{f}_{\rho,p}(x)] = \int\frac{\gamma_{\rho,p}(x-\overline{x}-\xi)}{\gamma_{\rho,p}(\xi)}\mathbb{E}[g(\overline{x}+\xi)]\gamma_{\rho,p}(\xi)d\xi &= \int\gamma_{\rho,p}(x-\overline{x}-\xi)\mathbb{E}[g(\overline{x}+\xi)]d\xi \\ &= \int\mathbb{E}\left[g(\overline{x}+\xi)\right]\cdot\gamma_{\rho,p}(x-\overline{x}-\xi)d\xi = \nabla\hat{f}_{\rho,p}(x)\end{aligned}$$

To show the second statement, let $v = x-\bar{x}$. By definition,

$$\left\|\widetilde{\nabla}^g_{\bar{x},p}\widehat{f}_{\rho,p}(x)\right\|_q^q = \left(\frac{\gamma_{\rho,p}(v-\xi)}{\gamma_{\rho,p}(\xi)}\right)^q\|g(\bar{x}+\xi)\|_q^q.$$

Taking expectation first with respect to the randomness of g and using $\mathbb{E}_g\|g(z)\|_q^q \le G^q$, we get

$$\begin{aligned}\mathbb{E}\left\|\widetilde{\nabla}^g_{\overline{x},p}\widehat{f}_{\rho,p}(x)\right\|_q^q &\le G^q\mathbb{E}_{\xi\sim\gamma_{\rho,p}}\left[\left(\frac{\gamma_{\rho,p}(v-\xi)}{\gamma_{\rho,p}(\xi)}\right)^q\right] = G^q\int\frac{\gamma_{\rho,p}(v-\xi)^q}{\gamma_{\rho,p}(\xi)^{q-1}}d\xi \\ &= G^q\left(\frac{(2\Gamma(1+1/p))^d}{\Gamma(1+d/p)}(2\rho^2)^{d/2}\Gamma(1+d/2)\right)^{-1}\int\exp\left(\frac{\|\xi\|_p^2-p\|v-\xi\|_p^2}{2(p-1)\rho^2}\right)d\xi\end{aligned}$$

Use the fact that $f(x)=\frac{1}{2}\|x\|_p^2$ is $(p-1)$-smooth with respect to the p norm, by Theorem 5.8 of [Bec17], we have

$$f(\lambda x+(1-\lambda)y)\ge\lambda f(x)+(1-\lambda)f(y)-\frac{p-1}{2}\lambda(1-\lambda)\|x-y\|_p^2,\quad\forall x,y\in\mathbb{R}^d,\ \lambda\in[0,1].$$

Let $\lambda=(p-1)/p$, $x=\xi-\frac{1}{\lambda}v$ and $y=\xi$ one can get

$$\begin{aligned}f(\xi-v)=f(\lambda x+(1-\lambda)y)&\ge\lambda f(x)+(1-\lambda)f(y)-\frac{p-1}{2}\lambda(1-\lambda)\|x-y\|_p^2 \\ &=\frac{p-1}{p}f(\xi-\frac{1}{\lambda}v)+\frac{1}{p}f(\xi)-\frac{(p-1)^2}{2p^2}\|\frac{p}{p-1}v\|_p^2\end{aligned}$$

By rearranging and multiplying both side by $p/(p-1)$, this yields

$$\frac{\|\xi\|_p^2}{2(p-1)}-\frac{p\|\xi-v\|_p^2}{2(p-1)}\le-\frac{\|\xi-\frac{1}{\lambda}v\|_p^2}{2}+\frac{p\|v\|_p^2}{2(p-1)}$$

Using this inequality we have

$$\begin{aligned}\mathbb{E}\left\|\tilde{\nabla}^g_{\bar{x}}\widehat{f}_\rho(x)\right\|_q^q &\le G^q\left(\frac{(2\Gamma(1+1/p))^d}{\Gamma(1+d/p)}(2\rho^2)^{d/2}\Gamma(1+d/2)\right)^{-1}\exp\left(\frac{p\|v\|_p^2}{2(p-1)\rho^2}\right)\int\exp\Big(-\frac{\|\xi-\frac{1}{\lambda}v\|_p^2}{2\rho^2}\Big)d\xi \\ &\le G^q\exp\left(\frac{p\|v\|_p^2}{2(p-1)\rho^2}\right)\le 3G^q\end{aligned}$$

where the last inequality uses that $\|v\|_p\le\rho$. □

We also prove the following bounds on higher-order moments for two important quantities.

Lemma 2. *If $x,x'\in B_{\overline{x}}(\rho/m)$ for $m\ge 2$, then*

$$\begin{aligned}&\mathbb{E}_{\xi\sim\mathcal{GG}_p(0,\rho^2)}\left[\left(\frac{\gamma_{\rho,p}(x-\overline{x}-\xi)}{\gamma_{\rho,p}(\xi)}\right)^m\right]\le e^{p-1} \\ &\mathbb{E}_{\xi\sim\mathcal{GG}_p(0,\rho^2)}\left[\left|\frac{\gamma_{\rho,p}(x-\overline{x}-\xi)-\gamma_{\rho,p}(x'-\overline{x}-\xi)}{\gamma_{\rho,p}(\xi)}\right|^m\right]\le e^{p-1}\left(\frac{C_p\sqrt{m}\|x-x'\|_p}{\rho}\right)^m\end{aligned}$$

for some constant $C_p>0$.

Proof. Denote $v=x-\bar{x}$ and $v'=x'-\bar{x}$. By definition,

$$\begin{aligned}\mathbb{E}_{\xi\sim\mathcal{GG}_p(0,\rho^2)}\left[\left(\frac{\gamma_{\rho,p}(x-\overline{x}-\xi)}{\gamma_{\rho,p}(\xi)}\right)^m\right]&=\mathbb{E}_{\xi\sim\mathcal{GG}_p(0,\rho^2)}\left[\exp\left(\frac{m(\|\xi\|_p^2-\|v-\xi\|_p^2)}{2\rho^2}\right)\right] \\ &\le\exp\left(\frac{m(p-1)\|v\|_p^2}{2\rho^2}\right)\cdot\mathbb{E}_{\xi\sim\mathcal{GG}_p(0,\rho^2)}\left[\exp\left(\frac{m}{\rho^2}\langle\nabla\frac{1}{2}\|\xi\|_p^2,v\rangle\right)\right]\end{aligned}$$

where the inequality is by the smoothness of $\frac{1}{2}\|\cdot\|_p^2$. Since $\|v\|_p \le \rho/m$, we have, with Fact 5,

$$\begin{aligned}\mathbb{E}_{\xi\sim\mathcal{GG}_p(0,\rho^2)}\left[\left(\frac{\gamma_{\rho,p}(x-\overline{x}-\xi)}{\gamma_{\rho,p}(\xi)}\right)^m\right] &\le \exp\left(\frac{m(p-1)\|v\|_p^2}{2\rho^2}\right)\cdot\mathbb{E}_{\xi\sim\mathcal{GG}_p(0,\rho^2)}\left[\exp\left(\frac{m}{\rho^2}\langle\nabla\frac{1}{2}\|\xi\|_p^2, v\rangle\right)\right]\\ &\le \exp\left(\frac{p-1}{2m}\right)\exp\left(\frac{(p-1)m^2}{2\rho^2}\|v\|_p^2\right) \le \exp\left(\frac{p-1}{2m}+\frac{p-1}{2}\right)\\ &\le e^{p-1}\end{aligned}$$

For the second inequality, we define the line segment $v_t := v' + t(v-v')$ for $t\in[0,1]$. By the fundamental theorem of calculus,

$$\frac{\gamma_{\rho,p}(v-\xi)-\gamma_{\rho,p}(v'-\xi)}{\gamma_{\rho,p}(\xi)} = \int_0^1 \frac{\gamma_{\rho,p}(v_t-\xi)}{\gamma_{\rho,p}(\xi)}\frac{\langle\nabla\Phi(\xi-v_t),\, v-v'\rangle}{\rho^2}\,dt,$$

where $\Phi(z) := \frac{1}{2}\|z\|_p^2$. Taking absolute values and applying Jensen's inequality (convexity of $|\cdot|^m$) gives

$$\left|\frac{\gamma_{\rho,p}(v-\xi)-\gamma_{\rho,p}(v'-\xi)}{\gamma_{\rho,p}(\xi)}\right|^m \le \int_0^1 \left(\frac{\gamma_{\rho,p}(v_t-\xi)}{\gamma_{\rho,p}(\xi)}\right)^m\left(\frac{|\langle\nabla\Phi(\xi-v_t),\, v-v'\rangle|}{\rho^2}\right)^m dt$$

Taking expectations with respect to $\xi\sim\mathcal{GG}_p(0,\rho^2)$ and applying Fubini's theorem (as the integral is nonnegative) we get

$$\mathbb{E}_\xi\left[\left|\frac{\gamma_{\rho,p}(v-\xi)-\gamma_{\rho,p}(v'-\xi)}{\gamma_{\rho,p}(\xi)}\right|^m\right] \le \int_0^1 \mathbb{E}_\xi\left[\left(\frac{\gamma_{\rho,p}(v_t-\xi)}{\gamma_{\rho,p}(\xi)}\right)^m\left(\frac{|\langle\nabla\Phi(\xi-v_t), v-v'\rangle}{\rho^2}\right)^m\right]dt$$

Now, let $z=\xi-v_t$, which implies that $\xi = z+v_t$. Since Φ is even, it holds that $\gamma(v_t-\xi)=\gamma(-z)=\gamma(z)$. Therefore, letting $w = v-v' = x-x'$,

$$\begin{aligned}\mathbb{E}_\xi\left[\left(\frac{\gamma_{\rho,p}(v_t-\xi)}{\gamma_{\rho,p}(\xi)}\right)^m\left(\frac{|\langle\nabla\Phi(\xi-v_t), v-v'\rangle}{\rho^2}\right)^m\right]dt &= \int_{\mathbb{R}^d}\frac{\gamma_{\rho,p}(\xi)\gamma_{\rho,p}\,(v_t-\xi)^m}{\gamma_{\rho,p}\,(\xi)^m}\left(\frac{|\langle\nabla\Phi(\xi-v_t), w\rangle|}{\rho^2}\right)^m d\xi\\ &= \int_{\mathbb{R}^d}\gamma_{\rho,p}(z)\left(\frac{\gamma_{\rho,p}(z)}{\gamma_{\rho,p}(z+v_t)}\right)^{m-1}\left(\frac{|\langle\nabla\Phi(z), w\rangle|}{\rho^2}\right)^m dz\\ &= \mathbb{E}_{z\sim\mathcal{GG}_p(0,\rho^2)}\left[\exp\left(\frac{(m-1)(\Phi(z+v_t)-\Phi(z))}{\rho^2}\right)\left(\frac{|\langle\nabla\Phi(z), w\rangle|}{\rho^2}\right)^m\right]\end{aligned}$$

By the $(p-1)$-smoothness of Φ, it holds that

$$\Phi(z+v_t)-\Phi(z) \le \langle\nabla\Phi(z), v_t\rangle + \frac{p-1}{2}||v_t||_p^2.$$

Therefore

$$\begin{aligned}&\mathbb{E}_{z\sim\mathcal{GG}_p(0,\rho^2)}[\exp(\frac{(m-1)(\Phi(z+v_t)-\Phi(z))}{\rho^2})(\frac{|\langle\nabla\Phi(z), w\rangle|}{\rho^2})^m]\\ &\le \exp(\frac{(m-1)(p-1)}{2\rho^2}||v_t||_p^2)\cdot\mathbb{E}_z[\exp(\frac{m-1}{\rho^2}\langle\nabla\Phi(z), v_t\rangle)(\frac{|\langle\nabla\Phi(z), w\rangle|}{\rho^2})^m]\end{aligned}$$

Letting

$$A = \frac{m-1}{\rho^2}\langle\nabla\Phi(z), v_t\rangle, X = \frac{\langle\nabla\Phi(z), w\rangle}{\rho^2}$$

we have that by Cauchy-Schwarz, $\mathbb{E}[e^A|X|^m] \leq \sqrt{\mathbb{E}[e^{2A}]}\sqrt{\mathbb{E}[|X|^{2m}]}$. We bound the two terms separately. For the first term, using Fact 5, it holds that

$$\mathbb{E}[e^{2A}] = \mathbb{E}_z[\exp(\frac{\langle \nabla\Phi(z), 2(m-1)v_t\rangle}{\rho^2})] \leq \exp(\frac{2(p-1)(m-1)^2}{\rho^2}||v_t||_p^2) \leq \exp(\frac{2(p-1)(m-1)^2}{m^2})$$

where we used $||v_t||_p \leq (1-t)||v'||_p + t||v||_p \leq \rho/m$. Therefore,

$$\sqrt{\mathbb{E}[e^{2A}]} \leq \exp(\frac{(p-1)(m-1)^2}{m^2})$$

We now bound the second term. First, we show that X is sub-Gaussian. To prove this, we first prove that $Y(\xi) = \langle \nabla\Phi(\xi), v - v'\rangle$ is sub-Gaussian. First, since $\gamma_{\rho,p}$ is even and $\nabla\Phi$ is odd, it can be shown that $Y(-\xi) = -Y(\xi)$, which means that $\mathbb{E}[Y(\xi)] = 0$. Again, letting $w = v - v'$, and using Fact 5 with $u = \lambda\rho^2 w$ for $\lambda \in \mathbb{R}$, it holds that

$$\mathbb{E}_\xi[\exp(\lambda Y)] = \mathbb{E}_\xi\left[\exp\left(\frac{1}{\rho^2}\langle \nabla\Phi(\xi), \lambda\rho^2 w\rangle\right)\right] \leq \exp(\frac{p-1}{2\rho^2}\lambda^2\rho^4||w||_p^2)$$

where we use the fact that $||w||_p \leq 2\rho/m$. By point 4 of proposition 2.5.2 of [Ver22], we have that Y is sub-gaussian with $K_4^2 := \frac{(p-1)\rho^2||w||_p^2}{2}$. By Markov's inequality, we have that

$$\Pr(Y \geq t) = \Pr(e^{\lambda Y} \geq e^{\lambda t}) \leq e^{-\lambda t}\mathbb{E}[e^{\lambda Y}] \leq \exp(K_4^2\lambda^2 - \lambda t)$$

Minimizing the exponent over $\lambda \geq 0$ gives that $\Pr(Y \geq t) \leq \exp(-t^2/4K_4^2)$. Applying the same argument with $-Y$ instead of Y, it holds that

$$\Pr(|Y| \geq t) \leq 2\exp(-\frac{t^2}{4K_4^2}) := 2\exp(-\frac{t^2}{K_1^2})$$

so property 1 of proposition 2.5.2 of [Ver22] holds with $K_1 := 2K_4$. Now, it also holds that for all $k \geq 2$,

$$\mathbb{E}|Y|^k \leq \int_0^\infty kt^{k-1}\Pr(|Y| \geq t)dt \leq \int_0^\infty 2kt^{k-1}\exp(-\frac{t^2}{K_1^2})dt = kK_1^k\Gamma(\frac{k}{2}) \leq kK_1^k \cdot (\frac{k}{2})^{\frac{k}{2}}$$

which implies that $(\mathbb{E}|Y|^k)^{1/k} \leq K_1\sqrt{2k} := K_2\sqrt{k}$ by property 2 of proposition 2.5.2 of [Ver22], so $K_2 = K_1\sqrt{2} = 2K_4\sqrt{2}$. By this property, it holds that

$$\sqrt{\mathbb{E}|X|^{2m}} = \frac{\sqrt{\mathbb{E}|Y|^{2m}}}{\rho^{2m}} \leq \frac{(K_2\sqrt{2m})^m}{\rho^{2m}} = \frac{(4K_4\sqrt{m})^m}{\rho^{2m}} \leq (\frac{4\sqrt{p-1}\rho||v-v'||_p}{\sqrt{2}}\sqrt{m})^m\frac{1}{\rho^{2m}} = (\frac{C_p\sqrt{m}||v-v'||_p}{\rho})^m$$

where $C_p = 2\sqrt{2(p-1)}$. Therefore,

$$\mathbb{E}[e^A|X|^m] \leq \exp(\frac{(p-1)(m-1)^2}{m^2})(\frac{C_p\sqrt{m}||w||_p}{\rho})^m.$$

Thus,

$$\begin{aligned}
&\mathbb{E}_{z\sim\mathcal{GG}_p(0,\rho^2)}[\exp(\frac{(m-1)(\Phi(z+v_t)-\Phi(z))}{\rho^2})(\frac{|\langle\nabla\Phi(z),w\rangle|}{\rho^2})^m]\\
&\leq \exp(\frac{(m-1)(p-1)}{2\rho^2}||v_t||_p^2)\cdot\exp(\frac{(p-1)(m-1)^2}{m^2})(\frac{C_p\sqrt{m}||w||_p}{\rho})^m\\
&\leq \exp(\frac{(m-1)(p-1)}{2m^2})\exp(\frac{(p-1)(m-1)^2}{m^2})(\frac{C_p\sqrt{m}||w||_p}{\rho})^m\\
&= \exp((p-1)\frac{(m-1)(m-\frac{1}{2})}{m^2})(\frac{C_p\sqrt{m}||w||_p}{\rho})^m\\
&\leq e^{p-1}(\frac{C_p\sqrt{m}||v-v'||_p}{\rho})^m.
\end{aligned}$$

Integrating over $t\in[0,1]$ completes the proof. □

3 ℓ_p Stochastic Bias-Reduced Methods

In this section, we introduce a class of ℓ_p-optimal-distance-convergence (ODC$_p$) algorithms, detailing the required convergence guarantee and giving a concrete algorithm for computing a low-bias estimator of an order-p uniformly convex function F. We then use these algorithms in conjunction with multilevel Monte Carlo (MLMC) methods to use them for various applications. We assume that F can be represented in the form $F=f+\psi$, where ψ is a simple function and f satisfies Assumption 1.

Assumption 1. *The function $f:\mathcal{X}\to\mathbb{R}$ is convex, where $\mathcal{X}$ is a closed convex set. We assume that there is an unbiased subgradient estimator $\hat{\nabla}f$ that has bounded q_{th}-moment $\mathbb{E}[\|\hat{\nabla}f(x)\|_q^q]\leq G^q$ for all $x\in\mathcal{X}$.*

3.1 ODC$_p$ algorithms

Our estimator uses an arbitrary optimization routine for F as a black-box component, provided that it achieves a sufficiently fast convergence rate toward $\operatorname{argmin}_{x\in\mathcal{X}}F(x)$. We describe the required convergence property below.

Definition 3 (ODC$_p$ algorithm)**.** *For $p\geq 2$, an ℓ_p-optimal-distance-convergence (ODC$_p$) algorithm takes as input an unbiased stochastic subgradient oracle $\hat{\nabla}f$ for a function f satisfying Assumption 1, a simple function ψ, and an oracle budget $T\geq 1$. Suppose that $F=f+\psi$ is μ-uniformly convex of order p with minimizer $x^\star$. The algorithm returns a point x using at most T oracle calls, satisfying*

$$\mathbb{E}\|x-x^\star\|_p^p\leq\frac{cG^q}{\mu^qT^{1/(p-1)}},$$

where $c>0$ is a constant.

As in the work of [ACJ+21], we leverage the multilevel Monte Carlo (MLMC) technique of Blanchet and Glynn [BG15] to construct a low-bias, bounded-dual-moment estimator of the ℓ_p-Moreau gradient from an ODC$_p$ algorithm, where we define the ℓ_p-Moreau envelope to be

$$f_{p,\lambda}(y):=\min_x\{f(x)+\frac{\lambda}{p}\|x-y\|_p^p\}$$

For $p > 2$, however, the map from the proximal point to the Moreau gradient is nonlinear. We therefore apply the MLMC construction directly to the Moreau-gradient approximations, rather than first constructing a low-bias estimator of the proximal point. Fix $y \in \mathcal{X}$ and $\lambda > 0$, and define $\psi(x) := \frac{\lambda}{p}\|x - y\|_p^p$. For every $j \geq 0$, let $x_j := \mathrm{ODC}_p\big(\widehat{\nabla} f, \psi, 2^j\big)$, and define the corresponding approximate Moreau gradient

$$g_j := \lambda J_p(y - x_j), \qquad J_p(z) := \big(|z_k|^{p-2} z_k\big)_{k=1}^d.$$

Let $\mathbb{P}(J = j) = 2^{-j}$ for $j \in \{1, 2, \dots\}$, and define

$$g' := g_0 + \begin{cases} 2^J\big(g_J - g_{J-1}\big), & J \leq j_{\max}, \\ 0, & J > j_{\max}. \end{cases} \tag{1}$$

where $j_{\max} = \lfloor \log_2 T_{\max} \rfloor$.

Algorithm 1 ℓ_p-MorGradEst

Input: $\widehat{\nabla} f, y, \lambda, \delta, \sigma^q, \mathcal{X}$
Parameters: δ: desired bias, σ^q: desired q-th moment error
$T_{\max} = \left\lceil (\frac{2C_1 G}{\min\{\delta, \sigma\}})^{p(p-1)} \right\rceil$ (see (4))
and $N = \left\lceil 2^p C_3 \frac{G^p}{\sigma^p} T_{\max}^{\frac{p-2}{p-1}} \right\rceil$ (see (5))
for $i = 1, \dots, N$ **do**
    $\hat{g}^{(i)} \leftarrow$ draw of estimator (1)
end for
return $\widehat{\nabla} f_{p,\lambda}(y) = \frac{1}{N} \sum_{i=1}^N \hat{g}^{(i)}$

Lemma 3. *Let f and $\widehat{\nabla} f$ satisfy Assumption 1, let $y \in \mathcal{X}$, $p > 2$, and let $\lambda > 0$. Let $\psi(x) = \frac{\lambda}{p}\|x - y\|_p^p$, suppose $F = f + \psi$ has minimizer x_* over $\mathcal{X}$, and let $T_{\max} \in \mathbb{N}$ with $T_{\max} \geq 1$. For any ODC_p algorithm on $\mathcal{X}$ with constant c, let g' be the estimator in (1), where J is independent of all oracle randomness. The following statements hold, for some constant $C_p > 0$ depending only on p and c:*

$$\big\|\mathbb{E} g' - g_*\big\|_q \leq C_p G T_{\max}^{-\frac{1}{p(p-1)}} \quad \text{and} \quad \mathbb{E}\|g' - \mathbb{E} g'\|_q^q \leq C_p G^q T_{\max}^{\frac{p-2}{(p-1)^2}},$$

where $g_* = \lambda J_p(y - x_*) = \nabla f_{p,\lambda}(y)$. Moreover, the expected number of stochastic-gradient evaluations required to compute g' is $O(1 + \log T_{\max})$.

Proof. Since $\psi(y) = 0$ and x_* is a minimizer over $\mathcal{X}$,

$$f(x_*) + \frac{\lambda}{p}\|x_* - y\|_p^p \leq f(y).$$

By Jensen's inequality, it holds that

$$\|\mathbb{E}\widehat{\nabla} f(x)\|_q^q \leq \mathbb{E}\|\widehat{\nabla} f(x)\|_q^q \leq G^q.$$

Since $\mathbb{E}\widehat{\nabla} f(x) \in \partial f(x)$, this implies that f is G-Lipschitz with respect to $\|\cdot\|_p$ on $\mathcal{X}$. Moreover, by convexity and Hölder's inequality,

$$f(y) - f(x_*) \leq \Big\langle \mathbb{E}\widehat{\nabla} f(y), y - x_* \Big\rangle \leq \|\mathbb{E}\widehat{\nabla} f(y)\|_q \|y - x_*\|_p \leq G\|y - x_*\|_p.$$

Therefore,

$$\frac{\lambda}{p}\|x_* - y\|_p^p \le G\|x_* - y\|_p,$$

which implies that

$$\|y - x_*\|_p \le p^{1/(p-1)} \left(\frac{G}{\lambda}\right)^{1/(p-1)}.$$

Since F is $2^{2-p}\lambda$-uniformly convex of order p, the ODC guarantee gives

$$\mathbb{E}\|x_j - x_*\|_p^p \le c \left(\frac{G}{2^{2-p}\lambda}\right)^q 2^{-\frac{j}{p-1}} = c\, 2^{(p-2)q} \left(\frac{G}{\lambda}\right)^q 2^{-\frac{j}{p-1}}.$$

By Lyapunov's inequality, for $0 < s \le p$, it holds that

$$\mathbb{E}\|x_j - x_*\|_p^s \le \left(\mathbb{E}\|x_j - x_*\|_p^p\right)^{s/p} \le C_p \left(\frac{G}{\lambda}\right)^{\frac{s}{p-1}} 2^{-\frac{js}{p(p-1)}}.$$

Moreover, it holds that

$$\|J_p(u) - J_p(v)\|_q \le (p-1)(\|u\|_p + \|v\|_p)^{p-2}\|u - v\|_p$$

for all $u, v \in \mathbb{R}^d$. We now apply this inequality with $u = y - x_j$ and $v = y - x_*$. Then,

$$\begin{aligned}\|y - x_*\|_p + \|y - x_j\|_p &\le 2\|y - x_*\|_p + \|x_j - x_*\|_p \\ &\le 2p^{1/(p-1)} \left(\frac{G}{\lambda}\right)^{1/(p-1)} + \|x_j - x_*\|_p.\end{aligned}$$

Using that $(s+t)^{p-2} \le C_p(s^{p-2} + t^{p-2})$ for nonnegative s, t and some constant C_p, we have

$$\|g_j - g_*\|_q \le C_p\lambda \left[\left(\frac{G}{\lambda}\right)^{\frac{p-2}{p-1}} \|x_j - x_*\|_p + \|x_j - x_*\|_p^{p-1}\right]. \tag{2}$$

Taking expectations in (2) and using the Lyapunov bound with $s = 1$ and $s = p - 1$ yields

$$\|\mathbb{E}g_j - g_*\|_q \le \mathbb{E}\|g_j - g_*\|_q \le C_p G \left(2^{-\frac{j}{p(p-1)}} + 2^{-\frac{j}{p}}\right) \le C_p G\, 2^{-\frac{j}{p(p-1)}}, \tag{3}$$

where we use $2^{-j/p} \le 2^{-j/[p(p-1)]}$ for $j \ge 0$ and $p > 2$. Moreover, it holds that

$$\|g_j - g_*\|_q^q \le C_p\lambda^q \left[\left(\frac{G}{\lambda}\right)^{\frac{(p-2)q}{p-1}} \|x_j - x_*\|_p^q + \|x_j - x_*\|_p^p\right],$$

where we used $(p-1)q = p$. Absorbing constants into C_p and taking expectations yields

$$\mathbb{E}\|g_j - g_*\|_q^q \le C_p G^q \left(2^{-\frac{j}{(p-1)^2}} + 2^{-\frac{j}{p-1}}\right) \le C_p G^q\, 2^{-\frac{j}{(p-1)^2}},$$

because $p > 2$ implies $2^{-j/(p-1)} \le 2^{-j/(p-1)^2}$ for $j \ge 0$. Therefore, for $j \ge 1$,

$$\mathbb{E}\|g_j - g_{j-1}\|_q^q \le 2^{q-1}\left(\mathbb{E}\|g_j - g_*\|_q^q + \mathbb{E}\|g_{j-1} - g_*\|_q^q\right) \le C_p G^q\, 2^{-\frac{j}{(p-1)^2}}.$$

On the other hand,

$$\mathbb{E}g' = \mathbb{E}g_0 + \sum_{j=1}^{j_{\max}} 2^{-j} 2^j \mathbb{E}(g_j - g_{j-1}) = \mathbb{E}g_{j_{\max}}.$$

Since $2^{j_{\max}} > T_{\max}/2$, combining this with (3) implies that

$$\|\mathbb{E}g' - g_*\|_q \le C_p G T_{\max}^{-\frac{1}{p(p-1)}}.$$

Since

$$g' - g_* = (g_0 - g_*) + \mathbf{1}[J \le j_{\max}]\, 2^J (g_J - g_{J-1}),$$

it holds that

$$\begin{aligned}
\mathbb{E}\|g' - g_*\|_q^q &\le 2^{q-1}\mathbb{E}\|g_0 - g_*\|_q^q + 2^{q-1}\mathbb{E}\left\|\mathbf{1}[J \le j_{\max}]\, 2^J (g_J - g_{J-1})\right\|_q^q \\
&\le 2^{q-1}\mathbb{E}\|g_0 - g_*\|_q^q + 2^{q-1}\sum_{j=1}^{j_{\max}} 2^{-j} 2^{qj}\mathbb{E}\|g_j - g_{j-1}\|_q^q \\
&\le C_p G^q \left(1 + \sum_{j=1}^{j_{\max}} 2^{\left(\frac{1}{p-1} - \frac{1}{(p-1)^2}\right)j}\right) = C_p G^q \left(1 + \sum_{j=1}^{j_{\max}} 2^{\frac{(p-2)j}{(p-1)^2}}\right)
\end{aligned}$$

where the second inequality follows by the law of total expectation and the independence of J from the oracle randomness. Since $p > 2$, the geometric sum gives

$$\mathbb{E}\|g' - g_*\|_q^q \le C_p G^q T_{\max}^{\frac{p-2}{(p-1)^2}}.$$

Therefore,

$$\begin{aligned}
\mathbb{E}\|g' - \mathbb{E}g'\|_q^q &\le 2^{q-1}\mathbb{E}\|g' - g_*\|_q^q + 2^{q-1}\|\mathbb{E}g' - g_*\|_q^q \le C_p G^q \left(T_{\max}^{\frac{p-2}{(p-1)^2}} + T_{\max}^{-\frac{q}{p(p-1)}}\right) \\
&\le C_p G^q T_{\max}^{\frac{p-2}{(p-1)^2}}.
\end{aligned}$$

Moreover, the expected number of stochastic-gradient evaluations is at most

$$1 + \sum_{j=1}^{j_{\max}} 2^{-j}(2^j + 2^{j-1}) = 1 + \frac{3}{2} j_{\max} = O(1 + \log T_{\max}),$$

since level j is drawn with probability 2^{-j} and requires at most $2^j + 2^{j-1}$ oracle calls, while the base term g_0 requires at most one oracle call. This finishes the proof. □

Theorem 1. *Let f and $\widehat{\nabla} f$ satisfy Assumption 1, let $y \in \mathcal{X}$, and let $\lambda, \delta, \sigma > 0$ and $p > 2$. The procedure*

$$\ell_p\text{-MorGradEst}\big(\widehat{\nabla} f, y, \lambda, \delta, \sigma^q, \mathcal{X}\big)$$

outputs an estimator $\widehat{\nabla} f_{p,\lambda}(y)$ satisfying

$$\left\|\mathbb{E}\,\widehat{\nabla} f_{p,\lambda}(y) - \nabla f_{p,\lambda}(y)\right\|_q \le \delta \quad \text{and} \quad \mathbb{E}\left\|\widehat{\nabla} f_{p,\lambda}(y) - \nabla f_{p,\lambda}(y)\right\|_q^q \le \sigma^q,$$

using $N_{\widehat{\nabla} f}$ stochastic-gradient computations, where

$$\mathbb{E}N_{\widehat{\nabla} f} = O\left(\left[1 + \frac{G^p}{\sigma^p}\left(\max\left\{1, \frac{G}{\min\{\delta, \sigma\}}\right\}\right)^{p(p-2)}\right]\left[1 + \log\max\left\{1, \frac{G}{\min\{\delta, \sigma\}}\right\}\right]\right).$$

Proof. Write the algorithm's output as $\widehat{\nabla} f_{p,\lambda}(y) = \frac{1}{N}\sum_{i=1}^{N} g'^{(i)}$, with $g'^{(i)}$ i.i.d. copies of the MLMC estimator. Note that from Lemma 3, we have

$$\|\mathbb{E}g' - \nabla f_{p,\lambda}(y)\|_q \le C_1 G T_{\max}^{-\frac{1}{p(p-1)}} \quad , \quad \mathbb{E}\|g' - \mathbb{E}g'\|_q^q \le C_2 G^q T_{\max}^{\frac{p-2}{(p-1)^2}}. \tag{4}$$

Since $T_{\max} \ge \left(\frac{2C_1 G}{\min\{\delta,\sigma\}}\right)^{p(p-1)}$, it holds that

$$\|\mathbb{E}g' - \nabla f_{p,\lambda}(y)\|_q \le \frac{\min\{\delta,\sigma\}}{2}.$$

Since $\mathbb{E}\widehat{\nabla} f_{p,\lambda}(y) = \mathbb{E}g'$, it holds that

$$\|\mathbb{E}\widehat{\nabla} f_{p,\lambda}(y) - \nabla f_{p,\lambda}(y)\|_q \le \frac{\min\{\delta,\sigma\}}{2} \le \frac{\delta}{2} \le \delta.$$

The random vectors $g'^{(i)} - \mathbb{E}g'^{(i)}$ are i.i.d. and centered. Since $1 < q \le 2$, the von Bahr–Esseen inequality applied coordinatewise, together with the lemma, gives, for some constant C_{vBE},

$$\mathbb{E}\left\|\frac{1}{N}\sum_{i=1}^{N}\left(g'^{(i)} - \mathbb{E}g'^{(i)}\right)\right\|_q^q \le \frac{C_{\mathrm{vBE}}}{N^{q-1}}\mathbb{E}\|g' - \mathbb{E}g'\|_q^q \le \frac{C_{\mathrm{vBE}} C_2 G^q T_{\max}^{\frac{p-2}{(p-1)^2}}}{N^{q-1}}.$$

Since

$$N \ge 2^p (C_{\mathrm{vBE}} C_2)^{p-1} \frac{G^p}{\sigma^p} T_{\max}^{\frac{p-2}{p-1}} = 2^p C_3 \frac{G^p}{\sigma^p} T_{\max}^{\frac{p-2}{p-1}}, \qquad C_3 := (C_{\mathrm{vBE}} C_2)^{p-1}, \tag{5}$$

it holds that

$$\frac{C_{\mathrm{vBE}} C_2 G^q T_{\max}^{\frac{p-2}{(p-1)^2}}}{N^{q-1}} \le \left(\frac{\sigma}{2}\right)^q.$$

Therefore,

$$\mathbb{E}\left\|\widehat{\nabla} f_{p,\lambda}(y) - \mathbb{E}\widehat{\nabla} f_{p,\lambda}(y)\right\|_q^q \le \frac{\sigma^q}{2^q}.$$

Thus,

$$\begin{aligned}\mathbb{E}\left\|\widehat{\nabla} f_{p,\lambda}(y) - \nabla f_{p,\lambda}(y)\right\|_q^q &\le 2^{q-1}\mathbb{E}\left\|\widehat{\nabla} f_{p,\lambda}(y) - \mathbb{E}\widehat{\nabla} f_{p,\lambda}(y)\right\|_q^q + 2^{q-1}\left\|\mathbb{E}\widehat{\nabla} f_{p,\lambda}(y) - \nabla f_{p,\lambda}(y)\right\|_q^q \\ &\le \frac{\sigma^q}{2} + 2^{q-1}\left(\frac{\min\{\delta,\sigma\}}{2}\right)^q \le \sigma^q\end{aligned}$$

Using $\lceil x \rceil \le 1 + x$, it holds that

$$\begin{aligned}T_{\max} &\le 1 + \left(\frac{2C_1 G}{\min\{\delta,\sigma\}}\right)^{p(p-1)} \le \left(1 + (2C_1)^{p(p-1)}\right)\left(\max\left\{1, \frac{G}{\min\{\delta,\sigma\}}\right\}\right)^{p(p-1)} \\ &= C_p \left(\max\left\{1, \frac{G}{\min\{\delta,\sigma\}}\right\}\right)^{p(p-1)}\end{aligned}$$

for some constant $C_p \geq 1$. Consequently,

$$1 + \log T_{\max} \leq (1 + \log C_p) + p(p-1) \log \max\left\{1, \frac{G}{\min\{\delta, \sigma\}}\right\}$$
$$\leq C_p' \left(1 + \log \max\left\{1, \frac{G}{\min\{\delta, \sigma\}}\right\}\right)$$

for some constant C_p'. Since N is deterministic, the expected total number of stochastic-gradient computations satisfies

$$\mathbb{E} N_{\widehat{\nabla} f} \leq N\left(1 + \frac{3}{2} j_{\max}\right) = O\big(N(1 + \log T_{\max})\big).$$

Moreover,

$$N \leq 1 + 2^p C_3 \frac{G^p}{\sigma^p} T_{\max}^{\frac{p-2}{p-1}} \leq 1 + C_p'' \frac{G^p}{\sigma^p} \left(\max\left\{1, \frac{G}{\min\{\delta, \sigma\}}\right\}\right)^{p(p-2)}$$

for some constant C_p''. Combining these bounds gives

$$\mathbb{E} N_{\widehat{\nabla} f} = O\left(\left[1 + \frac{G^p}{\sigma^p}\left(\max\left\{1, \frac{G}{\min\{\delta, \sigma\}}\right\}\right)^{p(p-2)}\right]\left[1 + \log \max\left\{1, \frac{G}{\min\{\delta, \sigma\}}\right\}\right]\right),$$

which finishes the proof. □

We now provide a concrete ODC$_p$ algorithm. Our Algorithm 9 extends Algorithm 3 of [Nes09] to a composite setting. Following [ACJ+21], to show that Algorithm 9 is an ODC$_p$ algorithm, we first establish a convergence guarantee for the single-epoch Algorithm 2.

Throughout this section, we use the global prox function

$$d_{\bar{x},R}(x) := \frac{1}{pR^p}\|x - \bar{x}\|_p^p, \qquad x \in \mathcal{X},$$

where $R > 0$. Let $\mu_p > 0$ denote the p-uniform convexity constant of $u \mapsto \frac{1}{p}\|u\|_p^p$ with respect to $\|\cdot\|_p$. Consequently, $d_{\bar{x},R}$ is μ_p/R^p-uniformly convex on all of $\mathcal{X}$.

$$V_{z,R,\beta,c}(s) = \max_{x \in \mathcal{X}} \left\{\langle s, x - z\rangle - c\psi(x) - \beta d_{z,R}(x)\right\}. \tag{6}$$

$$\pi_{z,R,\beta,c}(s) = \arg\max_{x \in \mathcal{X}} \left\{\langle s, x - z\rangle - c\psi(x) - \beta d_{z,R}(x)\right\}. \tag{7}$$

where $\psi : \mathcal{X} \to \mathbb{R}$ is μ-uniformly convex of order p, and $c, \beta > 0$. We also assume $z \in \mathcal{X}$.

Algorithm 2 Dual Averaging Method $DAM(\hat{\nabla} f, \psi, \mathcal{X}, \bar{x}, \gamma, R, N)$

Set $x_0 = \bar{x}$, $s_0 = 0 \in E^\star$, $c_0 = 0$ and $\beta = \gamma N^{1/q}$
for $i = 0, 1, \ldots, N-1$ **do**
  $c_{i+1} = c_i + 1$
  $x_i = \pi_{\bar{x},R,\beta,c_{i+1}}(-s_i)$
  Query a stochastic gradient $\hat{\nabla} f(x_i)$, and set $s_{i+1} = s_i + \hat{\nabla} f(x_i)$
end for
return $\bar{x}_N(\bar{x}, R) = \frac{1}{N}\sum_{i=0}^{N-1} x_i$

Algorithm 3 EpochDA($\hat{\nabla} f, \psi, \mathcal{X}, x_0, R_0, N$)

Set $y_0 = x_0, \tau = p - 1$

For each $k \geq 1$, define $N_k = \left\lceil \frac{2^p p^p G^p}{\mu^p \mu_p R_{k-1}^{p(p-1)}} \right\rceil$. Define $m(N) = \max\left\{k \geq 0 : \sum_{j=1}^{k} N_j \leq N\right\}$.

for $k = 1, \ldots, m(N)$ **do**

  $R_k^p = 2^{-k} R_0^p$, $\gamma_k = \frac{G R_{k-1}}{\mu_p^{1/p}}$.

  $y_k = DAM(\hat{\nabla} f, \psi, \mathcal{X}, y_{k-1}, \gamma_k, R_{k-1}, N_k)$

end for

return $\hat{x}_N = y_{m(N)}$.

Lemma 4. *(Lemma 1 of [Nes09], Lemma 1 of [IN14]) Let $E^\star$ be the dual space of E. The following statements hold:*

- *If $\beta_1 \leq \beta_2$, then*
$$V_{z,R,\beta_1,c}(s) \geq V_{z,R,\beta_2,c}(s)$$
- *The function $V_{z,R,\beta,c}$ is convex and differentiable on $E^\star$. Moreover,*
$$\left\|V'_{z,R,\beta,c}(s_1) - V'_{z,R,\beta,c}(s_2)\right\|_p \leq \left(\frac{R^p}{\beta\mu_p}\right)^{q-1} \|s_1 - s_2\|_q^{q-1}, \qquad \forall s_1, s_2 \in E^\star.$$
- *For any $s \in E^*$,*
$$V'_{z,R,\beta,c}(s) + z = \pi_{z,R,\beta,c}(s) \in \mathcal{X}$$

One might notice that $V_{z,R,\beta,c}(s), \pi_{z,R,\beta,c}(s)$ are defined without the term $\psi(x)$ in [IN14]. However, the same result will be achieved by completely following the proof in [Nes09].

Lemma 5. *Let $f : \mathcal{X} \to \mathbb{R}$ and $\hat{\nabla} f$ satisfy Assumption 1. For any $k \geq 1, N \geq 1$ and $x \in \mathcal{X}$, Algorithm 2 guarantees that*

$$\sum_{i=0}^{N-1} \left[\langle \hat{\nabla} f(x_i), x_i - x\rangle + \psi(x_i) - \psi(x)\right] \leq \beta d_{\bar{x},R}(x) + \frac{R^q}{q\mu_p^{q-1}\beta^{q-1}} \sum_{i=0}^{N-1} \|\hat{\nabla} f(x_i)\|_q^q. \tag{8}$$

Proof. Because $d_{\bar{x},R}(x) = \frac{1}{pR^p}\|x - \bar{x}\|_p^p$ is μ_p/R^p-uniformly convex, $c\psi(x) + \beta d_{\bar{x},R}(x)$ is at least $\beta\mu_p/R^p$-uniformly convex. For each $i = 0, \ldots, N-1$, since $s_{i+1} = s_i + \hat{\nabla} f(x_i)$, the standard smoothness inequality for $V_{\bar{x},R,\beta,c_{i+1}}$ gives

$$V_{\bar{x},R,\beta,c_{i+1}}(-s_{i+1}) \leq V_{\bar{x},R,\beta,c_{i+1}}(-s_i) - \left\langle \hat{\nabla} f(x_i), V'_{\bar{x},R,\beta,c_{i+1}}(-s_i)\right\rangle + \frac{R^q}{q\mu_p^{q-1}\beta^{q-1}} \|\hat{\nabla} f(x_i)\|_q^q. \tag{9}$$

By the definition of x_i, $V'_{\bar{x},R,\beta,c_{i+1}}(-s_i) + \bar{x} = x_i$. Therefore,

$$V_{\bar{x},R,\beta,c_{i+1}}(-s_{i+1}) \leq V_{\bar{x},R,\beta,c_{i+1}}(-s_i) - \langle \hat{\nabla} f(x_i), x_i - \bar{x}\rangle + \frac{R^q}{q\mu_p^{q-1}\beta^{q-1}} \|\hat{\nabla} f(x_i)\|_q^q. \tag{10}$$

Since $c_{i+1} = c_i + 1$, and x_i maximizes the objective defining $V_{\bar{x},R,\beta,c_{i+1}}(-s_i)$, we also have

$$V_{\bar{x},R,\beta,c_{i+1}}(-s_i) \leq V_{\bar{x},R,\beta,c_i}(-s_i) - \psi(x_i).$$

Combining the previous two inequalities yields

$$V_{\bar{x},R,\beta,c_{i+1}}(-s_{i+1}) \le V_{\bar{x},R,\beta,c_i}(-s_i) - \langle \hat{\nabla} f(x_i), x_i - \bar{x}\rangle - \psi(x_i) + \frac{R^q}{q\mu_p^{q-1}\beta^{q-1}} \|\hat{\nabla} f(x_i)\|_q^q \tag{11}$$

Summing over $i = 0, \dots, N-1$, we obtain

$$\sum_{i=0}^{N-1} \Big[\langle \hat{\nabla} f(x_i), x_i - \bar{x}\rangle + \psi(x_i)\Big] \le V_{\bar{x},R,\beta,c_0}(-s_0) - V_{\bar{x},R,\beta,c_N}(-s_N) + \frac{R^q}{q\mu_p^{q-1}\beta^{q-1}} \sum_{i=0}^{N-1} \|\hat{\nabla} f(x_i)\|_q^q. \tag{12}$$

Since $s_0 = 0, c_0 = 0$ and

$$d_{\bar{x},R}(x) \ge 0, \qquad d_{\bar{x},R}(\bar{x}) = 0, \qquad \bar{x} \in \mathcal{X},$$

we have

$$V_{\bar{x},R,\beta,c_0}(-s_0) = \max_{x\in\mathcal{X}} \{-\beta d_{\bar{x},R}(x)\} = 0$$

For any fixed $x \in \mathcal{X}$, the definition of V gives

$$\begin{aligned} -V_{\bar{x},R,\beta,c_N}(-s_N) &\le \langle s_N, x - \bar{x}\rangle + N\psi(x) + \beta d_{\bar{x},R}(x) \\ &= \sum_{i=0}^{N-1} \langle \hat{\nabla} f(x_i), x - \bar{x}\rangle + N\psi(x) + \beta d_{\bar{x},R}(x). \end{aligned} \tag{13}$$

Substituting this into (12) and rearranging proves (8). □

Lemma 6. *Let $F = f + \psi$, and let $x^\star \in \arg\min_{x\in\mathcal{X}} F(x)$. Assume that the stochastic oracle satisfies*

$$\mathbb{E}[\hat{\nabla} f(x_i) \mid \mathcal{F}_i] = \nabla f(x_i) \quad \textit{and} \quad \mathbb{E}[\|\hat{\nabla} f(x_i)\|_q^q \mid \mathcal{F}_i] \le G^q,$$

where $\mathcal{F}_i$ denotes the history before the fresh oracle call at iteration i. Suppose that $\bar{x}$ satisfies $\mathbb{E}\|\bar{x} - x^\star\|_p^p \le R^p$. Then the output of Algorithm 2 satisfies

$$\mathbb{E}[F(\bar{x}_N(\bar{x}, R)) - F(x^\star)] \le \frac{1}{N^{1/p}} \left[\frac{\gamma}{p} + \frac{R^q G^q}{q\gamma^{q-1}\mu_p^{q-1}}\right], \tag{14}$$

$$\mathbb{E}\|\bar{x}_N(\bar{x}, R) - x^\star\|_p^p \le \frac{p}{\mu N^{1/p}} \left[\frac{\gamma}{p} + \frac{R^q G^q}{q\gamma^{q-1}\mu_p^{q-1}}\right]. \tag{15}$$

Proof. Apply Lemma 5 with $x = x^\star$. Since $d_{\bar{x},R}(x^\star) = \frac{\|\bar{x}-x^\star\|_p^p}{pR^p}$, we have $\mathbb{E}[d_{\bar{x},R}(x^\star)] \le \frac{1}{p}$. Moreover, conditional unbiasedness of the stochastic oracle gives

$$\mathbb{E}\left[\left\langle \hat{\nabla} f(x_i), x_i - x^\star\right\rangle\right] = \mathbb{E}\left[\langle \nabla f(x_i), x_i - x^\star\rangle\right] \ge \mathbb{E}[f(x_i) - f(x^\star)], \tag{16}$$

where the last inequality follows from convexity of f. Therefore,

$$\mathbb{E}\Big[\langle \hat{\nabla} f(x_i), x_i - x^\star\rangle + \psi(x_i) - \psi(x^\star)\Big] \ge \mathbb{E}[F(x_i) - F(x^\star)]. \tag{17}$$

Taking expectations in (8), using $\beta = \gamma N^{1/q}$, and the moment bound $\mathbb{E}\|\hat{\nabla} f(x_i)\|_q^q \le G^q$, gives

$$\frac{1}{N} \sum_{i=0}^{N-1} \mathbb{E}[F(x_i) - F(x^\star)] \le \frac{1}{N^{1/p}} \left[\frac{\gamma}{p} + \frac{R^q G^q}{q\gamma^{q-1}\mu_p^{q-1}}\right]. \tag{18}$$

By convexity of F,

$$F(\bar{x}_N(\bar{x}, R)) = F\left(\frac{1}{N}\sum_{i=0}^{N-1} x_i\right) \le \frac{1}{N}\sum_{i=0}^{N-1} F(x_i).$$

Plugging this into (15) proves (14). Finally, since $F = f + \psi$ is μ-uniformly convex of order p,

$$F(x) - F(x^\star) \ge \frac{\mu}{p}\|x - x^\star\|_p^p.$$

Hence

$$\mathbb{E}\|\bar{x}_N(\bar{x}, R) - x^\star\|_p^p \le \frac{p}{\mu}\mathbb{E}[F(\bar{x}_N(\bar{x}, R)) - F(x^\star)],$$

which proves (15). □

Corollary 2. *Under the assumptions of Lemma 6, choose* $\gamma = \frac{GR}{\mu_p^{1/p}}$. *Then*

$$\mathbb{E}[F(\bar{x}_N(\bar{x}, R)) - F(x^\star)] \le \frac{GR}{\mu_p^{1/p} N^{1/p}}, \tag{19}$$

$$\mathbb{E}\|\bar{x}_N(\bar{x}, R) - x^\star\|_p^p \le \frac{GRp}{\mu\mu_p^{1/p} N^{1/p}}. \tag{20}$$

Theorem 3. *Assume* $\|y_0 - x^\star\|_p^p \le R_0^p$. *Then the sequence* $\{y_k\}_{k=0}^{m(N)}$ *generated by Algorithm 3 satisfies*

$$\mathbb{E}\|y_k - x^\star\|_p^p \le R_k^p = 2^{-k} R_0^p, \qquad k = 0, 1, \ldots, m(N).$$

Moreover, for every completed epoch $k \ge 1$,

$$\mathbb{E}[F(y_k) - F(x^\star)] \le \frac{\mu}{p} R_k^p.$$

Proof. We first prove by induction that $\mathbb{E}\|y_k - x^\star\|_p^p \le R_k^p$. For $k = 0$, this follows from the assumption $\|y_0 - x^\star\|_p^p \le R_0^p$. Now assume that for some $k \ge 1$, $\mathbb{E}\|y_{k-1} - x^\star\|_p^p \le R_{k-1}^p$. By Corollary 2, this gives

$$\mathbb{E}[F(y_k) - F(x^\star)] \le \frac{GR_{k-1}}{\mu_p^{1/p} N_k^{1/p}}.$$

By the definition of N_k, $N_k \ge \frac{2^p p^p G^p}{\mu^p \mu_p R_{k-1}^{p(p-1)}}$. Therefore, $N_k^{1/p} \ge \frac{2pG}{\mu\mu_p^{1/p} R_{k-1}^{p-1}}$. Substituting this bound yields

$$\mathbb{E}[F(y_k) - F(x^\star)] \le \frac{GR_{k-1}}{\mu_p^{1/p}} \cdot \frac{\mu\mu_p^{1/p} R_{k-1}^{p-1}}{2pG} = \frac{\mu}{2p} R_{k-1}^p = \frac{\mu}{p} R_k^p. \tag{21}$$

Since F is μ-uniformly convex of order p,

$$F(y_k) - F(x^\star) \ge \frac{\mu}{p}\|y_k - x^\star\|_p^p.$$

Taking expectations gives

$$\mathbb{E}\|y_k - x^\star\|_p^p \le R_k^p.$$

which completes the induction. Since $\tau = p-1$, then $N_k = \left\lceil \frac{2^p p^p G^p}{\mu^p \mu_p R_0^{p(p-1)}} 2^{(k-1)\tau} \right\rceil$. Let $m = m(N)$. By maximality of m, $N < \sum_{k=1}^{m+1} N_k$. Then,

$$N < \frac{2^p p^p G^p}{\mu^p \mu_p R_0^{p(p-1)}} \sum_{k=0}^{m} 2^{k\tau} + (m+1) \leq \frac{2^p p^p G^p}{\mu^p \mu_p R_0^{p(p-1)}} \frac{2^{(m+1)\tau}}{2^\tau - 1} + (m+1). \tag{22}$$

Since $\tau \geq 1$ and $m \geq 1$, $m+1 \leq 2^m \leq 2^{m\tau}$. Hence

$$N \leq \left[\frac{2^p p^p G^p}{\mu^p \mu_p R_0^{p(p-1)}} \frac{2^\tau}{2^\tau - 1} + 1 \right] 2^{m\tau}.$$

Therefore,

$$2^{-m} \leq \left[\frac{2^p p^p G^p}{\mu^p \mu_p R_0^{p(p-1)}} \frac{2^\tau}{(2^\tau - 1)N} + \frac{1}{N} \right]^{1/\tau}.$$

From the epoch guarantee,

$$\mathbb{E}[F(\hat{x}_N) - F(x^\star)] \leq \frac{\mu}{p} R_m^p = \frac{\mu}{p} R_0^p 2^{-m}.$$

Therefore,

$$\mathbb{E}[F(\hat{x}_N) - F(x^\star)] \leq \frac{\mu}{p} R_0^p \left[\frac{2^p G^p}{\mu^p \mu_p R_0^{p(p-1)}} \frac{2^\tau}{(2^\tau - 1)N} + \frac{1}{N} \right]^{1/\tau}.$$

Using $R_0 \leq p^{q-1} \left(\frac{G}{\mu}\right)^{1/(p-1)}$, both terms are bounded by $C_p \frac{G^q}{\mu^{1/(p-1)} N^{1/(p-1)}}$, for a constant C_p depending only on p. Thus,

$$\mathbb{E}[F(\hat{x}_N) - F(x^\star)] \leq C_p \frac{G^q}{\mu^{1/(p-1)} N^{1/(p-1)}}.$$

Finally, uniform convexity gives

$$\mathbb{E}\|\hat{x}_N - x^\star\|_p^p \leq \frac{1}{\mu} \mathbb{E}[F(\hat{x}_N) - F(x^\star)],$$

and hence

$$\mathbb{E}\|\hat{x}_N - x^\star\|_p^p \leq C_p \frac{G^q}{\mu^q N^{1/(p-1)}}.$$

□

Corollary 4. *Algorithm 3 is an* ODC_p *algorithm with a constant* C_p *depending only on* p.

3.2 Projection-Efficient ℓ_p-Lipschitz Optimization

We combine our ℓ_p-Moreau envelope gradient estimator with an accelerated gradient method to reduce the number of expensive constrained updates over $\mathcal{X}$. We assume that $f : \mathbb{R}^d \to \mathbb{R}$ and its stochastic subgradient oracle satisfy Assumption 1 on $\mathbb{R}^d$, and that the initial point $x_0 \in \mathcal{X}$ satisfies $\|x_0 - x^\star\|_p \leq R$. The Moreau envelope is defined over $\mathbb{R}^d$, and its gradient is estimated using ODC_p ℓ_p balls. Only the outer updates involve the complicated set $\mathcal{X}$.

Our approach is inspired by that of Algorithm 3 in [ACJ+21], which involves applying a variant of Nesterov's accelerated gradient descent method on the Moreau envelope $f_\lambda(y) = \min_x\{f(x)+\frac{\lambda}{2}\|x-y\|_2^2\}$ in the Euclidean setting. Importantly, it may be seen that projections onto $\mathcal{X}$ are not needed for computing the Euclidean Moreau envelope itself, so that accurate approximations of ∇f_λ allow the outer accelerated method to find an $O(\epsilon)$-suboptimal point of f_λ using $T = O\left(\sqrt{\lambda R^2/\epsilon}\right)$ projections onto $\mathcal{X}$. To ensure that this point is also $O(\epsilon)$-suboptimal for the original objective f, one must choose $\lambda = G^2/\epsilon$, which yields the projection complexity $O(GR/\epsilon)$.

Algorithm 4 Stochastic AGD with ℓ_p Moreau envelope

Parameters $\lambda, M > 0$ and $T \geq 1$
$r_\lambda = 2(G/\lambda)^{1/(p-1)}$
$a_0 = 1,\ A_0 = 0,\ x_0 = z_0 = y_0 = x_{\text{init}}$
Bias tolerances $\{\delta_t\}_{t=1}^T$ and moment bounds $\{\sigma_t^q\}_{t=1}^T$
for $t = 0, \dots, T-1$ **do**
  $A_{t+1} = 2^{p+q-1}(p-1)a_{t+1}^q, \quad A_{t+1} = A_t + a_{t+1}$
  $y_{t+1} \leftarrow \dfrac{A_t}{A_{t+1}}x_t + \dfrac{a_{t+1}}{A_{t+1}}z_t$
  $\nabla f'_{p,\lambda}(y_{t+1}) \leftarrow \ell_p$-MorGradEst$(\widehat{\nabla} f, y_{t+1}, \lambda, \delta_{t+1}, \sigma_{t+1}^q, \mathbb{B}_{r_\lambda}(y_{t+1}))$
  $x_{t+1} \leftarrow \arg\min_{x\in\mathcal{X}}\left\{\langle \nabla f'_{p,\lambda}(y_{t+1}), x - y_{t+1}\rangle + \dfrac{M}{p}\|x - y_{t+1}\|_p^p\right\}$
  $z_{t+1} \leftarrow \arg\min_{z\in\mathcal{X}}\left\{\dfrac{a_{t+1}}{M}\langle \nabla f'_{p,\lambda}(y_{t+1}), z\rangle + \omega_p(z, z_t)\right\}$
end for
return x_T

Throughout this section, define

$$\psi(u) = \frac{1}{p}\|u - x_{\text{init}}\|_p^p, \qquad \omega_p(u,v) = \psi(u) - \psi(v) - \langle \nabla\psi(v), u - v\rangle.$$

Lemma 7. *Let $\mathcal{X} \subseteq \mathbb{B}_R(0)$ be a nonempty closed convex set. Let $f : \mathbb{R}^d \to \mathbb{R}$ and $\widehat{\nabla} f$ satisfy Assumption 1. Let $x_0 = x_{\text{init}} \in \mathcal{X}$ satisfy $\|x_0 - x^\star\|_p \leq R$ for some $x^\star \in \mathcal{X}$. For any $\epsilon > 0$ and integer $T \geq 1$, choose $\lambda, M > 0$ such that*

$$L_\lambda = (p-1)\lambda^{q-1}G^{2-q}, \qquad M \geq 2L_\lambda^{p/2}\left(\frac{T}{\epsilon}\right)^{(p-2)/2}.$$

For $t = 0, \dots, T-1$, set

$$\delta_{t+1} = \frac{\epsilon}{2R}, \qquad \sigma_{t+1}^q = \frac{q\epsilon M^{1/(p-1)}a_{t+1}}{2^{q-1}A_{t+1}}. \tag{23}$$

Then the iterates of Algorithm 4 satisfy, for $t = 0, \dots, T-1$,

$$\begin{aligned}&\frac{A_{t+1}}{M}\mathbb{E}[f_{p,\lambda}(x_{t+1}) - f_{p,\lambda}(x^\star)] + \mathbb{E}[\omega_p(x^\star, z_{t+1})] \\ &\quad\leq \frac{A_t}{M}\mathbb{E}[f_{p,\lambda}(x_t) - f_{p,\lambda}(x^\star)] + \mathbb{E}[\omega_p(x^\star, z_t)] + \frac{3a_{t+1}}{M}\epsilon.\end{aligned} \tag{24}$$

Lemma 8 ([Zăl83])**.** *For any $x, y \in \mathbb{R}^n$ and $p \geq 2$, $\omega_p(y, x) \geq \frac{1}{2^{p+1}}\|y - x\|_p^p$.*

Lemma 9 ([Lin17, Section 12]). *For any vectors $a, b \in \mathbb{R}^n$ and $p \geq 2$,*

$$\left\langle |b|^{p-2}b - |a|^{p-2}a, b - a \right\rangle \geq 2^{2-p}\|b - a\|_p^p,$$

where $|a|^{p-2}a = (|a_1|^{p-2}a_1, \ldots, |a_n|^{p-2}a_n)$.

Lemma 10. *[ABJS26] With $A_0 = 0$, suppose that $a_{t+1} > 0$, $A_{t+1} = A_t + a_{t+1}$, and $A_{t+1} = 2^{p+q-1}(p-1)a_{t+1}^q$. Then, for all $t \geq 0$,*

$$A_{t+1}^{1/p} \geq A_t^{1/p} + \frac{1}{p[2^{p+q-1}(p-1)]^{1/q}}.$$

Consequently, for every $T \geq 1$,

$$A_T^{1/p} \geq A_0^{1/p} + \frac{T}{p[2^{p+q-1}(p-1)]^{1/q}}.$$

3.3 Proof of Lemma 7

Proof. The proof is generally inspired by those of [ABJS26, ACJ+21]. First, $x_t, z_t, y_{t+1} \in \mathcal{X}$ by construction and the convexity of $\mathcal{X}$. By the convexity of $f_{p,\lambda}$ and the identity $a_{t+1}(y_{t+1} - z_t) = A_t(x_t - y_{t+1})$, we have

$$\frac{A_{t+1}}{M}\left(f_{p,\lambda}(y_{t+1}) - f_{p,\lambda}(x^\star)\right) \leq \frac{A_t}{M}\left(f_{p,\lambda}(x_t) - f_{p,\lambda}(x^\star)\right) + \frac{a_{t+1}}{M}\left\langle \nabla f_{p,\lambda}(y_{t+1}), z_t - x^\star \right\rangle. \tag{25}$$

The optimality condition for z_{t+1} and the three-point identity give

$$\frac{a_{t+1}}{M}\left\langle \nabla f'_{p,\lambda}(y_{t+1}), z_{t+1} - x^\star \right\rangle \leq \omega_p(x^\star, z_t) - \omega_p(x^\star, z_{t+1}) - \omega_p(z_{t+1}, z_t). \tag{26}$$

By the L_λ-smoothness of $f_{p,\lambda}$ and the choice of M, for all $u, v \in \mathbb{R}^d$,

$$\begin{aligned} f_{p,\lambda}(u) &\leq f_{p,\lambda}(v) + \langle \nabla f_{p,\lambda}(v), u - v\rangle + \frac{L_\lambda}{2}\|u - v\|_p^2 \\ &\leq f_{p,\lambda}(v) + \langle \nabla f_{p,\lambda}(v), u - v\rangle + \frac{M}{2p}\|u - v\|_p^p + \frac{\epsilon}{T}. \end{aligned} \tag{27}$$

On the other hand, by Young's inequality we have,

$$\begin{aligned} &\left\langle \nabla f_{p,\lambda}(y_{t+1}) - \nabla f'_{p,\lambda}(y_{t+1}), x_{t+1} - y_{t+1} \right\rangle \\ &\quad \leq \frac{M}{2p}\|x_{t+1} - y_{t+1}\|_p^p + \frac{2^{q-1}}{qM^{q-1}}\left\|\nabla f_{p,\lambda}(y_{t+1}) - \nabla f'_{p,\lambda}(y_{t+1})\right\|_q^q. \end{aligned}$$

By the convexity of $\mathcal{X}$,

$$\frac{A_t}{A_{t+1}}x_t + \frac{a_{t+1}}{A_{t+1}}z_{t+1} \in \mathcal{X}$$

With the optimality of x_{t+1} and (27), we obtain

$$\begin{aligned} \frac{A_{t+1}}{M}\left(f_{p,\lambda}(x_{t+1}) - f_{p,\lambda}(y_{t+1})\right) \leq \frac{a_{t+1}}{M}\left\langle \nabla f'_{p,\lambda}(y_{t+1}), z_{t+1} - z_t \right\rangle + \frac{a_{t+1}^p}{pA_{t+1}^{p-1}}\|z_{t+1} - z_t\|_p^p \\ + \frac{A_{t+1}\epsilon}{MT} + \frac{2^{q-1}A_{t+1}}{qM^q}\left\|\nabla f_{p,\lambda}(y_{t+1}) - \nabla f'_{p,\lambda}(y_{t+1})\right\|_q^q. \end{aligned} \tag{28}$$

Now, by Lemma 8, $\omega_p(u,v) \geq \frac{2^{2-p}}{p}\|u-v\|_p^p$. Moreover, the definition of A_{t+1} implies

$$\frac{a_{t+1}^p}{pA_{t+1}^{p-1}} = \frac{1}{p[2^{p+q-1}(p-1)]^{p-1}} \leq \frac{2^{2-p}}{p}.$$

Thus

$$\frac{a_{t+1}^p}{pA_{t+1}^{p-1}}\|z_{t+1}-z_t\|_p^p \leq \omega_p(z_{t+1},z_t).$$

Combining (25), (26), and (28), the terms involving $\langle \nabla f'_{p,\lambda}(y_{t+1}), z_{t+1}-z_t\rangle$ cancel. Taking conditional expectations, applying theorem 1, and using $\|z_t - x^\star\|_p \leq 2R$, we obtain

$$\begin{aligned}\frac{A_{t+1}}{M}\mathbb{E}[f_{p,\lambda}(x_{t+1}) - f_{p,\lambda}(x^\star)] + \mathbb{E}[\omega_p(x^\star, z_{t+1})] \leq \frac{A_t}{M}\mathbb{E}[f_{p,\lambda}(x_t) - f_{p,\lambda}(x^\star)] + \mathbb{E}[\omega_p(x^\star, z_t)] \\ + \frac{2Ra_{t+1}}{M}\delta_{t+1} + \frac{A_{t+1}\epsilon}{MT} + \frac{2^{q-1}A_{t+1}}{qM^q}\sigma_{t+1}^q.\end{aligned} \tag{29}$$

To bound the last three terms, observe that

$$A_k^{1/p} - A_{k-1}^{1/p} \leq \frac{a_k}{A_k^{1/q}} = \frac{1}{[2^{p+q-1}(p-1)]^{1/q}}.$$

Since $A_0 = 0$, summing this inequality gives

$$\frac{A_{t+1}}{a_{t+1}} = [2^{p+q-1}(p-1)]^{1/q}A_{t+1}^{1/p} \leq t+1 \leq T.$$

Substituting (23), we therefore have

$$\frac{2Ra_{t+1}}{M}\delta_{t+1} + \frac{A_{t+1}\epsilon}{MT} + \frac{2^{q-1}A_{t+1}}{qM^q}\sigma_{t+1}^q = \frac{2a_{t+1}\epsilon}{M} + \frac{A_{t+1}\epsilon}{MT} \leq \frac{3a_{t+1}\epsilon}{M}.$$

Combining this with (29) proves (24). □

Theorem 5. *Under the conditions of Lemma 7, let $x^\star \in \operatorname{argmin}_{x\in\mathcal{X}} f(x)$. For any $0 < \varepsilon \leq GR$, with the parameter choices specified below, Algorithm 4 computes $x_T \in \mathcal{X}$ such that $\mathbb{E}[f(x_T)] \leq f(x^\star) + \varepsilon$, using*

$$T = O\left(\left(\frac{GR}{\varepsilon}\right)^{2p/(p+2)}\right)$$

outer iterations and $N_{\widehat{\nabla} f}$ stochastic subgradient computations, where

$$\mathbb{E}[N_{\widehat{\nabla} f}] = O\left(\left(\frac{GR}{\varepsilon}\right)^{p(p-1)}\left(1+\log\frac{GR}{\varepsilon}\right)\right).$$

Proof. Set $\epsilon = \varepsilon/5$. Since $A_0 = 0$, summing Lemma 7 from $t = 0$ to $T-1$ gives

$$\begin{aligned}\frac{A_T}{M}\mathbb{E}[f_{p,\lambda}(x_T) - f_{p,\lambda}(x^\star)] + \mathbb{E}[\omega_p(x^\star, z_T)] \\ \leq \omega_p(x^\star, z_0) + \sum_{t=0}^{T-1}\frac{3a_{t+1}}{M}\epsilon = \omega_p(x^\star, z_0) + \frac{3A_T}{M}\epsilon.\end{aligned}$$

Using $\omega_p(x^\star, z_T) \geq 0$, we obtain

$$\mathbb{E}[f_{p,\lambda}(x_T) - f_{p,\lambda}(x^\star)] \leq \frac{M}{A_T}\omega_p(x^\star, z_0) + 3\epsilon.$$

By the G-Lipschitz property of f, for any $y \in \mathbb{R}^d$,

$$0 \leq f(y) - f_{p,\lambda}(y) \leq \sup_{r \geq 0}\left\{Gr - \frac{\lambda}{p}r^p\right\} = \frac{G^q}{q\lambda^{q-1}}.$$

Since $f_{p,\lambda}(x^\star) \leq f(x^\star)$,

$$\mathbb{E}[f(x_T)] - f(x^\star) \leq \frac{M}{A_T}\omega_p(x^\star, z_0) + \frac{G^q}{q\lambda^{q-1}} + 3\epsilon.$$

Choose $\lambda = \left(\frac{G^q}{q\epsilon}\right)^{p-1}$. Then

$$\frac{G^q}{q\lambda^{q-1}} = \epsilon, \qquad L_\lambda = (p-1)\lambda^{q-1}G^{2-q} = \frac{p-1}{q}\frac{G^2}{\epsilon}.$$

By Lemma 10, we have

$$A_T \geq \frac{T^p}{p^p[2^{p+q-1}(p-1)]^{p-1}}.$$

Moreover, since $z_0 = x_{\text{init}}$,

$$\omega_p(x^\star, z_0) = \frac{1}{p}\|x^\star - x_{\text{init}}\|_p^p \leq \frac{R^p}{p}.$$

Taking

$$M = 2L_\lambda^{p/2}\left(\frac{T}{\epsilon}\right)^{(p-2)/2},$$

we therefore have

$$\frac{M}{A_T}\omega_p(x^\star, z_0) \leq \frac{2p^{p-1}[2^{p+q-1}(p-1)]^{p-1}L_\lambda^{p/2}R^p}{\epsilon^{(p-2)/2}T^{(p+2)/2}}.$$

Thus it suffices to choose

$$T = \left\lceil \left(2p^{p-1}[2^{p+q-1}(p-1)]^{p-1}\right)^{2/(p+2)}\left(\frac{L_\lambda R^2}{\epsilon}\right)^{p/(p+2)}\right\rceil.$$

Since $L_\lambda = \Theta(G^2/\epsilon)$, this gives

$$T = \Theta\left(\left(\frac{GR}{\epsilon}\right)^{2p/(p+2)}\right), \qquad \frac{M}{A_T}\omega_p(x^\star, z_0) \leq \epsilon.$$

Consequently,

$$\mathbb{E}[f(x_T)] - f(x^\star) \leq \epsilon + \epsilon + 3\epsilon = 5\epsilon = \varepsilon.$$

To finish, we bound the expected total number of stochastic subgradient oracle calls. Recall that

$$\delta_t = \frac{\epsilon}{2R}, \qquad \sigma_t^q = \frac{q\epsilon M^{1/(p-1)}a_t}{2^{q-1}A_t}, \qquad t = 1, \ldots, T.$$

The choices of T and M above imply $M = \Theta\big(\frac{\epsilon T^p}{R^p}\big)$. Since $p/q = p-1$ and $(q-1)(p-1) = 1$,

$$\sigma_t^p = \frac{q^{p-1}}{2}\epsilon^{p-1} M \left(\frac{a_t}{A_t}\right)^{p-1}.$$

As shown in the proof of Lemma 7, $A_t/a_t \le t \le T$. Hence

$$\sigma_t^p \ge \frac{q^{p-1}\epsilon^{p-1}M}{2T^{p-1}} = \Theta\left(\frac{\epsilon^p T}{R^p}\right).$$

Together with $\delta_t = \epsilon/(2R)$ and $T \ge 1$, this implies, uniformly over $t = 1, \dots, T$,

$$\min\{\delta_t, \sigma_t\} = \Theta(\epsilon/R), \qquad 1 + \log\frac{G}{\min\{\delta_t,\sigma_t\}} = O\left(1 + \log\frac{GR}{\epsilon}\right).$$

Also,

$$\sum_{t=1}^{T}\frac{G^p}{\sigma_t^p} = \frac{2G^p}{q^{p-1}\epsilon^{p-1}M}\sum_{t=1}^{T}\left(\frac{A_t}{a_t}\right)^{p-1} \le \frac{2G^pT^p}{q^{p-1}\epsilon^{p-1}M} = O\left(\left(\frac{GR}{\epsilon}\right)^p\right).$$

Thus, by theorem 1, the summation of stochastic gradient oracle is

$$\begin{aligned}
\mathbb{E}[N_{\widehat{\nabla} f}] &= O\left(T + \sum_{t=1}^{T}\frac{G^p}{\sigma_t^p}\left(\frac{G}{\min\{\delta_t,\sigma_t\}}\right)^{p(p-2)}\left(1+\log\frac{G}{\min\{\delta_t,\sigma_t\}}\right)\right)\\
&= O\left(T + \left(\frac{GR}{\epsilon}\right)^{p(p-1)}\left(1+\log\frac{GR}{\epsilon}\right)\right)\\
&= O\left(\left(\frac{GR}{\varepsilon}\right)^{p(p-1)}\left(1+\log\frac{GR}{\varepsilon}\right)\right),
\end{aligned}$$

where the last equality uses $\epsilon = \varepsilon/5$ and the bound on T. □

3.4 Convex Composite Minimization

Consider the convex composite optimization problem

$$\min_{x\in\mathcal{X}} \Psi(x) := \Lambda(x) + f(x), \tag{30}$$

where Λ is (L,κ)-weakly smooth with respect to $\|\cdot\|_p$ and f satisfies Assumption 1. We assume that there exists $x^\star \in \arg\min_{x\in\mathcal{X}}\Psi(x)$ such that $\|x_0 - x^\star\|_p \le R$. A closely related setting was studied in [DG24], who considered complementary composite minimization with Λ being (L,κ)-weakly smooth and f being uniformly convex. Their generalized AGD+ framework admits an ϵ-additive optimality gap in $O\left((LR^\kappa/\epsilon)^{\frac{p}{\kappa p - p + \kappa}}\right)$ iterations when specialized to ℓ_p geometry.

In this section, we extend the stochastic composite AGD framework of [ACJ+21] to the ℓ_p setting using Bregman proximal steps and our reduced-bias gradient estimator. Our method handles a nonsmooth stochastic component f without requiring uniform convexity, while preserving the accelerated outer-loop structure.

Lemma 11. *(Lemma 1 of [DG24]) Suppose Λ is (L,κ)-weakly smooth with respect to $||\cdot||_p$ for some $\kappa \in (1,2]$. Then, for $\delta > 0$, if $M \geq (\frac{2(p-\kappa)}{p\kappa\delta})^{\frac{p-\kappa}{\kappa}} L^{\frac{p}{\kappa}}$, then for all $x, y \in \mathcal{X}$, it holds that*

$$\Lambda(y) \leq \Lambda(x) + \langle \nabla\Lambda(x), y - x\rangle + \frac{M}{p}||y-x||_p^p + \frac{\delta}{2}.$$

Definition 4. *Let $D_h(z,y) := h(z) - h(y) - \langle \nabla h(y), z - y\rangle$. For $F(z) := f(z) + \lambda D_h(z,y)$, define*

$$P_{f,\lambda}^{\epsilon,h}(y) := \text{any } x \in \overline{\mathcal{X}} \text{ such that } \mathbb{E}[F(x)] \leq \min_{z\in\overline{\mathcal{X}}} F(z) + \epsilon. \tag{31}$$

Algorithm 5 Stochastic composite accelerated gradient descent

Input: A problem of the form (30) with $\Lambda, f, \nabla\Lambda, \widehat{\nabla} f$.
Parameters: Step size parameters $\beta_k = M\gamma_k^{p-1}$ and $\gamma_k = \frac{p}{k+p-1}$, iteration number N, approximation parameters $\{\epsilon_k, \delta_k, \sigma_k^q\}$, and $x_0 = v_0$ where $x_0 \in X$ satisfying $\|x_0 - x^*\|_p \leq R$.
Set $\bar{X} = X \cap B_p(v_0, R)$ and $h(z) = \frac{2^{p-2}}{p}\|z - v_0\|_p^p$.
for $k = 1, 2, \cdots, N$ **do**
  $y_k = (1-\gamma_k)x_{k-1} + \gamma_k v_{k-1}$
  $\bar{v}_k = P^{\epsilon_k,h}_{\bar{\Lambda}_k + f, \beta_k}(v_{k-1})$ for $\bar{\Lambda}_k(v) := \Lambda(y_k) + \langle \nabla\Lambda(y_k), v - y_k\rangle$
  $\psi_k(z) := \beta_k D_h(z, v_{k-1}) + \bar{\Lambda}_k(z)$
  $\widehat{g}_k = -\frac{1}{\beta_k}\, \ell_p\text{-MorGradEst}\Big(\widehat{\nabla} f, v_0, 2^{p-2}\beta_k, \beta_k\delta_k, \beta_k^q\sigma_k^q, \bar{X}\Big)$
  $v_k = \arg\min_{v\in\bar{X}} \{h(v) - \langle \widehat{g}_k, v\rangle\}$
  $x_k = (1-\gamma_k)x_{k-1} + \gamma_k \bar{v}_k$
end for
return x_N

Lemma 12. *Let f and $\widehat{\nabla} f$ satisfy Assumption 1, $p > 2$, and let*

$$v_k^\star = \arg\min_{z\in\bar{X}}\{f(z) + \psi_k(z)\}.$$

The procedure

$$\widehat{g}_k = -\frac{1}{\beta_k}\, \ell_p\text{-MorGradEst}\Big(\widehat{\nabla} f, \psi_k, v_0, 2^{p-2}\beta_k, \beta_k\delta_k, \beta_k^q\sigma_k^q, R, \bar{X}\Big)$$

outputs an estimator satisfying

$$\big\|\mathbb{E}\widehat{g}_k - \nabla h(v_k^\star)\big\|_q \leq \delta_k \quad \text{and} \quad \mathbb{E}\big\|\widehat{g}_k - \nabla h(v_k^\star)\big\|_q^q \leq \sigma_k^q,$$

using $N_{\widehat{\nabla} f}$ stochastic-gradient computations, where

$$\mathbb{E}N_{\widehat{\nabla} f} \leq N\left(1 + \frac{3}{2}\lfloor \log_2 T_{\max}\rfloor\right)$$

The parameters are

$$T_{\max} = \left\lceil 2\left(\frac{2^{p-1}(p-1)c^{1/p}R^{p-2}(G/\beta_k)^{1/(p-1)}}{\min\{\delta_k, \sigma_k\}}\right)^{p(p-1)}\right\rceil$$

and

$$N = \left\lceil \frac{2^{p^2+p+1}(p-1)^p c}{\sigma_k^p} \left(\frac{1+2^{1/(p-1)^2}}{1-2^{-(p-2)/(p-1)^2}} \right)^{p-1} R^{p(p-2)} \left(\frac{G}{\beta_k} \right)^q T_{\max}^{(p-2)/(p-1)} \right\rceil,$$

where c is the constant in the ODC_p *guarantee.*

Proof. The proof follows the MLMC argument in Lemma 3 and the averaging argument in Theorem 1. Since $f + \psi_k$ is β_k-uniformly convex, the ODC iterates satisfy

$$\mathbb{E}\|x_j - v_k^\star\|_p^p \le c(G/\beta_k)^q 2^{-j/(p-1)}.$$

Moreover, on $B_p(v_0, R)$,

$$\|\nabla h(u) - \nabla h(v)\|_q \le 2^{2p-4}(p-1)R^{p-2}\|u - v\|_p.$$

Consequently, the original MLMC estimator g', after normalization $Z = -g'/\beta_k$, satisfies

$$\|\mathbb{E}Z - \nabla h(v_k^\star)\|_q \le 2^{p-2+1/[p(p-1)]}(p-1)c^{1/p}R^{p-2}(G/\beta_k)^{1/(p-1)}T_{\max}^{-1/[p(p-1)]}.$$

Evaluating the geometric series in the same moment calculation gives

$$\mathbb{E}\|Z - \mathbb{E}Z\|_q^q \le \frac{2^{3q-2}\left(1+2^{1/(p-1)^2}\right)}{1-2^{-(p-2)/(p-1)^2}} \left[2^{p-2}(p-1)c^{1/p}R^{p-2}(G/\beta_k)^{1/(p-1)} \right]^q T_{\max}^{(p-2)/(p-1)^2}.$$

The choice of $T_{\max}$ makes the bias at most $\min\{\delta_k, \sigma_k\}/2$, and the choice of N makes the centered qth moment of the averaged estimator at most $(\sigma_k/2)^q$. Combining these bounds as in Theorem 1 proves the guarantees. Finally, each MLMC sample uses at most

$$1 + \sum_{j=1}^{\lfloor \log_2 T_{\max} \rfloor} 2^{-j}(2^j + 2^{j-1}) = 1 + \frac{3}{2}\lfloor \log_2 T_{\max} \rfloor$$

oracle calls in expectation, yielding the stated cost. □

Proposition 6. *Suppose x^* minimizes Ψ such that $\|x_0 - x^*\|_p \le R$, and let the iterates be generated by Algorithm 5. Suppose M satisfies Lemma 11 with approximation error $\eta > 0$. Choose*

$$\sigma_k^q = \frac{qR^p}{2N}, \qquad \delta_k = \frac{R^{p-1}}{4N}, \qquad \epsilon_k = \frac{p^{p-2}MR^p}{(k+p-1)^{p-1}N},$$

where also $\gamma_k = p/(k+p-1)$ and $\beta_k = M\gamma_k^{p-1}$. Further, for all $k \ge 2$, let $\Gamma_k = (1-\gamma_k)\Gamma_{k-1}$, where $\Gamma_1 = 1$. Then

$$\begin{aligned}\mathbb{E}[\Psi(x_N) - \Psi(x^*)] &\le \left(2^{p-1} + 2 + \frac{1}{p}\right)\frac{Mp^pR^p}{(N+p-1)^p} + \Gamma_N \sum_{k=1}^N \frac{\eta}{2\Gamma_k} \\ &= \left(2^{p-1} + 2 + \frac{1}{p}\right)\frac{Mp^pR^p}{(N+p-1)^p} + \frac{\eta(N+p)}{2(p+1)}.\end{aligned}$$

Proof. Let

$$\Phi_k(v) = \bar{\Lambda}_k(v) + f(v) + \beta_k D_h(v, v_{k-1}), \qquad v_k^\star = \arg\min_{v \in \bar{X}} \Phi_k(v).$$

Recall that

$$D_h(u,v) \geq \frac{1}{p}\|u-v\|_p^p.$$

By Lemma 11, the definitions of x_k, y_k, and $\beta_k = M\gamma_k^{p-1}$, we obtain

$$\begin{aligned}\Lambda(x_k) &\leq (1-\gamma_k)\Lambda(x_{k-1}) + \gamma_k\bar{\Lambda}_k(\bar{v}_k) + \frac{M\gamma_k^p}{p}\|\bar{v}_k - v_{k-1}\|_p^p + \frac{\eta}{2} \\ &\leq (1-\gamma_k)\Lambda(x_{k-1}) + \gamma_k\left[\bar{\Lambda}_k(\bar{v}_k) + \beta_k D_h(\bar{v}_k, v_{k-1})\right] + \frac{\eta}{2}.\end{aligned}$$

Adding the convexity inequality for f gives

$$\Psi(x_k) \leq (1-\gamma_k)\Psi(x_{k-1}) + \gamma_k\Phi_k(\bar{v}_k) + \frac{\eta}{2}.$$

For any fixed $u \in \bar{X}$, convexity of Λ and optimality of $v_k^\star$ give

$$\Psi(u) \geq \Phi_k(u) - \beta_k D_h(u, v_{k-1}), \qquad \Phi_k(u) \geq \Phi_k(v_k^\star) + \beta_k D_h(u, v_k^\star).$$

Consequently,

$$\begin{aligned}\Psi(x_k) - \Psi(u) \leq (1-\gamma_k)[\Psi(x_{k-1}) - \Psi(u)] + \gamma_k[\Phi_k(\bar{v}_k) - \Phi_k(v_k^\star)] \\ + \gamma_k\beta_k[D_h(u, v_{k-1}) - D_h(u, v_k^\star)] + \frac{\eta}{2}.\end{aligned}$$

Now, define

$$D_0 = D_h(u, v_0), \qquad D_k = \mathbb{E}\left[h(u) - h(v_k) - \langle \widehat{g}_k, u - v_k\rangle\right], \quad k \geq 1.$$

Optimality of v_k in the mirror update implies

$$h(u) - h(v_k) - \langle \widehat{g}_k, u - v_k\rangle \geq D_h(u, v_k) \geq 0.$$

Moreover, Young's inequality and $D_h(v_k, v_k^\star) \geq \|v_k - v_k^\star\|_p^p/p$ imply

$$\begin{aligned}&h(u) - h(v_k) - \langle \widehat{g}_k, u - v_k\rangle \\ &\quad \leq D_h(u, v_k^\star) + \langle \widehat{g}_k - \nabla h(v_k^\star), v_k^\star - u\rangle + \frac{1}{q}\|\widehat{g}_k - \nabla h(v_k^\star)\|_q^q.\end{aligned}$$

Conditioning on $\mathcal{F}_{k-1}$ and applying Lemma 12, together with $\|v_k^\star - u\|_p \leq 2R$, therefore gives

$$D_k \leq \mathbb{E}D_h(u, v_k^\star) + 2R\delta_k + \frac{\sigma_k^q}{q} = \mathbb{E}D_h(u, v_k^\star) + \frac{R^p}{N}$$

Also, $\mathbb{E}D_h(u, v_{k-1}) \leq D_{k-1}$. By the definition of the approximate proximal step,

$$\mathbb{E}[\Phi_k(\bar{v}_k) - \Phi_k(v_k^\star)] \leq \epsilon_k.$$

Taking expectations in the preceding recursion yields

$$\begin{aligned}\mathbb{E}[\Psi(x_k) - \Psi(u)] \leq (1-\gamma_k)\mathbb{E}[\Psi(x_{k-1}) - \Psi(u)] + \gamma_k\epsilon_k \\ + \gamma_k\beta_k(D_{k-1} - D_k) + \frac{\gamma_k\beta_k R^p}{N} + \frac{\eta}{2}.\end{aligned}$$

For $u, v \in \bar{X}$, Hölder's inequality gives

$$D_h(u,v) \le \frac{2^{p-2}}{p} R^p + \frac{2^{p-2}(p-1)}{p} R^p + 2^{p-2} R^p = 2^{p-1} R^p.$$

Hence, including $k = 0$,

$$0 \le D_k \le \left(2^{p-1} + 1\right) R^p.$$

Since $\gamma_1 = 1$, dividing the recursion by Γ_k and summing gives, with $a_k = \gamma_k \beta_k / \Gamma_k$,

$$\begin{aligned}\mathbb{E}[\Psi(x_N) - \Psi(u)] \le \Gamma_N \sum_{k=1}^{N} a_k (D_{k-1} - D_k) \\ + \frac{(1+1/p) R^p \Gamma_N}{N} \sum_{k=1}^{N} a_k + \Gamma_N \sum_{k=1}^{N} \frac{\eta}{2\Gamma_k},\end{aligned}$$

where we used $\epsilon_k = \beta_k R^p / (pN)$. By Bernoulli's inequality,

$$\frac{a_{k+1}}{a_k} = \frac{k+p}{k} \left(1 - \frac{1}{k+p}\right)^p \ge 1.$$

Therefore,

$$\begin{aligned}\sum_{k=1}^{N} a_k (D_{k-1} - D_k) &= a_1 D_0 + \sum_{k=1}^{N-1} (a_{k+1} - a_k) D_k - a_N D_N \\ &\le (2^{p-1} + 1) R^p a_N,\end{aligned}$$

and $\sum_{k=1}^{N} a_k \le N a_N$. Using

$$\Gamma_N a_N = \gamma_N \beta_N = M \gamma_N^p = \frac{M p^p}{(N+p-1)^p},$$

we conclude that

$$\mathbb{E}[\Psi(x_N) - \Psi(u)] \le \left(2^{p-1} + 2 + \frac{1}{p}\right) \frac{M p^p R^p}{(N+p-1)^p} + \Gamma_N \sum_{k=1}^{N} \frac{\eta}{2\Gamma_k}.$$

Finally, the recurrence for Γ_k gives

$$\Gamma_N \sum_{k=1}^{N} \frac{1}{\Gamma_k} = \frac{N+p}{p+1}.$$

Setting $u = x^*$ finishes the proof. □

Theorem 7. *Suppose the conditions of Proposition 6 hold and let Λ be (L, κ)-weakly smooth with $\kappa \in (1, 2]$. For a target accuracy $0 < \varepsilon \le L R^\kappa$, choose*

$$\eta = \frac{\varepsilon}{N}, \qquad M = \left[\frac{2(p-\kappa)}{p \kappa \eta}\right]^{\frac{p-\kappa}{\kappa}} L^{\frac{p}{\kappa}}.$$

With N chosen sufficiently large such that

$$N = \Theta\left(\left(\frac{LR^\kappa}{\varepsilon}\right)^{\frac{p}{\kappa p - p + \kappa}}\right),$$

Algorithm 5, using EpochDA for the inner computations, returns an iterate satisfying $\mathbb{E}[\Psi(x_N)] \leq \Psi(x^*) + \varepsilon$. The number of evaluations of $\nabla\Lambda$ is

$$N_{\nabla\Lambda} = N = O\left(\left(\frac{LR^\kappa}{\varepsilon}\right)^{\frac{p}{\kappa p - p + \kappa}}\right),$$

and the expected number of stochastic-gradient evaluations, including both inner computations, is

$$\mathbb{E}N_{\nabla f} = \widetilde{O}\left(N + \left(\frac{GR}{\varepsilon}\right)^p N^{p(p-2)}\right).$$

Proof. By Proposition 6,

$$\mathbb{E}[\Psi(x_N) - \Psi(x^*)] \leq \left(2^{p-1} + 2 + \frac{1}{p}\right)\frac{Mp^pR^p}{(N+p-1)^p} + \frac{\eta(N+p)}{2(p+1)}.$$

Choose $\eta = \varepsilon/N$. Since $N + p \leq (p+1)N$, the second term is at most $\varepsilon/2$. Substituting this choice into the definition of M gives

$$M = \left[\frac{2(p-\kappa)N}{p\kappa\varepsilon}\right]^{\frac{p-\kappa}{\kappa}} L^{\frac{p}{\kappa}}.$$

Now choose N sufficiently large such that

$$N \geq \left[2p^p\left(2^{p-1} + 2 + \frac{1}{p}\right)\left(\frac{2(p-\kappa)}{p\kappa}\right)^{\frac{p-\kappa}{\kappa}}\right]^{\frac{\kappa}{\kappa p - p + \kappa}}\left(\frac{LR^\kappa}{\varepsilon}\right)^{\frac{p}{\kappa p - p + \kappa}}.$$

Using $N + p - 1 \geq N$, this ensures

$$\left(2^{p-1} + 2 + \frac{1}{p}\right)\frac{Mp^pR^p}{(N+p-1)^p} \leq \frac{\varepsilon}{2}.$$

Therefore, $\mathbb{E}[\Psi(x_N) - \Psi(x^*)] \leq \varepsilon$. Each outer iteration evaluates $\nabla\Lambda$ once, so $N_{\nabla\Lambda} = N$. For the stochastic-gradient complexity, at each outer iteration k, substitute

$$\delta_k = \frac{R^{p-1}}{4N}, \qquad \sigma_k = R^{p-1}\left(\frac{q}{2N}\right)^{1/q}$$

into Lemma 12. For fixed $p > 2$, δ_k and σ_k scale as N^{-1} and $N^{-1/q}$, respectively, and $\min\{\delta_k, \sigma_k\} = \delta_k$. Consequently, the MLMC cost at iteration k is

$$\widetilde{O}\left(1 + \left(\frac{G}{\beta_k R^{p-1}}\right)^p N^{p^2-p-1}\right).$$

To compute $\bar{v}_k$, we choose $z_0 \in \arg\min_{z\in\bar{X}} \psi_k(z)$ and set $R_0 = (pG/\beta_k)^{1/(p-1)}$. An expected suboptimality of at most ϵ_k can then be achieved using $O(1 + G^p/(\beta_k\epsilon_k^{p-1}))$ stochastic subgradient calls, where $\epsilon_k = \beta_k R^p/(pN)$. The total cost at iteration k therefore satisfies

$$\mathbb{E}N_k = \widetilde{O}\left(1 + \left(\frac{G}{\beta_k R^{p-1}}\right)^p N^{p^2-p-1}\right).$$

Since $\beta_k = M\gamma_k^{p-1}$ and $\gamma_k \geq 1/N$, summing over the N outer iterations gives

$$\mathbb{E}N_{\nabla f} = \widetilde{O}\left(N + \left(\frac{G}{MR^{p-1}}\right)^p N^{2p(p-1)}\right).$$

Finally, the choices of M and N imply $M = \Theta(\varepsilon N^p/R^p)$. Substituting this relation yields

$$\mathbb{E}N_{\nabla f} = \widetilde{O}\left(N + \left(\frac{GR}{\varepsilon}\right)^p N^{p(p-2)}\right),$$

which finishes the proof. □

3.5 ℓ_p Accelerated Proximal Point Methods

In this section, we establish the guarantees of our stochastic accelerated ℓ_p proximal point method. We first analyze the NextLambda procedure, showing that it controls the proximal point movement while requiring only logarithmically many BROO calls. We then establish a potential decrease bound using the ℓ_p Bregman divergence and our reduced-bias Moreau gradient estimator. Combining these results, we obtain convergence and stochastic gradient query complexity guarantees for the accelerated proximal point method. We define

$$\tilde{P}_{f,\lambda}^{\varphi}(y) \in \{x \in X : \mathbb{E}[F(x)] \leq F(P_{f,\lambda}(y)) + \varphi, F(x) = f(x) + \frac{\lambda}{p}||x-y||_p^p\}$$

Algorithm 6 Stochastic accelerated proximal point method

Initialize $x_0 = v_0 = 0$, $A_0 \geq 0$, gradient estimator $\hat{\nabla} f$, function NextLambda
for $k = 0, 1, \ldots,$ **do**
  $\lambda_{k+1} = \text{NextLambda}(x_k, v_k, A_k)$
  Solve $(A_k + a_{k+1})^{p-1} = 2^{p-1}\lambda_{k+1}a_{k+1}^p$ for a_{k+1}
  $A_{k+1} = A_k + a_{k+1}$
  $y_k = \frac{A_k}{A_{k+1}}x_k + \frac{a_{k+1}}{A_{k+1}}v_k$
  $x_{k+1} = \tilde{P}_{f,\lambda_{k+1}}^{\varphi_{k+1}}(y_k)$
  $g_{k+1} = \text{MORGRADEST}(\hat{\nabla} f, y_k, \lambda_{k+1}, \delta_{k+1}, \sigma_{k+1}^q, X \cap \mathbb{B}_r(y_k))$
  $v_{k+1} = \text{argmin}_{v \in X}\{\frac{1}{2}a_{k+1}\langle g_{k+1}, v\rangle + D_\Phi(v, v_k)\}$, $\Phi(x) = \frac{1}{p}||x||_p^p$
  return x_{k+1} if $A_{k+1} \geq A_{\max}$ or $k+1 = K_{\max}$
end for

3.5.1 NextLambda Algorithm

Algorithm 7 NextLambda(x, v, A)

Input: $x, v \in \mathbb{R}^d$, $A \geq 0$, $\lambda_{\min}, \lambda_{\max}$, G, R, r
For $\lambda' > 0$, let $a_{\lambda'} > 0$ satisfy $(A+a_{\lambda'})^{p-1} = 2^{p-1}\lambda' a_{\lambda'}^p$, where $A_{\lambda'} = A+a_{\lambda'}$ and $y_{\lambda'} = \frac{A}{A_{\lambda'}}x + \frac{a_{\lambda'}}{A_{\lambda'}}v$
Define $\Delta(\lambda) = \left\|\mathcal{O}_{\lambda,r/17}(y_\lambda) - y_\lambda\right\|_p$.
$\lambda \leftarrow \lambda_{\max}$
while $\lambda \geq \lambda_{\min}$ and $\Delta(\lambda) \leq 13r/16$ **do** $\lambda \leftarrow \lambda/2$
end while
if $\lambda < \lambda_{\min}$ **then return** 2λ
end if
$\lambda_u = 2\lambda$, $\lambda_\ell = \lambda$, $\lambda_m = \sqrt{\lambda_u \lambda_\ell}$
if $\Delta(\lambda_\ell) \leq 15r/16$ **then return** λ_ℓ
end if
while $\Delta(\lambda_m) \notin [13r/16, 15r/16]$ and $\log \frac{\lambda_u}{\lambda_\ell} > \frac{r}{8\left(R + \frac{1}{p-1}\left(\frac{G}{\lambda_\ell}\right)^{1/(p-1)}\right)}$ **do**
    if $\Delta(\lambda_m) < 13r/16$ **then** $\lambda_u = \lambda_m$
    else $\lambda_\ell = \lambda_m$
    end if
    $\lambda_m = \sqrt{\lambda_u \lambda_\ell}$
end while
return λ_m

Here, we discuss our implementation of the subroutine NextLambda (inspired by [ACJ+21] and [CJJS21]), such that for $\hat{x}_{k+1} = P_{f,\lambda}(y_k)$, it holds that $\|\hat{x}_{k+1} - y_k\|_p \leq r$ and either $\|\hat{x}_{k+1} - y_k\|_p \geq 3r/4$ or $\lambda < 2\lambda_{\min}$. We let $J_p(z) = \nabla(\frac{1}{p}||z||_p^p) = (|z_i|^{p-2}z_i)_i$ and $\hat{x}_\lambda = \text{argmin}_x\{f(x) + \frac{\lambda}{p}||x - y_\lambda||_p^p\}$. Assume that f is G-Lipschitz with respect to the ℓ_p norm.

Definition 5. *We say that a mapping $\mathcal{O}_{\lambda,\rho}(\cdot) : \mathcal{X} \to \mathbb{R}^d$ is a Ball Regularized Optimization Oracle of radius r for f if, for every query point $\overline{x}$, regularization parameter $\lambda > 0$, and desired accuracy $\rho > 0$, it 4*

$\tilde{x} = \mathcal{O}_{\lambda,\rho}(\overline{x}) \in \mathcal{X} \cap \mathbb{B}_r(\overline{x})$ *such that*

$$f(\tilde{x}) + \frac{\lambda}{p}\|\tilde{x} - \overline{x}\|_p^p \leq \min_{x \in \mathbb{B}_r(\overline{x})} \left\{ f(x) + \frac{\lambda}{p}\|x - \overline{x}\|_p^p \right\} + \frac{\lambda 2^{2-p}}{p}\rho^p.$$

Lemma 13. *Let $\hat{x}_\lambda = \text{argmin}_z\{f(z) + \frac{\lambda}{p}||z - y_\lambda||_p^p\}$. For all $\lambda > 0$, it holds that*

$$||\hat{x}_\lambda - y_\lambda||_p \leq (\frac{G}{\lambda})^{1/(p-1)}$$

Proof. Let $\delta_\lambda = \hat{x}_\lambda - y_\lambda$. We know that for some $g_\lambda \in \partial f(\hat{x}_\lambda)$, we have that $g_\lambda + \lambda J_p(\delta_\lambda) = 0$. Since f is G-Lipschitz, it holds that $||g_\lambda||_q \leq G$, which implies that $\lambda||J_p(\delta_\lambda)||_q \leq G$. Since $||J_p(z)||_q = ||z||_p^{p-1}$, it holds that $\lambda\|\delta_\lambda\|_p^{p-1} \leq G$ from which the stated claim follows. □

Lemma 14. *Let $A \geq 0$ and $x, v \in \mathbb{R}^d$ satisfy $\|x - v\|_p \leq 2R$. For every $\lambda > 0$, let $a_\lambda > 0$ be the unique solution of $(A + a_\lambda)^{p-1} = 2^{p-1}\lambda a_\lambda^p$, and define $y_\lambda = \frac{Ax + a_\lambda v}{A + a_\lambda}$. Then*

$$\left\| \frac{d}{d\lambda} y_\lambda \right\|_p \leq \frac{R}{\lambda}.$$

Proof. Denote $\alpha_\lambda = a_\lambda/(A+a_\lambda)$. Taking logarithmic derivatives on both sides of $(A+a_\lambda)^{p-1} = 2^{p-1}\lambda a_\lambda^p$, we get that

$$(p-1)\frac{a'_\lambda}{A+a_\lambda} = \frac{1}{\lambda} + p\frac{a'_\lambda}{a_\lambda} \implies a'_\lambda = -\frac{a_\lambda(A+a_\lambda)}{\lambda(a_\lambda+(pA))}$$

Since $a_\lambda + pA \geq A + a_\lambda$, this means that $|a'| \leq a/\lambda$. Now, notice that

$$|\alpha'_\lambda| = |\frac{Aa'_\lambda}{(A+a_\lambda)^2}| \leq \frac{Aa_\lambda}{\lambda(A+a_\lambda)^2} \leq \frac{1}{2\lambda}$$

and since $y_\lambda = (1-\alpha_\lambda)x + \alpha_\lambda v$, this implies that

$$||\frac{d}{d\lambda}y_\lambda||_p = |\alpha'_\lambda| \cdot ||v-x||_p \leq \frac{R}{\lambda}$$

as $\|v-x\|_p \leq 2R$. □

Lemma 15. *Let $f : \mathbb{R}^d \to \mathbb{R}$ be convex and G-Lipschitz, let y_λ be continuously differentiable for $\lambda > 0$. For every $0 < \lambda_1 \leq \lambda_2$, it holds that*

$$|\|\delta_{\lambda_2}\|_p - \|\delta_{\lambda_1}\|_p| \leq \int_{\lambda_1}^{\lambda_2} \left\|\frac{d}{d\lambda}y_\lambda\right\|_p d\lambda + \frac{1}{p-1}\int_{\lambda_1}^{\lambda_2} \frac{\|\delta_\lambda\|_p}{\lambda}\, d\lambda.$$

Proof. We first smooth f and use a nondegenerate approximation of the p-power regularizer. Let

$$f_\sigma(x) = \mathbb{E}[f(x+\nu_\sigma)], \qquad \nu_\sigma \sim \mathcal{N}(0, \sigma^2 I),$$

and, for $\epsilon > 0$, define $\phi_\epsilon(z) = \frac{1}{p}\sum_{i=1}^d \left[(z_i^2+\epsilon^2)^{p/2} - \epsilon^p\right]$. Let

$$\hat{x}_\lambda^{\sigma,\epsilon} = \operatorname*{argmin}_x \left\{f_\sigma(x) + \lambda\phi_\epsilon(x-y_\lambda)\right\}, \qquad \delta_\lambda^{\sigma,\epsilon} = \hat{x}_\lambda^{\sigma,\epsilon} - y_\lambda,$$

and define $\rho_\epsilon(z) = \left(\sum_{i=1}^d (z_i^2+\epsilon^2)^{p/2}\right)^{1/p}$. Notice that $\phi_\epsilon(z) = \frac{1}{p}\left(\rho_\epsilon(z)^p - d\epsilon^p\right)$ and $\rho_\epsilon(z) \to \|z\|_p$ as $\epsilon \to 0$. Since f_σ is smooth and convex and ϕ_ϵ has a positive definite Hessian, $\hat{x}_\lambda^{\sigma,\epsilon}$ is uniquely defined and differentiable in λ. Let $J_{p,\epsilon}(z) = \nabla\phi_\epsilon(z)$, so that

$$[J_{p,\epsilon}(z)]_i = z_i(z_i^2+\epsilon^2)^{p/2-1}.$$

The Hessian of ϕ_ϵ is diagonal, with $[\nabla^2\phi_\epsilon(z)]_{ii} = (z_i^2+\epsilon^2)^{p/2-2}\left[\epsilon^2 + (p-1)z_i^2\right] > 0$. Now, the first-order optimality gives

$$\nabla f_\sigma(\hat{x}_\lambda^{\sigma,\epsilon}) + \lambda J_{p,\epsilon}(\delta_\lambda^{\sigma,\epsilon}) = 0.$$

Differentiating with respect to λ gives

$$\left[\nabla^2 f_\sigma(\hat{x}_\lambda^{\sigma,\epsilon}) + \lambda\nabla^2\phi_\epsilon(\delta_\lambda^{\sigma,\epsilon})\right]\frac{d}{d\lambda}\delta_\lambda^{\sigma,\epsilon} = -J_{p,\epsilon}(\delta_\lambda^{\sigma,\epsilon}) - \nabla^2 f_\sigma(\hat{x}_\lambda^{\sigma,\epsilon})\frac{d}{d\lambda}y_\lambda.$$

For brevity, write

$$H = \nabla^2 f_\sigma(\hat{x}_\lambda^{\sigma,\epsilon}), \qquad D = \nabla^2\phi_\epsilon(\delta_\lambda^{\sigma,\epsilon}).$$

By construction we have $H \succeq 0$ and $D \succ 0$, and

$$(H+\lambda D)\frac{d}{d\lambda}\delta_\lambda^{\sigma,\epsilon} = -J_{p,\epsilon}(\delta_\lambda^{\sigma,\epsilon}) - H\frac{d}{d\lambda}y_\lambda.$$

Since $\nabla \rho_\epsilon(z) = \frac{J_{p,\epsilon}(z)}{\rho_\epsilon(z)^{p-1}}$, we have $\frac{d}{d\lambda}\rho_\epsilon(\delta_\lambda^{\sigma,\epsilon}) = \frac{J_{p,\epsilon}(\delta_\lambda^{\sigma,\epsilon})^\top \frac{d}{d\lambda}\delta_\lambda^{\sigma,\epsilon}}{\rho_\epsilon(\delta_\lambda^{\sigma,\epsilon})^{p-1}}$. Therefore,

$$\left|\frac{d}{d\lambda}\rho_\epsilon(\delta_\lambda^{\sigma,\epsilon})\right| \le \frac{J_{p,\epsilon}(\delta_\lambda^{\sigma,\epsilon})^\top (H+\lambda D)^{-1} J_{p,\epsilon}(\delta_\lambda^{\sigma,\epsilon})}{\rho_\epsilon(\delta_\lambda^{\sigma,\epsilon})^{p-1}} + \frac{\left|J_{p,\epsilon}(\delta_\lambda^{\sigma,\epsilon})^\top (H+\lambda D)^{-1} H \frac{d}{d\lambda} y_\lambda\right|}{\rho_\epsilon(\delta_\lambda^{\sigma,\epsilon})^{p-1}}.$$

We first bound the first term. Since $H + \lambda D \succeq \lambda D$ and $D \succ 0$, we have $(H+\lambda D)^{-1} \preceq \frac{1}{\lambda} D^{-1}$. Hence

$$J_{p,\epsilon}(\delta_\lambda^{\sigma,\epsilon})^\top (H+\lambda D)^{-1} J_{p,\epsilon}(\delta_\lambda^{\sigma,\epsilon}) \le \frac{1}{\lambda} J_{p,\epsilon}(\delta_\lambda^{\sigma,\epsilon})^\top D^{-1} J_{p,\epsilon}(\delta_\lambda^{\sigma,\epsilon}).$$

For every coordinate,

$$\frac{[J_{p,\epsilon}(\delta_\lambda^{\sigma,\epsilon})]_i^2}{D_{ii}} = \frac{|\delta_{\lambda,i}^{\sigma,\epsilon}|^2 \left(|\delta_{\lambda,i}^{\sigma,\epsilon}|^2 + \epsilon^2\right)^{p/2}}{\epsilon^2 + (p-1)|\delta_{\lambda,i}^{\sigma,\epsilon}|^2} \le \frac{1}{p-1}\left(|\delta_{\lambda,i}^{\sigma,\epsilon}|^2 + \epsilon^2\right)^{p/2}.$$

Therefore,

$$J_{p,\epsilon}(\delta_\lambda^{\sigma,\epsilon})^\top D^{-1} J_{p,\epsilon}(\delta_\lambda^{\sigma,\epsilon}) \le \frac{\rho_\epsilon(\delta_\lambda^{\sigma,\epsilon})^p}{p-1},$$

and thus

$$J_{p,\epsilon}(\delta_\lambda^{\sigma,\epsilon})^\top (H+\lambda D)^{-1} J_{p,\epsilon}(\delta_\lambda^{\sigma,\epsilon}) \le \frac{\rho_\epsilon(\delta_\lambda^{\sigma,\epsilon})^p}{(p-1)\lambda}.$$

We next bound the second term. Let $S = D^{-1/2} H D^{-1/2}$. Since $H \succeq 0$, we have $S \succeq 0$, and

$$D^{-1/2} H (H+\lambda D)^{-1} J_{p,\epsilon}(\delta_\lambda^{\sigma,\epsilon}) = S(S+\lambda I)^{-1} D^{-1/2} J_{p,\epsilon}(\delta_\lambda^{\sigma,\epsilon}).$$

Every eigenvalue of $S(S+\lambda I)^{-1}$ is in $[0,1]$. Therefore,

$$\left\|D^{-1/2} H (H+\lambda D)^{-1} J_{p,\epsilon}(\delta_\lambda^{\sigma,\epsilon})\right\|_2 \le \left\|D^{-1/2} J_{p,\epsilon}(\delta_\lambda^{\sigma,\epsilon})\right\|_2.$$

It follows from Cauchy–Schwarz that

$$\left|J_{p,\epsilon}(\delta_\lambda^{\sigma,\epsilon})^\top (H+\lambda D)^{-1} H \frac{d}{d\lambda} y_\lambda\right| \le \left[J_{p,\epsilon}(\delta_\lambda^{\sigma,\epsilon})^\top D^{-1} J_{p,\epsilon}(\delta_\lambda^{\sigma,\epsilon})\right]^{1/2} \left[\left(\frac{d}{d\lambda} y_\lambda\right)^\top D \left(\frac{d}{d\lambda} y_\lambda\right)\right]^{1/2}.$$

We already showed that $J_{p,\epsilon}(\delta_\lambda^{\sigma,\epsilon})^\top D^{-1} J_{p,\epsilon}(\delta_\lambda^{\sigma,\epsilon}) \le \frac{\rho_\epsilon(\delta_\lambda^{\sigma,\epsilon})^p}{p-1}$. Moreover,

$$D_{ii} = \left(|\delta_{\lambda,i}^{\sigma,\epsilon}|^2 + \epsilon^2\right)^{p/2-2}\left[\epsilon^2 + (p-1)|\delta_{\lambda,i}^{\sigma,\epsilon}|^2\right] \le (p-1)\left(|\delta_{\lambda,i}^{\sigma,\epsilon}|^2 + \epsilon^2\right)^{(p-2)/2}.$$

Hence, with Hölder's inequality, we have

$$\left(\frac{d}{d\lambda} y_\lambda\right)^\top D \left(\frac{d}{d\lambda} y_\lambda\right) \le (p-1)\sum_i \left(|\delta_{\lambda,i}^{\sigma,\epsilon}|^2 + \epsilon^2\right)^{(p-2)/2} \left|\frac{d}{d\lambda} y_{\lambda,i}\right|^2 \le (p-1)\rho_\epsilon(\delta_\lambda^{\sigma,\epsilon})^{p-2} \left\|\frac{d}{d\lambda} y_\lambda\right\|_p^2$$

Combining the preceding two inequalities gives

$$\left|J_{p,\epsilon}(\delta_\lambda^{\sigma,\epsilon})^\top (H+\lambda D)^{-1} H \frac{d}{d\lambda} y_\lambda\right| \le \rho_\epsilon(\delta_\lambda^{\sigma,\epsilon})^{p-1} \left\|\frac{d}{d\lambda} y_\lambda\right\|_p.$$

Consequently,

$$\left|\frac{d}{d\lambda}\rho_\epsilon(\delta_\lambda^{\sigma,\epsilon})\right| \le \left\|\frac{d}{d\lambda}y_\lambda\right\|_p + \frac{\rho_\epsilon(\delta_\lambda^{\sigma,\epsilon})}{(p-1)\lambda}.$$

Integrating from λ_1 to λ_2 gives

$$\left|\rho_\epsilon(\delta_{\lambda_2}^{\sigma,\epsilon}) - \rho_\epsilon(\delta_{\lambda_1}^{\sigma,\epsilon})\right| \le \int_{\lambda_1}^{\lambda_2} \left\|\frac{d}{d\lambda}y_\lambda\right\|_p d\lambda + \frac{1}{p-1}\int_{\lambda_1}^{\lambda_2} \frac{\rho_\epsilon(\delta_\lambda^{\sigma,\epsilon})}{\lambda}\, d\lambda.$$

It remains to let $\sigma, \epsilon \to 0$. Since f is G-Lipschitz, f_σ is also G-Lipschitz and

$$|f_\sigma(x) - f(x)| \le G\mathbb{E}\|\nu_\sigma\|_p \longrightarrow 0$$

uniformly in x. Now, by the definition of $\hat{x}_\lambda^{\sigma,\epsilon}$,

$$f_\sigma(\hat{x}_\lambda^{\sigma,\epsilon}) + \lambda\phi_\epsilon(\delta_\lambda^{\sigma,\epsilon}) = f_\sigma(\hat{x}_\lambda^{\sigma,\epsilon}) + \lambda\phi_\epsilon(\hat{x}_\lambda^{\sigma,\epsilon} - y_\lambda) \le f_\sigma(y_\lambda) + \lambda\phi_\epsilon(y_\lambda - y_\lambda) = f_\sigma(y_\lambda) + \lambda\phi_\epsilon(0)$$

We now compute $\phi_\epsilon(0)$. Since $\rho_\epsilon(0)^p = \sum_{i=1}^d (0^2+\epsilon^2)^{p/2} = \sum_{i=1}^d \epsilon^p = d\epsilon^p$, we have $\phi_\epsilon(0) = \frac{1}{p}\left[\rho_\epsilon(0)^p - d\epsilon^p\right] = \frac{1}{p}\left[d\epsilon^p - d\epsilon^p\right] = 0$. Therefore,

$$f_\sigma(\hat{x}_\lambda^{\sigma,\epsilon}) + \lambda\phi_\epsilon(\delta_\lambda^{\sigma,\epsilon}) \le f_\sigma(y_\lambda)$$

Using $\phi_\epsilon(z) = \frac{1}{p}\left[\rho_\epsilon(z)^p - d\epsilon^p\right]$ again, we obtain

$$f_\sigma(\hat{x}_\lambda^{\sigma,\epsilon}) + \frac{\lambda}{p}\left[\rho_\epsilon(\delta_\lambda^{\sigma,\epsilon})^p - d\epsilon^p\right] \le f_\sigma(y_\lambda) \Rightarrow \frac{\lambda}{p}\left[\rho_\epsilon(\delta_\lambda^{\sigma,\epsilon})^p - d\epsilon^p\right] \le f_\sigma(y_\lambda) - f_\sigma(\hat{x}_\lambda^{\sigma,\epsilon})$$

Since f_σ is G-Lipschitz, $f_\sigma(y_\lambda) - f_\sigma(\hat{x}_\lambda^{\sigma,\epsilon}) \le G\left\|y_\lambda - \hat{x}_\lambda^{\sigma,\epsilon}\right\|_p \le G\|\delta_\lambda^{\sigma,\epsilon}\|_p$. Moreover,

$$\rho_\epsilon(\delta_\lambda^{\sigma,\epsilon})^p = \sum_{i=1}^d \left(|\delta_{\lambda,i}^{\sigma,\epsilon}|^2 + \epsilon^2\right)^{p/2} \ge \sum_{i=1}^d |\delta_{\lambda,i}^{\sigma,\epsilon}|^p = \|\delta_\lambda^{\sigma,\epsilon}\|_p^p.$$

Hence$\|\delta_\lambda^{\sigma,\epsilon}\|_p \le \rho_\epsilon(\delta_\lambda^{\sigma,\epsilon})$. Combining the preceding inequalities gives

$$\frac{\lambda}{p}\left[\rho_\epsilon(\delta_\lambda^{\sigma,\epsilon})^p - d\epsilon^p\right] \le G\rho_\epsilon(\delta_\lambda^{\sigma,\epsilon}) \Rightarrow \frac{\lambda}{p}\rho_\epsilon(\delta_\lambda^{\sigma,\epsilon})^p \le G\rho_\epsilon(\delta_\lambda^{\sigma,\epsilon}) + \frac{\lambda d}{p}\epsilon^p$$

Since $\lambda \in [\lambda_1, \lambda_2]$ with $\lambda_1 > 0$, this implies that $\rho_\epsilon(\delta_\lambda^{\sigma,\epsilon})$ is uniformly bounded over $\lambda \in [\lambda_1, \lambda_2]$ for sufficiently small σ, ϵ. Therefore the corresponding minimizers remain in a common bounded set. On every bounded set,

$$f_\sigma(x) + \lambda\phi_\epsilon(x - y_\lambda) \longrightarrow f(x) + \frac{\lambda}{p}\|x - y_\lambda\|_p^p$$

uniformly as $\sigma, \epsilon \to 0$. Since the limiting objective is strictly convex, its minimizer is unique. Also, $\hat{x}_\lambda^{\sigma,\epsilon} \longrightarrow \hat{x}_\lambda$ and $\delta_\lambda^{\sigma,\epsilon} \longrightarrow \delta_\lambda$ for every $\lambda \in [\lambda_1, \lambda_2]$. Moreover, $\rho_\epsilon(\delta_\lambda^{\sigma,\epsilon}) \longrightarrow \|\delta_\lambda\|_p$. The uniform bound above and $\lambda \ge \lambda_1 > 0$ allow us to apply dominated convergence to the integral. Taking $\sigma, \epsilon \to 0$ yields

$$|\|\delta_{\lambda_2}\|_p - \|\delta_{\lambda_1}\|_p| \le \int_{\lambda_1}^{\lambda_2} \left\|\frac{d}{d\lambda}y_\lambda\right\|_p d\lambda + \frac{1}{p-1}\int_{\lambda_1}^{\lambda_2} \frac{\|\delta_\lambda\|_p}{\lambda}\, d\lambda,$$

which proves the claim. □

Lemma 16. *Under the same context as Lemma 15, let* $\hat{\Delta}(\lambda) = \|\hat{x}_\lambda - y_\lambda\|_p$. *If* $\|x - v\|_p \le 2R$, *then for every* $0 < \lambda_1 \le \lambda_2$,

$$\left|\hat{\Delta}(\lambda_1) - \hat{\Delta}(\lambda_2)\right| \le \left[R + \frac{1}{p-1}\left(\frac{G}{\lambda_1}\right)^{1/(p-1)}\right] \log \frac{\lambda_2}{\lambda_1}.$$

Proof. By Lemma 15,

$$\left|\hat{\Delta}(\lambda_1) - \hat{\Delta}(\lambda_2)\right| \le \int_{\lambda_1}^{\lambda_2} \left\|\frac{d}{d\lambda} y_\lambda\right\|_p d\lambda + \frac{1}{p-1}\int_{\lambda_1}^{\lambda_2} \frac{\hat{\Delta}(\lambda)}{\lambda}\, d\lambda.$$

Lemma 14 gives $\left\|\frac{d}{d\lambda} y_\lambda\right\|_p \le \frac{R}{\lambda}$, and Lemma 13 gives

$$\hat{\Delta}(\lambda) \le \left(\frac{G}{\lambda}\right)^{1/(p-1)} \le \left(\frac{G}{\lambda_1}\right)^{1/(p-1)}$$

for every $\lambda \in [\lambda_1, \lambda_2]$. Therefore,

$$\left|\hat{\Delta}(\lambda_1) - \hat{\Delta}(\lambda_2)\right| \le \left[R + \frac{1}{p-1}\left(\frac{G}{\lambda_1}\right)^{1/(p-1)}\right] \int_{\lambda_1}^{\lambda_2} \frac{d\lambda}{\lambda},$$

which proves the claim. □

Lemma 17. *Let* $\hat{x}_\lambda = \operatorname{argmin}_x \left\{f(x) + \frac{\lambda}{p}\|x - y_\lambda\|_p^p\right\}$ *and* $\hat{\Delta}(\lambda) = \|\hat{x}_\lambda - y_\lambda\|_p$ *Define*

$$\Delta(\lambda) = \left\|\mathcal{O}_{\lambda,\rho}(y_\lambda) - y_\lambda\right\|_p.$$

If either $\hat{\Delta}(\lambda) \le r - \rho$ *or* $\Delta(\lambda) < r - \rho$, *then*

$$\operatorname{bprox}_{\lambda,r}^f(y_\lambda) = \operatorname{prox}_\lambda^f(y_\lambda) \quad \textit{and} \quad |\Delta(\lambda) - \hat{\Delta}(\lambda)| \le \rho.$$

where $\operatorname{bprox}_{\lambda,r}^f(y_\lambda) = x_{\lambda,r}^\star$ *and* $\operatorname{prox}_\lambda^f(y_\lambda) = \hat{x}_\lambda$

Proof. Let

$$F_\lambda(x) = f(x) + \frac{\lambda}{p}\|x - y_\lambda\|_p^p, \qquad x_{\lambda,r}^* = \operatorname*{argmin}_{x \in \mathbb{B}_r(y_\lambda)} F_\lambda(x).$$

By Definition 5,

$$F_\lambda(\mathcal{O}_{\lambda,\rho}(y_\lambda)) \le F_\lambda(x_{\lambda,r}^*) + \frac{\lambda 2^{2-p}}{p}\rho^p. \tag{32}$$

Case 1: $\hat{\Delta}(\lambda) \le r - \rho$. Since $\hat{x}_\lambda$ belongs to $\mathbb{B}_r(y_\lambda)$ and minimizes F_λ globally, we have $x_{\lambda,r}^* = \hat{x}_\lambda$. By the p-uniform convexity of $\frac{1}{p}\|\cdot\|_p^p$ and the global optimality of $\hat{x}_\lambda$,

$$F_\lambda(\mathcal{O}_{\lambda,\rho}(y_\lambda)) - F_\lambda(\hat{x}_\lambda) \ge \frac{\lambda 2^{2-p}}{p}\|\mathcal{O}_{\lambda,\rho}(y_\lambda) - \hat{x}_\lambda\|_p^p.$$

Combining this with (32) gives

$$\|\mathcal{O}_{\lambda,\rho}(y_\lambda) - \hat{x}_\lambda\|_p \le \rho. \tag{33}$$

Case 2: $\Delta(\lambda) < r - \rho$. By the definition of $\Delta(\lambda)$, $\mathcal{O}_{\lambda,\rho}(y_\lambda) \in \mathbb{B}_r(y_\lambda)$. Since $x^*_{\lambda,r}$ minimizes F_λ over this convex ball, the constrained first-order optimality condition and p-uniform convexity imply

$$F_\lambda(\mathcal{O}_{\lambda,\rho}(y_\lambda)) - F_\lambda(x^*_{\lambda,r}) \geq \frac{\lambda 2^{2-p}}{p} \|\mathcal{O}_{\lambda,\rho}(y_\lambda) - x^*_{\lambda,r}\|_p^p.$$

Together with (32), this yields

$$\|\mathcal{O}_{\lambda,\rho}(y_\lambda) - x^*_{\lambda,r}\|_p \leq \rho.$$

Consequently,

$$\|x^*_{\lambda,r} - y_\lambda\|_p \leq \|x^*_{\lambda,r} - \mathcal{O}_{\lambda,\rho}(y_\lambda)\|_p + \Delta(\lambda) \leq \rho + \Delta(\lambda) < r.$$

Thus $x^*_{\lambda,r}$ lies in the interior of $\mathbb{B}_r(y_\lambda)$. By convexity, an interior minimizer over the ball is also a global minimizer of F_λ. Hence $x^*_{\lambda,r} = \hat{x}_\lambda$, and (33) holds in this case as well. Therefore, in both cases,

$$\mathrm{bprox}^f_{\lambda,r}(y_\lambda) = \mathrm{prox}^f_\lambda(y_\lambda).$$

Finally, the reverse triangle inequality gives

$$|\Delta(\lambda) - \hat{\Delta}(\lambda)| \leq \|\mathcal{O}_{\lambda,\rho}(y_\lambda) - \hat{x}_\lambda\|_p \leq \rho.$$

□

Theorem 8. *Let $f : \mathbb{R}^d \to \mathbb{R}$ be convex and G-Lipschitz, let $x, v \in \mathbb{R}^d$ satisfy $\|x - v\|_p \leq 2R$, $0 < r \leq R$, and let*

$$\hat{x}_\lambda = \operatorname*{argmin}_x \left\{ f(x) + \frac{\lambda}{p}\|x - y_\lambda\|_p^p \right\}, \qquad \hat{\Delta}(\lambda) = \|\hat{x}_\lambda - y_\lambda\|_p.$$

*Given that $\lambda_{\max} \geq \frac{2^{p-1}G}{r^{p-1}}$ and $\lambda_{\min} \in (0, \lambda_{\max})$, Algorithm **??** returns $\lambda \in [\lambda_{\min}, \lambda_{\max}]$ such that $\hat{\Delta}(\lambda) \leq r$. Moreover, either*

- $\lambda < 2\lambda_{\min}$,
- $\lambda \in [2\lambda_{\min}, \lambda_{\max}]$ *and* $\hat{\Delta}(\lambda) \in \left[\frac{3r}{4}, r\right)$.

Furthermore, the overall BROO complexity is

$$O\left(\log \frac{\lambda_{\max}}{\lambda_{\min}} + \log\left(1 + \frac{R + \frac{1}{p-1}\left(\frac{G}{\lambda_{\min}}\right)^{1/(p-1)}}{r}\right)\right)$$

.

Proof. First, by Lemma 13,

$$\hat{\Delta}(\lambda_{\max}) \leq (\frac{G}{\lambda_{\max}})^{1/(p-1)} \leq (\frac{G}{2^{p-1}G/r^{p-1}})^{1/(p-1)} = \frac{r}{2}.$$

Since $\frac{r}{2} < r - \frac{r}{17}$, Lemma 17 with $\rho = r/17$ gives

$$\Delta(\lambda_{\max}) \leq \hat{\Delta}(\lambda_{\max}) + \frac{r}{17} \leq \frac{r}{2} + \frac{r}{17} < \frac{13r}{16}.$$

Therefore, the first halving loop executes at least once. Suppose first that the halving loop terminates because $\lambda < \lambda_{\min}$. Let λ_k denote this value and $\lambda_{k-1} = 2\lambda_k$ the preceding iterate. Since the loop body executed at λ_{k-1},

$$\lambda_{k-1} \geq \lambda_{\min}, \qquad \Delta(\lambda_{k-1}) \leq \frac{13r}{16}.$$

Because $\frac{13r}{16} < r - \frac{r}{17}$, Lemma 17 gives $\hat{\Delta}(\lambda_{k-1}) \leq \frac{13r}{16} + \frac{r}{17} < r$. The algorithm returns

$$\lambda_{k-1} = 2\lambda_k < 2\lambda_{\min},$$

so the first case holds. For the remainder of the proof, the halving loop terminates because $\Delta(\lambda_\ell) > \frac{13r}{16}$ while $\lambda_\ell \geq \lambda_{\min}$. Since the loop executed at the previous iterate $\lambda_u = 2\lambda_\ell$, $\Delta(\lambda_u) \leq \frac{13r}{16}$. If $\Delta(\lambda_\ell) \leq \frac{15r}{16}$,then $\Delta(\lambda_\ell) \in \left(\frac{13r}{16}, \frac{15r}{16}\right]$. Since $\frac{15r}{16} < r - \frac{r}{17}$, Lemma 17 gives

$$\hat{\Delta}(\lambda_\ell) \in \left(\frac{13r}{16} - \frac{r}{17}, \frac{15r}{16} + \frac{r}{17}\right] \subset \left[\frac{3r}{4}, r\right).$$

Thus $\hat{\Delta}(\lambda_\ell) < r$. It remains to consider the binary-search phase. At the beginning of this phase,

$$\Delta(\lambda_\ell) > \frac{15r}{16}, \qquad \Delta(\lambda_u) \leq \frac{13r}{16}.$$

The update rule preserves these two inequalities throughout the binary search. If the binary search terminates because

$$\Delta(\lambda_m) \in \left[\frac{13r}{16}, \frac{15r}{16}\right],$$

then, exactly as above, Lemma 17 gives $\hat{\Delta}(\lambda_m) \in \left[\frac{3r}{4}, r\right)$. Finally, suppose the binary search terminates because

$$\log \frac{\lambda_u}{\lambda_\ell} \leq \frac{r}{8\left(R + \frac{1}{p-1}\left(\frac{G}{\lambda_\ell}\right)^{1/(p-1)}\right)}.$$

Then, Lemma 17 gives $\hat{\Delta}(\lambda_u) \leq \frac{13r}{16} + \frac{r}{17} < \frac{7r}{8}$. We next show that $\hat{\Delta}(\lambda_\ell) > \frac{7r}{8}$: Suppose instead that $\hat{\Delta}(\lambda_\ell) \leq \frac{7r}{8}$. Since $\frac{7r}{8} < r - \frac{r}{17}$, Lemma 17 would imply

$$\Delta(\lambda_\ell) \leq \frac{7r}{8} + \frac{r}{17} < \frac{15r}{16},$$

contradicting the binary-search invariant. Therefore, $\hat{\Delta}(\lambda_u) < \frac{7r}{8} < \hat{\Delta}(\lambda_\ell)$. By Lemma 16, $\hat{\Delta}$ is continuous. It follows by the Intermediate Value that there exists $\lambda^* \in (\lambda_\ell, \lambda_u)$ such that $\hat{\Delta}(\lambda^*) = \frac{7r}{8}$. Since $\lambda_m = \sqrt{\lambda_u \lambda_\ell}$, we have $\left|\log \frac{\lambda_m}{\lambda^*}\right| \leq \frac{1}{2} \log \frac{\lambda_u}{\lambda_\ell}$. Applying Lemma 16 to λ_m and λ^*, and using $\min\{\lambda_m, \lambda^*\} \geq \lambda_\ell$, gives

$$\begin{aligned}\left|\hat{\Delta}(\lambda_m) - \hat{\Delta}(\lambda^*)\right| &\leq \left[R + \frac{1}{p-1}\left(\frac{G}{\lambda_\ell}\right)^{1/(p-1)}\right] \left|\log \frac{\lambda_m}{\lambda^*}\right| \\ &\leq \frac{1}{2}\left[R + \frac{1}{p-1}\left(\frac{G}{\lambda_\ell}\right)^{1/(p-1)}\right] \log \frac{\lambda_u}{\lambda_\ell} \leq \frac{r}{16}.\end{aligned}$$

Since $\hat{\Delta}(\lambda^*) = \frac{7r}{8}$, it follows that $\hat{\Delta}(\lambda_m) \in \left[\frac{13r}{16}, \frac{15r}{16}\right] \subset \left[\frac{3r}{4}, r\right)$. It remains to bound the number of BROO calls. The halving phase clearly requires at most $O\left(\log \frac{\lambda_{\max}}{\lambda_{\min}}\right)$ calls. For the binary-search

phase, initially $\log \frac{\lambda_u}{\lambda_\ell} = \log 2$. Since $\lambda_m = \sqrt{\lambda_u \lambda_\ell}$, each bisection halves the logarithmic interval width. Therefore, after k iterations, $\log \frac{\lambda_u}{\lambda_\ell} = \frac{\log 2}{2^k}$. The binary search terminates once

$$\log \frac{\lambda_u}{\lambda_\ell} \leq \frac{r}{8\left[R + \frac{1}{p-1}\left(\frac{G}{\lambda_\ell}\right)^{1/(p-1)}\right]}$$

Throughout the binary search, $\lambda_\ell \geq \lambda_{\min}$. Therefore, $\left(\frac{G}{\lambda_\ell}\right)^{1/(p-1)} \leq \left(\frac{G}{\lambda_{\min}}\right)^{1/(p-1)}$. Consequently

$$\frac{r}{8\left[R + \frac{1}{p-1}\left(\frac{G}{\lambda_\ell}\right)^{1/(p-1)}\right]} \geq \frac{r}{8\left[R + \frac{1}{p-1}\left(\frac{G}{\lambda_{\min}}\right)^{1/(p-1)}\right]}$$

Thus, it is sufficient to make the logarithmic interval width smaller than the right-hand side above. In particular, the binary search is guaranteed to have terminated once

$$\frac{\log 2}{2^k} \leq \frac{r}{8\left[R + \frac{1}{p-1}\left(\frac{G}{\lambda_{\min}}\right)^{1/(p-1)}\right]} \Rightarrow 2^k \geq \frac{8\log 2}{r}\left[R + \frac{1}{p-1}\left(\frac{G}{\lambda_{\min}}\right)^{1/(p-1)}\right]$$

Therefore, the number of binary-search iterations is at most

$$O\left(\log\left(1 + \frac{R + \frac{1}{p-1}\left(\frac{G}{\lambda_{\min}}\right)^{1/(p-1)}}{r}\right)\right)$$

Combining the halving and binary-search phases, the overall BROO complexity is

$$O\left(\log \frac{\lambda_{\max}}{\lambda_{\min}} + \log\left(1 + \frac{R + \frac{1}{p-1}\left(\frac{G}{\lambda_{\min}}\right)^{1/(p-1)}}{r}\right)\right)$$

. □

3.5.2 ℓ_p Proximal Point Method Guarantees

In the analysis, below, we define the filtration $\mathcal{F}_k = \sigma(x_1, v_1, A_1, \zeta_1, \ldots, x_k, v_k, A_k, \zeta_k)$, where ζ_i represents the internal randomness in NextLambda(x_i, v_i, A_i). Furthermore, we let $\hat{x}_k = P_{f,\lambda_k}(y_{k-1})$, i.e the exact proximal mapping for which x_k approximates.

Lemma 18. *Let $f : \mathcal{X} \to \mathbb{R}$ be convex. If $\mathcal{X} \in \mathbb{B}_R(x_0)$, it holds that*

$$\begin{aligned}
&\mathbb{E}[A_{k+1}(f(x_{k+1}) - f(x^*)) + 2D_\Phi(x^*, v_{k+1})] \\
&\leq A_k(f(x_k) - f(x^*)) + 2D_\Phi(x^*, v_k) + A_{k+1}\varphi_{k+1} - \frac{1 - 2^{1-q}}{q}\lambda_{k+1}A_{k+1}||\hat{x}_{k+1} - y_k||_p^p \\
&+ \frac{2^{2-q}a_{k+1}^q}{q}\sigma_{k+1}^q + 2a_{k+1}\delta_{k+1}R
\end{aligned}$$

Proof. Following a similar approach to [ACJ+21], we let $\hat{g}_k = \nabla f_{\lambda_k}(y_{k-1})$ and we bound from above and below the quantity $a_{k+1}\langle \hat{g}_{k+1}, v_k - x^* \rangle$. First, observe that

$$v_k - x^* = \hat{x}_{k+1} - x^* + \frac{A_k}{a_{k+1}}(\hat{x}_{k+1} - x_k) - \frac{A_{k+1}}{a_{k+1}}(\hat{x}_{k+1} - y_k)$$

which implies that

$$\begin{aligned}
\langle \hat{g}_{k+1}, v_k - x^* \rangle &= \langle \hat{g}_{k+1}, \hat{x}_{k+1} - x^* \rangle + \frac{A_k}{a_{k+1}}\langle \hat{g}_{k+1}, \hat{x}_{k+1} - x_k \rangle - \frac{A_{k+1}}{a_{k+1}}\langle \hat{g}_{k+1}, \hat{x}_{k+1} - y_k \rangle \\
&\geq f(\hat{x}_{k+1}) - f(x^*) + \frac{A_k}{a_{k+1}}(f(\hat{x}_{k+1}) - f(x_k)) + \frac{\lambda_{k+1}A_{k+1}}{a_{k+1}}||\hat{x}_{k+1} - y_k||_p^p \\
&= f(\hat{x}_{k+1}) - f(x^*) + \frac{A_k}{a_{k+1}}(f(\hat{x}_{k+1}) - f(x^*) - (f(x_k) - f(x^*))) + \frac{\lambda_{k+1}A_{k+1}}{a_{k+1}}||\hat{x}_{k+1} - y_k||_p^p \\
&= \frac{A_{k+1}}{a_{k+1}}(f(\hat{x}_{k+1}) - f(x^*)) - \frac{A_k}{a_{k+1}}(f(x_k) - f(x^*)) + \frac{\lambda_{k+1}A_{k+1}}{a_{k+1}}||\hat{x}_{k+1} - y_k||_p^p
\end{aligned}$$

where we used the fact that f is convex and that $\langle \hat{g}_{k+1}, \hat{x}_{k+1} - y_k \rangle = -\lambda||\hat{x}_{k+1} - y_k||_p^p$. We also have that

$$\mathbb{E}[f(x_{k+1})|\mathcal{F}_k] \leq \mathbb{E}[f(x_{k+1}) + \frac{\lambda_{k+1}}{p}||x_{k+1} - y_k||_p^p] \leq f(\hat{x}_{k+1}) + \frac{\lambda_{k+1}}{p}||\hat{x}_{k+1} - y_k||_p^p + \varphi_{k+1}$$

which implies that

$$\begin{aligned}
&a_{k+1}\langle \hat{g}_{k+1}, v_k - x^* \rangle \\
&\geq A_{k+1}(\mathbb{E}[f(x_{k+1})|\mathcal{F}_k] - \frac{\lambda_{k+1}}{p}||\hat{x}_{k+1} - y_k||_p^p - \varphi_{k+1} - f(x^*)) - A_k(f(x_k) - f(x^*)) \\
&+ \lambda_{k+1}A_{k+1}||\hat{x}_{k+1} - y_k||_p^p \\
&= A_{k+1}(\mathbb{E}[f(x_{k+1})|\mathcal{F}_k] - f(x^*)) - A_k(f(x_k) - f(x^*)) + (1 - \frac{1}{p})\lambda_{k+1}A_{k+1}||\hat{x}_{k+1} - y_k||_p^p - A_{k+1}\varphi_{k+1}
\end{aligned}$$

We now upper bound $a_{k+1}\langle \hat{g}_{k+1}, v_k - x^* \rangle$. We have that

$$v_{k+1} = \operatorname*{argmin}_{v \in X}\{\frac{1}{2}a_{k+1}\langle g_{k+1}, v \rangle + D_\Phi(v, v_k)\} = \operatorname*{argmin}_{v \in X}\{\frac{1}{2}a_{k+1}\langle g_{k+1}, v \rangle + \Phi(v) - \langle \nabla\Phi(v_k), v \rangle\}$$

By first-order optimality conditions, it holds that $0 \in \frac{1}{2}\hat{a}_{k+1}g_{k+1} + \nabla\Phi(v_{k+1}) - \nabla\Phi(v_k) + N_X(v_{k+1})$, where N_X denotes the normal cone, which implies that for all $u \in X$,

$$\langle \frac{1}{2}a_{k+1}g_{k+1} + \nabla\Phi(v_{k+1}) - \nabla\Phi(v_k), u - v_{k+1} \rangle \geq 0.$$

Therefore, it holds that

$$\frac{1}{2}a_{k+1}\langle g_{k+1}, v_{k+1} - x^* \rangle \leq \langle \nabla\Phi(v_{k+1}) - \nabla\Phi(v_k), x^* - v_{k+1} \rangle$$

Since $\langle \nabla\Phi(v_{k+1}) - \nabla\Phi(v_k), x^* - v_{k+1} \rangle = D_\Phi(x^*, v_k) - D_\Phi(x^*, v_{k+1}) - D_\Phi(v_{k+1}, v_k)$, it holds that

$$\frac{1}{2}a_{k+1}\langle g_{k+1}, v_{k+1} - x^* \rangle \leq D_\Phi(x^*, v_k) - D_\Phi(x^*, v_{k+1}) - D_\Phi(v_{k+1}, v_k)$$

Noting that $v_k - x^* = (v_k - v_{k+1}) + (v_{k+1} - x^*)$, we have that

$$\begin{aligned} a_{k+1}\langle \hat{g}_{k+1}, v_k - x^* \rangle &= a_{k+1}\langle g_{k+1}, v_{k+1} - x^* \rangle \\ &\quad + a_{k+1}\langle \hat{g}_{k+1} + g_{k+1} - \mathbb{E}[g_{k+1}|\mathcal{F}_k], v_k - v_{k+1} \rangle \\ &\quad + a_{k+1}\langle \mathbb{E}[g_{k+1}|\mathcal{F}_k] - \hat{g}_{k+1}, x^* - v_{k+1} \rangle \\ &\quad - a_{k+1}\langle g_{k+1} - \mathbb{E}[g_{k+1}|\mathcal{F}_k], v_k - x^* \rangle. \end{aligned}$$

By Hölder's inequality, we have that for all $a, b \in \mathbb{R}^d$ and $\eta > 0$, $\langle a, b \rangle \leq \frac{\eta^{1-q}}{q}\|a\|_q^q + \frac{\eta}{p}\|b\|_p^p$. By choosing $\eta = 2^{3-p}$, it holds that

$$\begin{aligned} a_{k+1}\langle \hat{g}_{k+1} + g_{k+1} - \mathbb{E}[g_{k+1}|\mathcal{F}_k], v_k - v_{k+1} \rangle &\leq \frac{2^{3-2q}a_{k+1}^q}{q}\|\hat{g}_{k+1} + g_{k+1} - \mathbb{E}[g_{k+1}|\mathcal{F}_k]\|_q^q \\ &\quad + \frac{2^{3-p}}{p}\|v_k - v_{k+1}\|_p^p. \end{aligned}$$

By Fact 2, we have that $-2D_\Phi(v_{k+1}, v_k) + \frac{2^{3-p}}{p}\|v_k - v_{k+1}\|_p^p \leq 0$, which implies that

$$\begin{aligned} a_{k+1}\langle \hat{g}_{k+1}, v_k - x^* \rangle &\leq 2D_\Phi(x^*, v_k) - 2\mathbb{E}[D_\Phi(x^*, v_{k+1})|\mathcal{F}_k] \\ &\quad + \frac{2^{3-2q}a_{k+1}^q}{q}\mathbb{E}\big[\|\hat{g}_{k+1} + g_{k+1} - \mathbb{E}[g_{k+1}|\mathcal{F}_k]\|_q^q \big| \mathcal{F}_k\big] \\ &\quad + a_{k+1}\mathbb{E}\big[\langle \mathbb{E}[g_{k+1}|\mathcal{F}_k] - \hat{g}_{k+1}, x^* - v_{k+1} \rangle \big| \mathcal{F}_k\big], \end{aligned}$$

where we used the fact that v_k and x^* are $\mathcal{F}_k$-measurable and therefore

$$\mathbb{E}\big[\langle g_{k+1} - \mathbb{E}[g_{k+1}|\mathcal{F}_k], v_k - x^* \rangle \big| \mathcal{F}_k\big] = 0.$$

It also holds that

$$\begin{aligned} \mathbb{E}\big[\|\hat{g}_{k+1} + g_{k+1} - \mathbb{E}[g_{k+1}|\mathcal{F}_k]\|_q^q \big| \mathcal{F}_k\big] &\leq 2^{q-1}\|\hat{g}_{k+1}\|_q^q + 2^{q-1}\mathbb{E}\big[\|g_{k+1} - \mathbb{E}[g_{k+1}|\mathcal{F}_k]\|_q^q \big| \mathcal{F}_k\big] \\ &\leq 2^{q-1}\sigma_{k+1}^q + 2^{q-1}\|\hat{g}_{k+1}\|_q^q. \end{aligned}$$

Moreover, by the bias guarantee and the fact that $\mathcal{X} \subseteq \mathbb{B}_R(x_0)$, we have

$$\begin{aligned} \mathbb{E}\big[\langle \mathbb{E}[g_{k+1}|\mathcal{F}_k] - \hat{g}_{k+1}, x^* - v_{k+1} \rangle \big| \mathcal{F}_k\big] &\leq \|\mathbb{E}[g_{k+1}|\mathcal{F}_k] - \hat{g}_{k+1}\|_q \mathbb{E}[\|x^* - v_{k+1}\|_p|\mathcal{F}_k] \\ &\leq 2\delta_{k+1}R. \end{aligned}$$

Therefore, we have that

$$\begin{aligned} a_{k+1}\langle \hat{g}_{k+1}, v_k - x^* \rangle &\leq 2D_\Phi(x^*, v_k) - 2\mathbb{E}[D_\Phi(x^*, v_{k+1})|\mathcal{F}_k] \\ &\quad + \frac{2^{2-q}a_{k+1}^q}{q}\big(\sigma_{k+1}^q + \|\hat{g}_{k+1}\|_q^q\big) + 2a_{k+1}\delta_{k+1}R \\ &= 2D_\Phi(x^*, v_k) - 2\mathbb{E}[D_\Phi(x^*, v_{k+1})|\mathcal{F}_k] \\ &\quad + \frac{2^{2-q}a_{k+1}^q}{q}\big(\sigma_{k+1}^q + \lambda_{k+1}^q\|\hat{x}_{k+1} - y_k\|_p^p\big) + 2a_{k+1}\delta_{k+1}R. \end{aligned}$$

This implies that

$$\begin{aligned} &A_{k+1}(\mathbb{E}[f(x_{k+1}) \mid \mathcal{F}_k] - f(x^*)) - A_k(f(x_k) - f(x^*)) + \frac{1}{q}\lambda_{k+1}A_{k+1}||\hat{x}_{k+1} - y_k||_p^p - A_{k+1}\varphi_{k+1} \\ &\leq 2D_\Phi(x^*, v_k) - 2\mathbb{E}[D_\Phi(x^*, v_{k+1}) \mid \mathcal{F}_k] + \frac{2^{2-q}a_{k+1}^q}{q}(\sigma_{k+1}^q + \lambda_{k+1}^q||\hat{x}_{k+1} - y_k||_p^p) + 2a_{k+1}\delta_{k+1}R. \end{aligned}$$

This implies that

$$\begin{aligned}
&\mathbb{E}[A_{k+1}(f(x_{k+1}) - f(x^*)) + 2D_\Phi(x^*, v_{k+1}) \mid \mathcal{F}_k] \\
&\leq A_k(f(x_k) - f(x^*)) + 2D_\Phi(x^*, v_k) - \frac{1}{q}\lambda_{k+1}A_{k+1}||\hat{x}_{k+1} - y_k||_p^p \\
&\quad + A_{k+1}\varphi_{k+1} + \frac{2^{2-q}a_{k+1}^q}{q}(\sigma_{k+1}^q + \lambda_{k+1}^q||\hat{x}_{k+1} - y_k||_p^p) + 2a_{k+1}\delta_{k+1}R \\
&\leq A_k(f(x_k) - f(x^*)) + 2D_\Phi(x^*, v_k) - \frac{1 - 2^{1-q}}{q}\lambda_{k+1}A_{k+1}||\hat{x}_{k+1} - y_k||_p^p + A_{k+1}\varphi_{k+1} \\
&\quad + \frac{2^{2-q}a_{k+1}^q}{q}\sigma_{k+1}^q + 2a_{k+1}\delta_{k+1}R,
\end{aligned}$$

where we used $A_{k+1}^{p-1} = 2^{p-1}\lambda_{k+1}a_{k+1}^p$ in the last inequality. □

Lemma 19. *Let $K \leq K_{\max}$ be the iteration in which the algorithm stops and let*

$$\bar{\varepsilon} \geq \max_{k<K_{\max}} \{\frac{A_{k+1}\varphi_{k+1}}{a_{k+1}} + \frac{2^{2-q}a_{k+1}^{q-1}\sigma_{k+1}^q}{q} + 2\delta_{k+1}R\}$$

We then have that

$$\begin{aligned}
&\mathbb{E}\left[A_K(f(x_K) - f(x^*) - \bar{\varepsilon}) + 2D_\Phi(x^*, v_K) + \frac{1 - 2^{1-q}}{q}\sum_{i=1}^k \lambda_i A_i||\hat{x}_i - y_{i-1}||_p^p\right] \\
&\leq A_0(f(x_0) - f(x^*)) + 3^{p-2}(p-1)R^p
\end{aligned}$$

Proof. Let

$$T_k := -\frac{1 - 2^{1-q}}{q}\lambda_{k+1}A_{k+1}||\hat{x}_{k+1} - y_k||_p^p + A_{k+1}\varphi_{k+1} + \frac{2^{2-q}a_{k+1}^q}{q}\sigma_{k+1}^q + 2a_{k+1}\delta_{k+1}R.$$

First, we have from Lemma 18 that

$$\mathbb{E}[A_{k+1}(f(x_{k+1} - f(x^*)) + 2D_\Phi(x^*, v_{k+1})] \leq A_k(f(x_k) - f(x^*)) + 2D_\Phi(x^*, v_k) + T_k.$$

Define

$$M_k = A_k(f(x_k) - f(x^*) - \bar{\varepsilon}) + 2D_\Phi(x^*, v_k) + \frac{1 - 2^{1-q}}{q}\sum_{i=1}^k \lambda_i A_i||\hat{x}_i - y_{i-1}||_p^p.$$

We will now prove that M_k is a supermartingale. It holds that

$$\begin{aligned}
\mathbb{E}[M_{k+1}|\mathcal{F}_k] &= \mathbb{E}[A_{k+1}(f(x_{k+1}) - f(x^*) - \overline{\varepsilon}) + 2D_\Phi(x^*, v_{k+1})] + \frac{1-2^{1-q}}{q}\sum_{i=1}^{k}\lambda_i A_i ||\hat{x}_i - y_{i-1}||_p^p \\
&+ \frac{1-2^{1-q}}{q}\lambda_{k+1}A_{k+1}||\hat{x}_{k+1} - y_k||_p^p \\
&\leq A_k(f(x_k) - f(x^*)) + 2D_\Phi(x^*, v_k) + T_k - A_{k+1}\overline{\varepsilon} + \frac{1-2^{1-q}}{q}\sum_{i=1}^{k}\lambda_i A_i ||\hat{x}_i - y_{i-1}||_p^p \\
&+ \frac{1-2^{1-q}}{q}\lambda_{k+1}A_{k+1}||\hat{x}_{k+1} - y_k||_p^p \\
&\leq A_k(f(x_k) - f(x^*)) + 2D_\Phi(x^*, v_k) + T_k - (A_k + a_{k+1})\overline{\varepsilon} + \frac{1-2^{1-q}}{q}\sum_{i=1}^{k}\lambda_i A_i ||\hat{x}_i - y_{i-1}||_p^p \\
&+ \frac{1-2^{1-q}}{q}\lambda_{k+1}A_{k+1}||\hat{x}_{k+1} - y_k||_p^p \\
&\leq M_k + T_k + \frac{1-2^{1-q}}{q}\lambda_{k+1}A_{k+1}||\hat{x}_{k+1} - y_k||_p^p - a_{k+1}\overline{\varepsilon} \\
&= M_k + A_{k+1}\varphi_{k+1} + \frac{2^{2-q}a_{k+1}^q\sigma_{k+1}^q}{q} + 2a_{k+1}\delta_{k+1}R - a_{k+1}\overline{\varepsilon}
\end{aligned}$$

By our setting of $\overline{\varepsilon}$, we have that

$$M_k + A_{k+1}\varphi_{k+1} + \frac{2^{2-q}a_{k+1}^q}{q}\sigma_{k+1}^q + 2a_{k+1}\delta_{k+1}R - a_{k+1}\overline{\varepsilon} \leq M_k$$

so M_k is a supermartingale. Therefore, by the optional stopping theorem, it holds that

$$\mathbb{E}[M_K] \leq M_0 = A_0(f(x_0) - f(x^*) - \overline{\varepsilon}) + 2D_\Phi(x^*, v_0)$$

Using that $v_0 = x_0$, we have that

$$\begin{aligned}
D_\Phi(x^*, v_0) = D_\Phi(x^*, x_0) &\leq \frac{p-1}{2}(||x^*||_p + 2||x_0||_p)^{p-2}||x^* - x_0||_p^2 \\
&\leq \frac{p-1}{2}(3||x_0||_p + ||x^* - x_0||_p)^{p-2}R^2 \\
&\leq \frac{3^{p-2}(p-1)}{2}R^p
\end{aligned}$$

which finishes the proof. □

Fact 1. *(Lemma 32 of [CJJ+ 20]) Let $\{B_k\}_{k\in\mathbb{N}}$ be a nonnegative, nondecreasing sequence such that $B_k \geq \sum_{i\in[k]}\alpha B_i$ for some $\alpha \in [0,1)$ and all k. Then for all k, $B_k \geq \exp(\alpha(k-1))B_1$.*

Lemma 20. *Fix any $k > 0$. Define*

$$T_\lambda := \sum_{i\leq k}\mathbf{1}_{\{\lambda_i < 2\lambda_{\min}\}}, \quad T_r := \sum_{i\leq k}\mathbf{1}_{\{\|\hat{x}_i - y_{i-1}\| \geq 3r/4\}}$$

Then, with probability one, for $\varepsilon < \frac{2^{q-1}-1}{3\cdot 2^{q-1}-1}$,

$$\frac{1}{A_k}\left(3C_pR^p - \frac{(1-2^{1-q})}{q}\sum_{i\le k}\lambda_i A_i\|\hat{x}_i - y_{i-1}\|_p^p\right) \le O\left(\min\left\{\frac{\lambda_{\min}R^p}{T_\lambda^p},\ \frac{R^p}{A_0}\exp\left(-\Omega(1)\frac{r^{p/(p+1)}}{R^{p/(p+1)}}T_r\right)\right\}\right)$$

See (39) *for a definition of* C_p.

Proof. If $3C_pR^p - \frac{(1-2^{1-q})}{q}\sum_{i\le k}\lambda_i A_i\|\hat{x}_i - y_{i-1}\|_p^p \le 0$, then the desired inequality trivially holds. Therefore, we only consider the case where $\frac{(1-2^{1-q})}{q}\sum_{i\le k}\lambda_i A_i\|\hat{x}_i - y_{i-1}\|_p^p \le 3C_pR^p$. We first prove that

$$A_t^{1/p} \ge \frac{1}{2p}\sum_{i=1}^{t}\lambda_i^{-1/p}. \tag{34}$$

Since $A_i^{p-1} = 2^{p-1}\lambda_i a_i^p$, it holds that $a_i = \frac{A_i^{(p-1)/p}}{2^{(p-1)/p}\lambda_i^{1/p}}$. Therefore,

$$a_i \ge \frac{1}{2^{(p-1)/p}}A_i^{(p-1)/p}\lambda_i^{-1/p} \ge \frac{1}{2}A_i^{(p-1)/p}\lambda_i^{-1/p}$$

Since $f(x) = x^{1/p}$ is concave for $p \ge 2$, it holds that

$$A_i^{1/p} - A_{i-1}^{1/p} \ge \frac{1}{p}A_i^{\frac{1}{p}-1}(A_i - A_{i-1}) = \frac{a_i}{pA_i^{(p-1)/p}} \ge \frac{\lambda_i^{-1/p}}{2p} \tag{35}$$

Telescoping (35) from $i = 1$ to T and using that $A_0 \ge 0$ yields (34). For $t \le k$, define

$$I_{r,t} = \left\{i \le t : ||\hat{x}_i - y_{i-1}||_p \ge \frac{3r}{4}\right\}$$

It holds that

$$\frac{1-2^{1-q}}{q}\sum_{i\in I_{r,t}}\lambda_i A_i(\frac{3r}{4})^p \le \frac{1-2^{1-q}}{q}\sum_{i\le t}\lambda_i A_i||\hat{x}_i - y_{i-1}||_p^p \le 3C_pR^p \implies \sum_{i\in I_{r,t}}\lambda_i A_i \le 3C_p(\frac{R}{r})^p(\frac{4}{3})^p\cdot\frac{q}{1-2^{1-q}}$$

Now, by a reverse Hölder inequality, it holds that for all $\theta \in (0,1)$,

$$\sum_i u_i v_i \ge (\sum_i u_i^\theta)^{1/\theta}\cdot(\sum_i v_i^{-\theta/(1-\theta)})^{-(1-\theta)/\theta}$$

Taking $u_i = A_i^{1/p}, v_i = \frac{1}{\lambda_i^{1/p}A_i^{1/p}}, \theta = p/(p+1)$, it holds that

$$\begin{aligned}\sum_{i\in I_{r,t}}\frac{1}{\lambda_i^{1/p}} \ge (\sum_{i\in I_{r,t}}A_i^{1/(p+1)})^{(p+1)/p}\cdot(\sum_{i\in I_{r,t}}\lambda_i A_i)^{-1/p} &\ge (3C_p\cdot\frac{q}{1-2^{1-q}}\cdot(\frac{4}{3})^p)^{-1/p}\cdot\frac{r}{R}\cdot(\sum_{i\in I_{r,t}}A_i^{1/(p+1)})^{(p+1)/p}\\ &\ge C_p'\cdot\frac{r}{R}\cdot(\sum_{i\in I_{r,t}}A_i^{1/(p+1)})^{(p+1)/p}\end{aligned}$$

wher we let $C_p' = (3C_p\cdot\frac{q}{1-2^{1-q}}\cdot(\frac{4}{3})^p)^{-1/p}$. Combining with (34) gives that

$$A_t^{1/(p+1)} \ge (\frac{C_p'}{2p})^{p/(p+1)}\cdot(\frac{r}{R})^{\frac{p}{p+1}}\sum_{i\in I_{r,t}}A_i^{1/(p+1)}$$

By Fact 1, it holds that

$$A_k^{1/(p+1)} \geq A_0^{1/(p+1)} \exp(\Omega(1)(\frac{r}{R})^{\frac{p}{p+1}} T_r) \implies \frac{1}{A_k} \leq \frac{1}{A_0} \exp(-\Omega(1)(\frac{r}{R})^{\frac{p}{p+1}} T_r) \tag{36}$$

Therefore,

$$\frac{1}{A_k}\left(3C_pR^p - \frac{1-2^{1-q}}{q}\sum_{i\leq k}\lambda_i A_i \|\hat{x}_i - y_{i-1}\|_p^p\right) \leq \frac{3C_pR^p}{A_k} \leq O(\frac{R^p}{A_0}\exp(-\Omega(1)(\frac{r}{R})^{\frac{p}{p+1}}T_r)) \tag{37}$$

Let $I_\lambda = \{i \leq k : \lambda_i < 2\lambda_{\min}\}$. Combining (34) with $\lambda_i < 2\lambda_{\min}$ yields

$$A_k^{1/p} \geq \frac{1}{2p}\cdot\frac{T_\lambda}{(2\lambda_{\min})^{1/p}} \implies \frac{1}{A_k} \leq O(\frac{\lambda_{\min}}{T_\lambda^p}) \implies \frac{3C_pR^p}{A_k} \leq O(\frac{R^p\lambda_{\min}}{T_\lambda^p}) \tag{38}$$

Combining (37) and (38) finishes the proof. □

Proposition 9. *Let $f : \mathcal{X} \to \mathbb{R}$ and $\widehat{\nabla} f$ satisfy Assumption 1, and let $\mathcal{X} \subseteq B_R(x_0)$. For a target accuracy $\varepsilon \leq GR$ let*

$$\varphi_k = \frac{\varepsilon}{48C_p\lambda_k^{q-1}a_k^{q-1}}, \quad \delta_k = \frac{\varepsilon}{96C_pR}, \quad \sigma_k^q = \frac{q\varepsilon}{48C_p2^{2-q}a_k^{q-1}}, \quad A_0 = \frac{R^{p-1}}{G}, \quad A_{\max} = 3C_p\frac{R^p}{\varepsilon}$$

see (39). If $\lambda_k \geq \lambda_{\min} \geq \frac{1}{A_{\max}} = \Omega\left(\frac{\varepsilon}{R^p}\right)$ for all $k \leq K_{\max}$, then lines 7 and 9 of Algorithm 6 have total complexity

$$\mathbb{E}N_{\widehat{\nabla}f} = O\left(K_{\max}\log\left(\frac{GR}{\varepsilon}\right) + \left(\frac{GR}{\epsilon}\right)^{p(p-1)}\log\left(\frac{GR}{\epsilon}\right)\right)$$

If in addition $\|P_{f,\lambda_k}(y_{k-1}) - y_{k-1}\|_p \geq 3r/4$ whenever $\lambda_k \geq 2\lambda_{\min}$ then for

$$K_{\max} = O\left(\left(\frac{R}{r}\right)^{\frac{p}{p+1}}\log\left(\frac{GR}{\varepsilon}\right) + \left(\frac{\lambda_{\min}R^p}{\varepsilon}\right)^{\frac{1}{p}}\right)$$

the algorithm's output x_K satisfies $f(x_K) - f(x^\star) \leq \varepsilon$ with probability at least $2/3$.

Proof. First, let us prove correctness of the algorithm. The settings of φ_k, δ_k and σ_k in the proposition guarantee that

$$\max_{k<K_{\max}}\left\{\frac{A_{k+1}\varphi_{k+1}}{a_{k+1}} + \frac{2^{2-q}a_{k+1}^{q-1}\sigma_{k+1}^q}{q} + 2\delta_{k+1}R\right\} = \frac{\varepsilon}{12C_p}$$

Therefore, applying Lemma 19 with $\bar{\varepsilon} = \varepsilon/(12C_p) \leq \varepsilon/(6C_p) \leq R^p/(2A_{\max})$ yields

$$\mathbb{E}\left[A_K(f(x_K) - f(x^\star)) + \frac{1-2^{1-q}}{q}\sum_{i\leq K}\lambda_i A_i\|\hat{x}_i - y_{i-1}\|_p^p\right] \leq 3^{p-2}R^p(p-1) + \bar{\varepsilon}\cdot\mathbb{E}A_K + A_0(f(x_0) - f(x^\star))$$

We now upper bound a_K in terms of A_{K-1} and λ_K. It holds that $(A_{K-1} + a_K)^{p-1} = 2^{p-1}\lambda_K a_K^p$. If $a_K > A_{K-1}$, then $A_{K-1} + a_K < 2a_K$, meaning that $a_K < 1/\lambda_K$ because $2^{p-1}\lambda_K a_K^p = (A_{K-1} + a_K)^{p-1} < (2a_K)^{p-1}$. Combining this with the $a_K < A_{K-1}$ case, it holds that

$$a_K \leq \max\{A_{K-1}, \frac{1}{\lambda_K}\} \implies a_K \leq A_{K-1} + \frac{1}{\lambda_K} \implies A_K \leq 2A_{K-1} + \frac{1}{\lambda_K}$$

Since $A_{K-1} \leq A_{\max}$ and $\lambda_K \geq 1/A_{\max}$, it holds that $A_K \leq 3A_{\max}$ with probability 1. Since $A_0 = R^{p-1}/G$ and f is G-Lipschitz, it holds that $A_0(f(x_0)-f(x^*)) \leq A_0 G \cdot ||x_0 - x^*||_p \leq A_0 GR = R^p$. Moreover, using the definition of $A_{\max}$, we have that $\overline{\varepsilon} A_K \leq \frac{\varepsilon}{12C_p} \cdot 3(3C_p \frac{R^p}{\varepsilon}) = \frac{3}{4}R^p$. Thus,

$$\begin{aligned}\mathbb{E}[A_K(f(x_K) - f(x^*)) + \frac{1-2^{1-q}}{q}\sum_{i\leq K}\lambda_i A_i ||\hat{x}_i - y_{i-1}||_p^p] &\leq A_0(f(x_0)-f(x^*)) + 3^{p-2}R^p(p-1) + \overline{\varepsilon}\mathbb{E}[A_K] \\ &\leq [3^{p-2}(p-1) + 1 + \frac{3}{4}]R^p\end{aligned}$$

Denote

$$C_p = 3^{p-2}(p-1) + \frac{7}{4} \tag{39}$$

Therefore, with probability at least 2/3,

$$\mathbb{E}[A_K(f(x_K) - f(x^*)) + \frac{1-2^{1-q}}{q}\sum_{i\leq K}\lambda_i A_i ||\hat{x}_i - y_{i-1}||_p^p] \leq 3C_p R^p \implies f(x_K) - f(x^*) \leq \frac{3C_p R^p}{A_K}$$

If $A_K \geq A_{\max}$, then we are done. Otherwise, $K = K_{\max}$ and by the assumption on NextLambda, it holds that $T_\lambda + T_r \geq K_{\max}$, which means that either $T_\lambda \geq K_{\max}/2$ or $T_r \geq K_{\max}/2$. If $T_\lambda \geq K_{\max}/2$, using (34) for the T_λ iterations where $\lambda_i < 2\lambda_{\min}$, we get that

$$A_K^{1/p} \geq \frac{T_\lambda}{2p(2\lambda_{\min})^{1/p}} \implies \frac{3C_p R^p}{A_K} \leq O\left(\frac{\lambda_{\min} R^p}{T_\lambda^p}\right) \leq \varepsilon$$

as long as $T_\lambda \geq \Omega((\frac{\lambda_{\min} R^p}{\varepsilon})^{1/p})$. This implies that it suffices to let $K_{\max} \geq \Omega((\frac{\lambda_{\min} R^p}{\varepsilon})^{1/p})$. If $T_r \geq K_{\max}/2$, then we can use that $A_K \geq A_0 \exp(\Omega(1)(\frac{r}{R})^{p/(p+1)} T_r)$ (from (36)) and that $R^p/A_0 = GR$ to say that

$$\frac{3C_p R^p}{A_K} \leq 3C_p GR \exp(-\Omega(1)(\frac{r}{R})^{p/(p+1)} T_r)$$

which is at most ε provided that $T_r = \Omega((\frac{R}{r})^{p/(p+1)} \log(\frac{GR}{\varepsilon}))$. Therefore, taking

$$K_{\max} = O\left(\left(\frac{R}{r}\right)^{p/(p+1)} \log\left(\frac{GR}{\varepsilon}\right) + \left(\frac{\lambda_{\min} R^p}{\varepsilon}\right)^{1/p}\right)$$

suffices to conclude that $f(x_K) - f(x^*) \leq \varepsilon$ with probability at least 2/3. Now, we prove the complexity guarantee. Each step of computing x_{k+1} on line 7 of Algorithm 6 requires $O(\frac{G^q}{\lambda_k \phi_k^{p-1}})$ queries by Definition 3 and the definition of an approximate proximal mapping. Moreover, by Theorem 1 it requires

$$O\left(\frac{G^p}{\sigma^p}\left(\frac{G}{\min\{\delta,\sigma\}}\right)^{p(p-2)}\left(1 + \log\frac{G}{\min\{\delta,\sigma\}}\right)\right)$$

in expectation to compute g_{k+1} in line 8 of Algorithm 6. Observe that

$$\sigma_k^p = (\sigma_k^q)^{p-1} = O(\frac{\varepsilon^{p-1}}{a_k^{(q-1)(p-1)}}) = O(\frac{\varepsilon^{p-1}}{a_k}) \implies \frac{G^p}{\sigma_k^p} = O(\frac{G^p a_k}{\varepsilon^{p-1}}).$$

Therefore, plugging in ϕ_k,δ_k and σ_k, we have

$$\begin{aligned}
\mathbb{E}N_{\widehat{\nabla} f} &= \sum_{k\in[K]} O\left(\frac{G^p}{\lambda_k \varphi_k^{p-1}}\right) + \sum_{k\in[K]} O([1+\frac{G^p}{\sigma_k^p}(\frac{G}{\min\{\delta,\sigma_k\}})^{p(p-2)}]\cdot \log\frac{G}{\min\{\delta,\sigma_k\}}) \\
&\le O(\frac{G^p}{\varepsilon^{p-1}}\sum_{k\in[K]} a_k) + O(\sum_{k\in[K]} \log\frac{G}{\min\{\delta,\sigma_k\}} + \frac{G^p}{\sigma_k^p}(\frac{G}{\min\{\delta,\sigma_k\}})^{p(p-2)}\log\frac{G}{\min\{\delta,\sigma_k\}}) \\
&\le O(\frac{G^p}{\varepsilon^{p-1}}\sum_{k\in[K]} a_k) + O(\sum_{k\in[K]} \log\frac{GR}{\varepsilon} + \frac{G^p a_k}{\varepsilon^{p-1}}(\frac{GR}{\varepsilon})^{p(p-2)}\log\frac{GR}{\varepsilon}) \\
&\le O(\frac{G^p}{\varepsilon^{p-1}}A_{\max}) + O(K_{\max}\log\frac{GR}{\varepsilon}) + O(\frac{G^p A_{\max}}{\varepsilon^{p-1}}(\frac{GR}{\varepsilon})^{p(p-2)}\log\frac{GR}{\varepsilon}) \\
&\le O((\frac{GR}{\varepsilon})^p) + O(K_{\max}\log\frac{GR}{\varepsilon}) + O((\frac{GR}{\varepsilon})^{p(p-1)}\log\frac{GR}{\varepsilon})
\end{aligned}$$

since $a_k \le A_K \le 3A_{\max} \le O(R^p/\varepsilon)$, $\min\{\delta_k,\sigma_k\} = \Omega(\varepsilon/R)$, and $\sum_{k\in[K]} a_k^{p-1} \le (\sum_{k\in[K]} a_k)^{p-1} \le O(A_K^{p-1})$ and $A_k = O(A_{\max})$. This finishes the proof.

□

4 ℓ_p Parallel Stochastic Convex Optimization: Upper Bounds

4.1 Preliminaries

We study the following parallel SCO problem, where access to the objective is only allowed through the stochastic gradient oracle.

Problem 1. *Let $f:\mathbb{R}^d\to\mathbb{R}$ be convex. Assume that there exists a stochastic gradient oracle g such that for all $x\in\mathbb{R}^d$, $\mathbb{E}[g(x)]\in\partial f(x)$ and $\mathbb{E}||g(x)||_q^q \le L^q$. Our goal is to produce $x\in\mathbb{R}^d$ such that*

$$\mathbb{E}[f(x)] \le \min_{x^*\in\mathbb{B}(R)} f(x^*) + \epsilon_{\text{opt}}$$

for some $\epsilon_{\text{opt}} > 0$.

We will use the query depth, computational depth, and the total number of stochastic gradient oracle queries as metrics for characterizing the efficiency of our algorithms. Following [CJJ+23], we present a few useful oracle definitions which will be useful in our analysis.

Definition 6 (Line search oracle)**.** *We say $\mathcal{O}_{\text{ls}}$ is a (Δ,λ)-line search oracle for $F:\mathbb{R}^d\to\mathbb{R}$ if given $\bar{x}\in\mathbb{R}^d$, $\mathcal{O}_{\text{ls}}$ returns $x\in\mathbb{R}^d$ with*

$$\left\|x - x^\star_{\bar{x},\lambda}\right\|_p \le \Delta.$$

Definition 7 (Ball optimization oracle)**.** *We say $\mathcal{O}_{\text{bo}}$ is a (ϕ,λ)-ball optimization oracle for $F:\mathbb{R}^d\to\mathbb{R}$ if given $\bar{x}\in\mathbb{R}^d$, $\mathcal{O}_{\text{bo}}$ returns $x\in\mathbb{R}^d$ with*

$$\mathbb{E}\left[F(x) + \frac{\lambda}{p}\|x-\bar{x}\|_p^p\right] \le F\left(x^\star_{\bar{x},\lambda}\right) + \frac{\lambda}{p}\left\|x^\star_{\bar{x},\lambda} - \bar{x}\right\|_p^p + \phi.$$

Definition 8 (Stochastic proximal oracle)**.** *We say $\mathcal{O}_{\text{sp}}$ is a (Δ,σ,λ)-stochastic proximal oracle for $F:\mathbb{R}^d\to\mathbb{R}$ if, given $\bar{x}\in\mathbb{R}^d$, $\mathcal{O}_{\text{sp}}$ returns a random vector $g\in\mathbb{R}^d$ such that*

$$\left\|\mathbb{E}g - g^\star_{\bar{x},\lambda}\right\|_q \le \Delta, \qquad \mathbb{E}\left\|g - \mathbb{E}g\right\|_q^q \le \sigma^q,$$

Proposition 10. *Let $F : \mathbb{R}^d \to \mathbb{R}$ be L-Lipschitz and convex, and let $x^\star \in \mathbb{B}_p(R)$. There is an algorithm* BallAccel *(Algorithm 6) taking parameters $r \in [0, R]$ and $\epsilon_{\text{opt}} \in (0, LR]$ with the following guarantee. Define*

$$\kappa := \frac{LR}{\epsilon_{\text{opt}}}, \qquad K := \left(\frac{R}{r}\right)^{\frac{p}{p+1}}, \qquad \lambda^\star := \frac{\epsilon_{\text{opt}} K^p}{R^p} \log^p \kappa.$$

For a universal constant $C_{\text{ba}} > 0$, BallAccel *runs in at most $C_{\text{ba}} K \log \kappa$ iterations and produces $x \in \mathbb{R}^d$ such that*

$$\mathbb{E}F(x) \le F(x^\star) + \epsilon_{\text{opt}}.$$

Moreover, in each iteration BallAccel *requires the following oracle calls (all for F).*

1. *At most $C_{\text{ba}} \log\left(\frac{R\kappa}{r}\right)$ calls to a $\left(\frac{r}{C_{\text{ba}}}, \lambda\right)$-line search oracle with values of $\lambda \in \left[\frac{\lambda^\star}{C_{\text{ba}}}, \frac{C_{\text{ba}} L}{r^{p-1}}\right]$.*
2. *A single call to a $\left(\frac{\lambda r^p}{C_{\text{ba}} \log^{p+1} \kappa}, \lambda\right)$-ball optimization oracle with $\lambda \in \left[\frac{\lambda^\star}{C_{\text{ba}}}, \frac{C_{\text{ba}} L}{r^{p-1}}\right]$.*
3. *A single call to a $\left(\frac{\epsilon_{\text{opt}}}{C_{\text{ba}} R}, \frac{\epsilon_{\text{opt}} K^{1/p}}{C_{\text{ba}} R}, \lambda\right)$-stochastic proximal oracle with $\lambda \in \left[\frac{\lambda^\star}{C_{\text{ba}}}, \frac{C_{\text{ba}} L}{r^{p-1}}\right]$.*

Proof.

- **Iteration Count**: Having $\lambda_{\min} = O(\lambda^\star)$ and $\epsilon = \epsilon_{\text{opt}}$, our analogous of BallAccel runs in at most $C_{\text{ba}} K \log \kappa$ iterations, where the value of $K_{\max}$ from Proposition 9 is $O(K \log \kappa)$.
- **Line search oracle parameters**: From Theorem 8, the line search oracle has complexity bounded by

$$O\left(\log \frac{\lambda_{\max}}{\lambda_{\min}} + \log\left(1 + \frac{R + \frac{1}{p-1}\left(\frac{G}{\lambda_{\min}}\right)^{1/(p-1)}}{r}\right)\right)$$

With $\lambda_{\max} = \frac{C_{\text{ba}} L}{r^{p-1}}$ and $\lambda_{\min} = \frac{\lambda^\star}{C_{\text{ba}}}$, we have, as $K \ge 1$,

$$\frac{\lambda_{\max}}{\lambda_{\min}} = \frac{C_{\text{ba}} L}{r^{p-1}} \times \frac{C_{\text{ba}}}{\lambda^\star} = \frac{C_{\text{ba}}^2 L R^p}{r^{p-1} \epsilon K^p \log^p \kappa} = \frac{C_{\text{ba}}^2 \kappa R^{p-1}}{r^{p-1} K^p \log^p \kappa} = \frac{C_{\text{ba}}^2 \kappa}{K^{1/p} \log^p \kappa} \le \frac{C_{\text{ba}}^2 \kappa}{\log^p \kappa}$$

Now, note that

$$\frac{1}{r}\left(\frac{G}{\lambda_{\min}}\right)^{\frac{1}{p-1}} = \frac{R}{r} \frac{\kappa^{1/(p-1)}}{K^{p/(p-1)} \log^{p/(p-1)} \kappa} = \frac{\kappa^{1/(p-1)}}{(R/r)^{1/(p^2-1)} \log^{p/(p-1)} \kappa} \le \kappa^{1/(p-1)} \le \kappa,$$

Therefore,

$$\log\left(1 + \frac{R + \frac{1}{p-1}\left(\frac{G}{\lambda_{\min}}\right)^{1/(p-1)}}{r}\right) \le \log\left(1 + \frac{R}{r} + \kappa\right) = O\left(\log\left(\frac{R\kappa}{r}\right)\right),$$

Together,

$$O\left(\log \frac{\lambda_{\max}}{\lambda_{\min}} + \log\left(1 + \frac{R + \frac{1}{p-1}\left(\frac{G}{\lambda_{\min}}\right)^{1/(p-1)}}{r}\right)\right) \le O\left(\log \frac{R\kappa}{r}\right)$$

- **Ball optimization oracle parameters**: By Lemma 18, we have that

$$\begin{aligned}
&\mathbb{E}[A_{k+1}(f(x_{k+1}) - f(x^*)) + 2D_\Phi(x^*, v_{k+1})] \\
&\le A_k(f(x_k) - f(x^*)) + 2D_\Phi(x^*, v_k) + A_{k+1}\varphi_{k+1} - \frac{1-2^{1-q}}{q}\lambda_{k+1}A_{k+1}||\hat{x}_{k+1} - y_k||_p^p \\
&+ \frac{2^{2-q}a_{k+1}^q}{q}\sigma_{k+1}^q + 2a_{k+1}\delta_{k+1}R \\
&\le A_k(f(x_k) - f(x^*)) + 2D_\Phi(x^*, v_k) + A_{k+1}\varphi_{k+1} - \frac{1}{q}\left(1-2^{1-q}\right)\lambda_{k+1}A_{k+1}\|\hat{x}_{k+1} - y_k\|_p^p \\
&\quad + \frac{2^{2-q}a_{k+1}^q}{q}\sigma_{k+1}^q + 2a_{k+1}\delta_{k+1}R.
\end{aligned}$$

  If our algorithm does not terminate, one of the following occurs:

  (1) $||\hat{x}_{k+1} - y_k||_p^p = \Omega(r^p)$
  (2) $\lambda_{k+1} = O(\lambda^\star)$

  If (1) occurs, then $-\frac{1-2^{1-q}}{q}\lambda_{k+1}A_{k+1}||\hat{x}_{k+1} - y_k||_p^p = -\Omega(\lambda_{k+1}A_{k+1}r^p)$. If we choose $\varphi_{k+1} \le \frac{\lambda_{k+1}r^p}{C_{\text{ba}}\log^{p+1}\kappa}$, then $A_{k+1}\varphi_{k+1}$ can be bounded by the first case. If (2) occurs, since Proposition 9 requires $A_{\max} \le O(R^p/\epsilon)$, with the fact that the total iteration is $K\log\kappa$ and $\varphi_{k+1} \le \frac{\lambda_{k+1}r^p}{C_{\text{ba}}\log^{p+1}\kappa}$ by design, we have

$$\sum_{k+1\le K} A_{k+1}\varphi_{k+1} \le O\left(K\log\kappa \cdot \frac{R^p}{\epsilon_{\text{opt}}} \cdot \frac{\lambda^\star r^p}{C_{\text{ba}}\log^{p+1}\kappa}\right) \le O(R^p)$$

  Here, the first term is the iteration count, the second term is due to an upper bound on A_{k+1}, and the third term is bounded since $\lambda_{k+1} = O(\lambda^\star)$.

- **Stochastic proximal oracle parameters**: In Proposition 9 we require $\sigma^q = O(\epsilon_{\text{opt}}/a_{k+1}^{q-1})$, using the relation that $A_{k+1}^{p-1} = 2^{p-1}\lambda_{k+1}a_{k+1}^p$, we have

$$\sigma^q = \frac{\epsilon_{\text{opt}}}{a_{k+1}^{q-1}} = \frac{\epsilon_{\text{opt}}\lambda_{k+1}^{1/p(p-1)}2^{\frac{1}{p}}}{A_{k+1}^{1/p}} = \Omega\left(\frac{\epsilon_{\text{opt}}^{1+1/p}\cdot(\lambda^\star)^{1/p(p-1)}}{R}\right) = \Omega\left(\frac{\epsilon_{\text{opt}}^{p/(p-1)}\cdot K^{1/(p-1)}\log^{1/(p-1)}\kappa}{R^{p/(p-1)}}\right)$$

  where the second equality uses $A_{k+1} \le O(R^p/\varepsilon_{\text{opt}})$. Hence, $\sigma = \tilde{\Omega}(\epsilon_{\text{opt}}K^{1/p}/R)$.

□

Lemma 21. *Let $\bar{x} \in \mathcal{X}$, $\lambda > 0$. Suppose a $\left(\frac{2^{2-p}\lambda r^p}{p\,3^{p+1}C_{\text{ba}}^p}, \lambda\right)$-ball optimization oracle returns $x \in \mathcal{X} \cap B_p(\bar{x}, r)$,, then*

$$\Pr\left(\|x - x_{\bar{x},\lambda}^\star\|_p \le \frac{r}{3C_{\text{ba}}}\right) \ge \frac{2}{3}. \tag{40}$$

Moreover, let $x_1, \ldots, x_m$ be m independent outputs of the ball optimization oracle. If

$$m = \lceil 20\log(1/\zeta)\rceil,$$

then, with probability at least $1-\zeta$, at least $0.51m$ of the points $x_1, \ldots, x_m$ satisfy

$$\|x_i - x_{\bar{x},\lambda}^\star\|_p \le \frac{r}{3C_{\text{ba}}}.$$

Consequently, applying Aggregate *(Algorithm 11) to $x_1, \dots, x_m$ returns a point $\widehat{x}$ satisfying*

$$\|\widehat{x} - x^\star_{\bar{x},\lambda}\|_p \le \frac{r}{C_{\mathrm{ba}}}$$

with probability at least $1 - \zeta$.

Proof. By uniformly convexity, the function $\Phi(x) := \frac{1}{p}\|x\|_p^p$ satisfies, for all $u, v \in \mathbb{R}^d$,

$$\Phi(u) \ge \Phi(v) + \langle J_p(v), u - v\rangle + \frac{2^{2-p}}{p}\|u - v\|_p^p, \tag{41}$$

where $J_p(v) = \left(|v_i|^{p-2}v_i\right)_{i=1}^d$. Therefore,

$$\frac{\lambda}{p}\|x - \bar{x}\|_p^p \ge \frac{\lambda}{p}\|y - \bar{x}\|_p^p + \lambda\langle J_p(y - \bar{x}), x - y\rangle + \frac{2^{2-p}\lambda}{p}\|x - y\|_p^p.$$

Since F is convex, for any $g \in \partial F(y)$, $F(x) \ge F(y) + \langle g, x - y\rangle$. Combining two inequalities gives

$$F(x) + \frac{\lambda}{p}\|x - \bar{x}\|_p^p \ge F(y) + \frac{\lambda}{p}\|y - \bar{x}\|_p^p + \langle g + \lambda J_p(y - \bar{x}), x - y\rangle + \frac{2^{2-p}\lambda}{p}\|x - y\|_p^p. \tag{42}$$

Taking $y = x^\star_{\bar{x},\lambda}$ and using the first-order optimality condition for $x^\star_{\bar{x},\lambda}$ gives

$$F(x) + \frac{\lambda}{p}\|x - \bar{x}\|_p^p \ge F(y) + \frac{\lambda}{p}\|y - \bar{x}\|_p^p + \frac{2^{2-p}\lambda}{p}\|x - y\|_p^p. \tag{43}$$

Combining (43) with the definition of $\left(\frac{\lambda r^p}{C_{\mathrm{ba}}}, \lambda\right)$-ball optimization oracle gives

$$\begin{aligned}\mathbb{E}\|x - x^\star_{\bar{x},\lambda}\|_p^p &\le \frac{p\,2^{p-2}}{\lambda}\mathbb{E}\left[F(x) + \frac{\lambda}{p}\|x - \bar{x}\|_p^p - F(x^\star_{\bar{x},\lambda}) - \frac{\lambda}{p}\|x^\star_{\bar{x},\lambda} - \bar{x}\|_p^p\right] \\ &\le \frac{p\,2^{p-2}}{\lambda}\cdot\frac{2^{2-p}\lambda r^p}{p\,3^{p+1}C_{\mathrm{ba}}^p} = \frac{r^p}{3^{p+1}C_{\mathrm{ba}}^p}.\end{aligned}$$

Therefore, by Markov's inequality,

$$\Pr\left(\|x - x^\star_{\bar{x},\lambda}\|_p > \frac{r}{3C_{\mathrm{ba}}}\right) = \Pr\left(\|x - x^\star_{\bar{x},\lambda}\|_p^p > \frac{r^p}{3^pC_{\mathrm{ba}}^p}\right) \le \frac{\mathbb{E}\|x - x^\star_{\bar{x},\lambda}\|_p^p}{r^p/(3^pC_{\mathrm{ba}}^p)} \le \frac{1}{3}.$$

Hence

$$\Pr\left(\|x - x^\star_{\bar{x},\lambda}\|_p \le \frac{r}{3C_{\mathrm{ba}}}\right) \ge \frac{2}{3}.$$

For $i = 1, \dots, m$, define

$$Z_i := \mathbf{1}\left\{\|x_i - x^\star_{\bar{x},\lambda}\|_p \le \frac{r}{3C_{\mathrm{ba}}}\right\}.$$

Then $Z_1, \dots, Z_m$ are independent Bernoulli random variables with $\mathbb{E}Z_i \ge \frac{2}{3}$. By a Chernoff bound, with probability at least $1 - \zeta$, at least $0.51m$ of the points $x_1, \dots, x_m$ satisfy

$$\|x_i - x^\star_{\bar{x},\lambda}\|_p \le \frac{r}{3C_{\mathrm{ba}}}.$$

Therefore, the preconditions of Fact 6 are satisfied, and Aggregate (Algorithm 11) returns $\widehat{x}$ such that

$$\|\widehat{x} - x^\star_{\bar{x},\lambda}\|_p \le \frac{r}{C_{\mathrm{ba}}}.$$

□

Algorithm 8 p-MorGradEst

Input: r-BROO $\mathcal{O}_{\lambda,\delta}(\cdot)$ of radius r, query point y, regularization $\lambda > 0$, bias $\beta > 0$, $\sigma > 0$
Set $T_{\max} = \max\left\{2, \frac{C_p \lambda^p r^{p(p-1)}}{\beta^p}\right\}, T_0 = \max\left\{1, \frac{C_p \lambda^p r^{p(p-1)} \log^{p-1}(2T_{\max})}{\sigma^p}\right\}$
$j_{\max} = \lfloor \log_2 T_{\max} \rfloor, \quad \delta_j = \frac{r}{(2^j T_0)^{1/p}}$
$x_0 = \mathcal{O}_{\lambda,\delta_0}(y)$, $g_0 = \lambda J_p(y - x_0)$
Sample $J \sim \mathrm{Geom}(1/2)$
if $J \le j_{\max}$ **then**
    $x_J = \mathcal{O}_{\lambda,\delta_J}(y)$, $x_{J-1} = \mathcal{O}_{\lambda,\delta_{J-1}}(y)$
    $g_J = \lambda J_p(y - x_J)$, $g_{J-1} = \lambda J_p(y - x_{J-1})$
    $\widehat{g} = g_0 + 2^J(g_J - g_{J-1})$
else
    $\widehat{g} = g_0$
end if
return $\widehat{g}$

Lemma 22. *Let $F : \mathcal{X} \to \mathbb{R}$ satisfy Assumption 1. For a query point $y \in \mathcal{X}$ and $\lambda, r > 0$, define*

$$x^\star \in \arg\min_{x \in \mathcal{X}} \left\{ F(x) + \frac{\lambda}{p}\|x - y\|_p^p \right\}, \qquad g = \lambda J_p(y - x^\star),$$

where $J_p(z) = (|z_i|^{p-2} z_i)_{i=1}^d$. Suppose that $\|x^\star - y\|_p \le r$. Let $\widehat{g}$ be the geometric estimator in Algorithm 8, run with these values of T_0 and $j_{\max}$ and with

$$\delta_j = \frac{r}{(2^j T_0)^{1/p}}, \qquad 0 \le j \le j_{\max},$$

Then, for a constant $C_p > 0$ depending only on p,

$$\|\mathbb{E}\widehat{g} - g\|_q \le \frac{C_p \lambda r^{p-1}}{\sqrt{2^{j_{\max}} T_0}}, \tag{44}$$

$$\mathbb{E}\|\widehat{g} - \mathbb{E}\widehat{g}\|_q^q \le \frac{C_p \lambda^q r^p}{T_0^{q/2}} \left(1 + \sum_{j=1}^{j_{\max}} 2^{-j(1-q/2)}\right). \tag{45}$$

Proof. Let $\Phi(u) = \|u\|_p^p / p$, so that

$$D_\Phi(u, v) = \Phi(u) - \Phi(v) - \langle J_p(v), u - v \rangle.$$

We first show that

$$\|J_p(u) - J_p(v)\|_q^2 \le C_p(\|u\|_p + \|v\|_p)^{p-2} D_\Phi(u, v). \tag{46}$$

For $p = 2$, this follows from $\|u - v\|_2^2 = 2D_\Phi(u, v)$. For $p > 2$, integrating the second derivative of $|a|^p/p$ gives

$$D_\Phi(u, v) = (p-1)\sum_{i=1}^d (u_i - v_i)^2 \int_0^1 (1-t)|v_i + t(u_i - v_i)|^{p-2}\, dt \ge c_p \sum_{i=1}^d (|u_i| + |v_i|)^{p-2}(u_i - v_i)^2$$

for a constant $c_p > 0$ depending only on p. Indeed, after rescaling so that $|a| + |b| = 1$, the integral $\int_0^1 (1-t)|b + t(a-b)|^{p-2}\, dt$ is continuous and strictly positive on this compact set, and therefore has a positive minimum. Also,

$$\left||u_i|^{p-2}u_i - |v_i|^{p-2}v_i\right| \le (p-1)(|u_i| + |v_i|)^{p-2}|u_i - v_i|.$$

Hölder's inequality therefore yields

$$\begin{aligned}\|J_p(u) - J_p(v)\|_q^2 &\le (p-1)^2 \left(\sum_{i=1}^d (|u_i| + |v_i|)^p\right)^{(p-2)/p} \sum_{i=1}^d (|u_i| + |v_i|)^{p-2}(u_i - v_i)^2 \\ &\le C_p(\|u\|_p + \|v\|_p)^{p-2} D_\Phi(u, v),\end{aligned}$$

which proves (46). By the convexity of F and optimality of $x^\star$, for every $x \in \mathcal{X}$,

$$F(x) + \frac{\lambda}{p}\|x - y\|_p^p - F(x^\star) - \frac{\lambda}{p}\|x^\star - y\|_p^p \ge \lambda D_\Phi(x - y, x^\star - y).$$

Since $\|x_j - y\|_p \le r$ and $\|x^\star - y\|_p \le r$, applying (46) with $u = x_j - y$ and $v = x^\star - y$ gives

$$\begin{aligned}\mathbb{E}\|g_j - g\|_q^2 &\le C_p \lambda r^{p-2} \mathbb{E}\left[F(x_j) + \frac{\lambda}{p}\|x_j - y\|_p^p - F(x^\star) - \frac{\lambda}{p}\|x^\star - y\|_p^p\right] \\ &\le \frac{C_p \lambda^2 r^{2p-2}}{2^j T_0}. \end{aligned} \tag{47}$$

Here we used $g_j = \lambda J_p(y - x_j)$, $g = \lambda J_p(y - x^\star)$, and $J_p(-u) = -J_p(u)$. By the definition of $\widehat{g}$,

$$\mathbb{E}\widehat{g} = \mathbb{E}g_0 + \sum_{j=1}^{j_{\max}} (\mathbb{E}g_j - \mathbb{E}g_{j-1}) = \mathbb{E}g_{j_{\max}}.$$

Consequently,

$$\|\mathbb{E}\widehat{g} - g\|_q \le \left(\mathbb{E}\|g_{j_{\max}} - g\|_q^2\right)^{1/2} \le \frac{C_p \lambda r^{p-1}}{\sqrt{2^{j_{\max}} T_0}},$$

which proves (44). Since $1 < q \le 2$, Jensen's inequality and $(p-1)q = p$ give

$$\mathbb{E}\|g_j - g\|_q^q \le \left(\mathbb{E}\|g_j - g\|_q^2\right)^{q/2} \le \frac{C_p \lambda^q r^p}{(2^j T_0)^{q/2}}.$$

It follows that

$$\mathbb{E}\|g_j - g_{j-1}\|_q^q \le 2^{q-1}\left(\mathbb{E}\|g_j - g\|_q^q + \mathbb{E}\|g_{j-1} - g\|_q^q\right) \le \frac{C_p \lambda^q r^p}{(2^j T_0)^{q/2}}.$$

Using the definition of $\widehat{g}$ once more,

$$\begin{aligned}\mathbb{E}\|\widehat{g} - \mathbb{E}\widehat{g}\|_q^q &\le 2^q \mathbb{E}\|\widehat{g} - g\|_q^q \\ &\le C_p \left(\mathbb{E}\|g_0 - g\|_q^q + \sum_{j=1}^{j_{\max}} 2^{j(q-1)} \mathbb{E}\|g_j - g_{j-1}\|_q^q\right) \\ &\le \frac{C_p \lambda^q r^p}{T_0^{q/2}} \left(1 + \sum_{j=1}^{j_{\max}} 2^{-j(1-q/2)}\right).\end{aligned}$$

This proves (45). The argument also applies after conditioning on the history before the call. □

Lemma 23. *Let F satisfy Assumption 1 with a minimizer $x^\star$. Let $\ell = \log\left(\frac{GR^2}{r\epsilon_{\text{opt}}}\right)$. Let*

$$\varphi_k = \frac{\lambda_k r^p}{C_p \ell^{p+1}} \quad , \quad \sigma_k^q = \frac{\lambda_k^q r^p}{C_p \ell^{p+1}} \quad , \quad \beta_k = \frac{\epsilon_{\text{opt}}}{C_p R} \quad , \quad A_0 = \frac{R^{p-1}}{G} \text{ and } A_{\max} = \frac{C_p R^p}{\epsilon_{\text{opt}}}.$$

Define $\widehat{x}_k := \operatorname{argmin}_{x\in\mathcal{X}} \left\{F(x) + \frac{\lambda_k}{p}\|x - y_{k-1}\|_p^p\right\}$, and assume that for each k we have $\|\widehat{x}_k - y_{k-1}\|_p \le r$ and that one of the following must occur:

1. $\lambda_k < 2\lambda_m := \frac{2\epsilon_{\text{opt}}}{r^{\frac{p^2}{p+1}} R^{\frac{p}{p+1}}} \ell^p$, *or*
2. $\|\widehat{x}_k - y_{k-1}\|_p \ge \frac{3}{4} r$.

Suppose $\lambda_k \ge \lambda_m$ for every k and

$$A_{k+1}^{p-1} = 2^{p-1}\lambda_{k+1} a_{k+1}^p, \qquad A_{k+1} = A_k + a_{k+1}.$$

Let $K \le K_{\max}$ be the stopping time of the algorithm, which stops when $A_k \ge A_{\max}$ or $k = K_{\max}$. Then, for sufficiently large C_p, there exist constants $c_{1,p}, c_{2,p} > 0$ depending only on p such that

$$\mathbb{E}\left[A_K\left(F(x_K) - F(x^\star) - c_{1,p}\epsilon_{\text{opt}}\right) + c_{2,p}\sum_{i\le K}\lambda_i A_i \|\widehat{x}_i - y_{i-1}\|_p^p\right] \le A_0\left(F(x_0) - F(x^\star)\right) + C_p R^p.$$

Proof. Define the filtration

$$\mathcal{F}_k = \sigma(x_1, v_1, A_1, \zeta_1, \ldots, x_k, v_k, A_k, \zeta_k),$$

where ζ_i is the internal randomness of NextLambda, and note that A_{k+1}, y_k, and $\widehat{x}_{k+1}$ are deterministic when conditioned on x_k, v_k, A_k, ζ_k. Following the proof of Lemma 19, we define

$$M_k = A_k\left(F(x_k) - F(x^\star) - \frac{\epsilon_{\text{opt}}}{20}\right) + 2D_\Phi(x^\star, v_k) + \frac{1-2^{1-q}}{2q}\sum_{i\le k}\lambda_i A_i\|\widehat{x}_i - y_{i-1}\|_p^p$$

and show it is a supermartingale adapted to the filtration $\mathcal{F}_k$. From Lemma 18, we have

$$\begin{aligned}\mathbb{E}[M_{k+1} \mid \mathcal{F}_k] \le{}& A_k(F(x_k) - F(x^\star)) + 2D_\Phi(x^\star, v_k) - \frac{1-2^{1-q}}{q}\lambda_{k+1}A_{k+1}\|\widehat{x}_{k+1} - y_k\|_p^p \\ &+ \mu_{k+1} - A_{k+1}\frac{\epsilon_{\text{opt}}}{20} + \frac{1-2^{1-q}}{2q}\sum_{i\le k+1}\lambda_i A_i\|\widehat{x}_i - y_{i-1}\|_p^p, \end{aligned} \tag{21}$$

where

$$\mu_{k+1} := A_{k+1}\varphi_{k+1} + \frac{2^{2-q}a_{k+1}^q}{q}\sigma_{k+1}^q + 2Ra_{k+1}\beta_{k+1}.$$

Substituting the values of φ_{k+1}, σ_{k+1}^q, and β_{k+1} into the definition of μ_{k+1} gives

$$\mu_{k+1} = \frac{\lambda_{k+1}A_{k+1}r^p}{C_p\ell^{p+1}} + \frac{2^{2-q}a_{k+1}^q\lambda_{k+1}^q r^p}{qC_p\ell^{p+1}} + \frac{2a_{k+1}\epsilon_{\text{opt}}}{C_p}.$$

Recall that $A_{k+1}^{p-1} = 2^{p-1}\lambda_{k+1}a_{k+1}^p$, therefore $a_{k+1}^q\lambda_{k+1}^q = \lambda_{k+1}A_{k+1}/2$ and

$$\mu_{k+1} = \frac{1+2^{1-q}/q}{C_p\ell^{p+1}}\lambda_{k+1}A_{k+1}r^p + \frac{2a_{k+1}\epsilon_{\text{opt}}}{C_p} = \frac{2^{(p-1)/p}(1+2^{1-q}/q)a_{k+1}\lambda_{k+1}^{1+1/p}A_{k+1}^{1/p}r^p}{C_p\ell^{p+1}} + \frac{2a_{k+1}\epsilon_{\text{opt}}}{C_p}.$$

If $\|\widehat{x}_{k+1} - y_k\|_p \geq 3r/4$, then for sufficiently large C_p and $\ell \geq 1$,

$$\mu_{k+1} \leq \frac{1 - 2^{1-q}}{2q} \lambda_{k+1} A_{k+1} \|\widehat{x}_{k+1} - y_k\|_p^p + \frac{a_{k+1}\epsilon_{\text{opt}}}{20}.$$

Else, if $\lambda_{k+1} < 2\lambda_m$, we have

$$\mu_{k+1} < \frac{2^{(p-1)/p}(1 + 2^{1-q}/q) a_{k+1} (2\lambda_m)^{1+1/p} A_{k+1}^{1/p} r^p}{C_p \ell^{p+1}} + \frac{2a_{k+1}\epsilon_{\text{opt}}}{C_p}.$$

Now, note that $\lambda_m A_{\max} = C_p (R/r)^{p^2/(p+1)} \ell^p \geq 1$ for sufficiently large C_p when $0 < r \leq R$ and $\ell \geq 1$, so the NextLambda lower bound $\lambda_k \geq \lambda_m$ implies $\lambda_k \geq 1/A_{\max}$. From the definition of the stopping time, $A_{K-1} \leq A_{\max}$. If $a_K > A_{\max}$, then $A_K = A_{K-1} + a_K \leq 2a_K$, and the weight equation gives

$$a_K^p = \frac{A_K^{p-1}}{2^{p-1}\lambda_K} \leq \frac{(2a_K)^{p-1} A_{\max}}{2^{p-1}} = a_K^{p-1} A_{\max},$$

a contradiction. Therefore $a_K \leq A_{\max}$, and $A_k \leq A_K \leq 2A_{\max}$ for every $k \leq K$. Thus, when $\lambda_{k+1} < 2\lambda_m$, the bound on μ_{k+1} becomes

$$\begin{aligned}
\mu_{k+1} &< \frac{2^{(p-1)/p}(1 + 2^{1-q}/q) a_{k+1} (2\lambda_m)^{1+1/p} (2A_{\max})^{1/p} r^p}{C_p \ell^{p+1}} + \frac{2a_{k+1}\epsilon_{\text{opt}}}{C_p} \\
&= a_{k+1}\epsilon_{\text{opt}} \left(\frac{4 \cdot 2^{1/p}(1 + 2^{1-q}/q)}{C_p^{(p-1)/p}} + \frac{2}{C_p} \right) \\
&\leq a_{k+1} \frac{\epsilon_{\text{opt}}}{20} \leq \frac{1 - 2^{1-q}}{2q} \lambda_{k+1} A_{k+1} \|\widehat{x}_{k+1} - y_k\|_p^p + a_{k+1} \frac{\epsilon_{\text{opt}}}{20},
\end{aligned}$$

where the penultimate inequality holds for sufficiently large C_p. Therefore, for every $k < K$, we have

$$\mu_{k+1} \leq \frac{1 - 2^{1-q}}{2q} \lambda_{k+1} A_{k+1} \|\widehat{x}_{k+1} - y_k\|_p^p + a_{k+1} \frac{\epsilon_{\text{opt}}}{20}.$$

Noting that $\mathbb{E}|M_k| < \infty$ and substituting the bound on μ_{k+1} into (21), we get

$$\mathbb{E}[M_{k+1} \mid \mathcal{F}_k] \leq A_k \left(F(x_k) - F(x^\star) - \frac{\epsilon_{\text{opt}}}{20} \right) + 2D_\Phi(x^\star, v_k) + \frac{1 - 2^{1-q}}{2q} \sum_{i \leq k} \lambda_i A_i \|\widehat{x}_i - y_{i-1}\|_p^p = M_k.$$

Therefore M_k is a supermartingale adapted to $\mathcal{F}_k$. Since $K \leq K_{\max}$ is a bounded stopping time adapted to $\mathcal{F}_k$, by the optional stopping theorem for supermartingales we have

$$\begin{aligned}
\mathbb{E}M_K \leq M_0 &= A_0 \left(F(x_0) - F(x^\star) - \frac{\epsilon_{\text{opt}}}{20} \right) + 2D_\Phi(x^\star, v_0) \\
&\leq A_0 (F(x_0) - F(x^\star)) + C_p R^p,
\end{aligned}$$

where $v_0 = x_0$, and $2D_\Phi(x^\star, x_0) \leq C_p R^p$ holds when the mirror potential is centered at x_0 (or equivalently after translating x_0 to the origin). This completes the proof. □

Proposition 11. *Let $F : \mathbb{R}^d \to \mathbb{R}$ be convex and L-Lipschitz with respect to $\|\cdot\|_p$, and let $x^\star \in \mathbb{B}_p(R)$. There is an implementation of BallAccel (see Proposition 10) taking parameters $r \in (0, R]$ and $\epsilon_{\text{opt}} \in (0, LR/e]$ with the following guarantee. Define κ, K, and $\lambda^\star$ as in Proposition 10. For a sufficiently large constant $C_{\text{ba}} > 0$ depending only on p, BallAccel runs in at most*

$C_{\mathrm{ba}} K \log\kappa \log\left(\frac{R\kappa}{r}\right)$ *iterations and produces* $x \in \mathbb{R}^d$ *such that* $\mathbb{E}F(x) \le F(x^\star) + \epsilon_{\mathrm{opt}}$. *For general* F, *the ball optimization oracles below are applied to*

$$x \mapsto F(x) + L(\|x\|_p - R)_+.$$

Their calls satisfy:

1. *Each iteration makes at most* $C_{\mathrm{ba}} \log^2\left(\frac{R\kappa}{r}\right)$ *calls to a* $\left(\frac{\lambda r^p}{C_{\mathrm{ba}}}, \lambda\right)$ *-ball optimization oracle, with* $\lambda \in \left[\frac{\lambda^\star}{C_{\mathrm{ba}}}, \frac{C_{\mathrm{ba}} L}{r^{p-1}}\right]$.
2. *For each* $j \in [\lceil \log_2 K + C_{\mathrm{ba}} \rceil]$, *at most* $C_{\mathrm{ba}}^2\, 2^{-j} K \log\kappa \log\left(\frac{R\kappa}{r}\right)$ *iterations query a* $\left(\frac{\lambda r^p}{C_p\, 2^j \log^{2p}(R\kappa/r)}, \lambda\right)$ *-ball optimization oracle, for some* $\lambda \in \left[\frac{\lambda^\star}{C_{\mathrm{ba}}}, \frac{C_{\mathrm{ba}} L}{r^{p-1}}\right]$.

Moreover, if F *admits a global minimizer in* $\mathbb{B}_p(R)$, *the same guarantees and oracle-call bounds hold for an implementation whose ball optimization oracles are applied directly to* F, *without adding a radial penalty and without projecting the final output.*

Proof. We follow the organization of Appendix C of [CJJ+23]. For the general implementation, it suffices to take

$$x^\star \in \operatorname*{argmin}_{x \in \mathbb{B}_p(R)} F(x).$$

Apply the method to $x \mapsto F(x) + L(\|x\|_p - R)_+$. This function is convex and $2L$-Lipschitz, and

$$F(x) + L(\|x\|_p - R)_+ \ge F\left(\frac{x}{\max\{1, \|x\|_p/R\}}\right) \ge F(x^\star).$$

Thus its objective error is nonnegative everywhere, and projecting the final output onto $\mathbb{B}_p(R)$ gives the required guarantee for F. Replacing L by $2L$ changes the bounds only by constant factors. For a stochastic-subgradient implementation, add a subgradient of $L(\|x\|_p - R)_+$ to the original oracle. Its dual norm is at most L, so Minkowski's inequality bounds the resulting q-th moment by $(2L)^q$. Initialize $x_0 = v_0 = 0$, using Lipschitz bound $2L$ in the initialization of A_0, and retain

$$A_i = A_{i-1} + a_i, \qquad A_i^{p-1} = 2^{p-1} \lambda_i a_i^p.$$

Use the mirror update over $\mathbb{B}_p(R)$ and define

$$\widehat{x}_i = \operatorname*{argmin}_{x \in \mathbb{R}^d} \left\{ F(x) + L(\|x\|_p - R)_+ + \frac{\lambda_i}{p} \|x - y_{i-1}\|_p^p \right\}.$$

Lemma 18 applies with the same $2Ra_i\beta_i$ term: its radius bound is needed only for $x^\star$ and the mirror iterates, while $\widehat{x}_i$ minimizes the objective over $\mathbb{R}^d$. Use oracle implementations that return points in their queried radius-r balls for the primal, line-search, and MorGradEst calls. Each execution stops at the first completed iteration with $A_i \ge A_{\max}$, or after $\lceil C_p K \log(R\kappa/r) \rceil$ iterations, where

$$A_{\max} = \frac{C_{\mathrm{ba}} R^p}{\epsilon_{\mathrm{opt}}}.$$

Here C_{ba} is sufficiently large and depends only on p. Write τ for this stopping index. An execution is accepted only if $A_\tau \ge A_{\max}$. We first bound the calls required in each iteration. The primal updates and the mirror constraint give

$$\|x_i\|_p \le R + ir, \qquad \|x_i - v_i\|_p \le 2R + ir.$$

Each execution has $O(K\log(R\kappa/r))$ iterations. Since $\|x_i - v_i\|_p \le 2R + ir$ and $Kr \le R$, we have

$$\|x_i - v_i\|_p \le 2R + O(Kr\log(R\kappa/r)) = O(R\log(R\kappa/r)).$$

Using this bound in Theorem 8 gives $O(\log(R\kappa/r))$ distance tests per iteration. The mirror iterates remain in $\mathbb{B}_p(R)$.

For each test, use independent calls with expected objective error $\lambda r^p/C_p$. By Lemma 21, $O(\log(R\kappa/r))$ repetitions followed by AGGREGATE make the conditional failure probability at most

$$\frac{1}{C_p\kappa K\lceil\log_2(128\kappa)\rceil\log^2(R\kappa/r)}.$$

A union bound over all tests and executions gives total failure probability at most $1/(C_p\kappa)$. Abort an execution before exceeding the stated test budget. When all tests are accurate, this budget is sufficient. This proves item 1 without function-value queries.

For the analysis, consider an execution without the caps on oracle calls, in which inaccurate distance tests are replaced by their exact outcomes. Fresh primal and MorGradEst randomness retains its conditional guarantees; no conditioning on future test accuracy or future caps is used. Theorem 8 then gives

$$\|\widehat{x}_i - y_{i-1}\|_p \le r, \qquad \lambda_i < 2\lambda_{\min} \quad \text{or} \quad \|\widehat{x}_i - y_{i-1}\|_p \ge \frac{3r}{4},$$

where $\lambda_{\min}$ differs from $\lambda^\star$ by a constant factor. For the additional calls, use the construction of Lemma 22 with

$$j_{\max} = \lceil\log_2 K\rceil, \qquad T_0 = C_{\mathrm{ba}}^{4p}\log^{2p}(R\kappa/r), \qquad \delta_j = \frac{r}{(2^jT_0)^{1/p}}.$$

That lemma and an independent primal call give

$$\begin{aligned}
\beta_i &\le \frac{C_p\lambda_i r^{p-1}}{\sqrt{KC_{\mathrm{ba}}^{4p}}\log^p(R\kappa/r)},\\
\sigma_i^q &\le \frac{C_p\lambda_i^q r^p}{C_{\mathrm{ba}}^{2pq}\log^{pq}(R\kappa/r)}\left(1 + \sum_{j=1}^{\lceil\log_2 K\rceil} 2^{-j(1-q/2)}\right),\\
\varphi_i &\le \frac{C_p\lambda_i r^p}{C_{\mathrm{ba}}^{4p}\log^{2p}(R\kappa/r)}.
\end{aligned}$$

The sum in the second bound is bounded by a constant depending only on p when $p > 2$, and is $O(\log(R\kappa/r))$ when $p = 2$. Using $a_i^q\lambda_i^q = \lambda_i A_i/2$ and Young's inequality, we obtain, for any sufficiently small fixed $c_p > 0$ and sufficiently large C_{ba},

$$A_i\varphi_i + \frac{2^{2-q}a_i^q}{q}\sigma_i^q + 2Ra_i\beta_i \le \frac{c_p\lambda_i A_i r^p}{\log^{p+1}(R\kappa/r)} + \frac{C_pR^p}{C_{\mathrm{ba}}^{2p^2}K^{p/2}\log(R\kappa/r)}.$$

If $\|\widehat{x}_i - y_{i-1}\|_p \ge 3r/4$, the first term is absorbed by a fixed fraction of the negative displacement term. Otherwise $\lambda_i < 2\lambda_{\min}$. The final-increment argument gives $A_\tau \le 2A_{\max}$, and $K^{p+1}r^p = R^p$ implies

$$\frac{\lambda_i A_i r^p}{a_i\log^{p+1}(R\kappa/r)} \le C_pC_{\mathrm{ba}}^{1/p}\epsilon_{\mathrm{opt}}\left(\frac{\log\kappa}{\log(R\kappa/r)}\right)^{p+1} \le C_pC_{\mathrm{ba}}^{1/p}\epsilon_{\mathrm{opt}}.$$

Choose c_p sufficiently small, using $\sum_{i\le\tau} a_i \le 2A_{\max}$, and then choose C_{ba} sufficiently large, using $\tau \le \lceil C_p K \log(R\kappa/r)\rceil$ and $K^{1-p/2} \le 1$. The accumulated errors are then bounded by a fixed multiple of R^p. Applying bounded stopping to the conditional potential inequality gives

$$\mathbb{E}\Bigg[A_\tau\big(F(x_\tau) + L(\|x_\tau\|_p - R)_+ - F(x^\star)\big) + \frac{1-2^{1-q}}{2q}\sum_{i=1}^{\tau}\lambda_i A_i \|\widehat{x}_i - y_{i-1}\|_p^p\Bigg] \le C_{0,p}R^p. \tag{48}$$

Both terms on the left are nonnegative. Since the error constants are chosen after C_{ba} and the iteration budget, $C_{0,p}$ can be fixed independently of those choices.

We next show that an execution is accepted with constant probability. On $\{A_\tau < A_{\max}\}$, $\tau = \lceil C_p K \log(R\kappa/r)\rceil$. If at least half the iterations have $\lambda_i < 2\lambda_{\min}$, the weight identity gives

$$A_\tau^{1/p} - A_0^{1/p} \ge \frac{1}{p2^{(p-1)/p}}\sum_{i=1}^{\tau}\lambda_i^{-1/p} \ge \frac{\tau}{C_p\lambda_{\min}^{1/p}}.$$

For a sufficiently large iteration budget, this implies $A_\tau \ge A_{\max}$, a contradiction. Thus at least half the iterations have displacement at least $3r/4$. Using

$$\lambda_i A_i = \frac{1}{2^{p-1}}\left(\frac{A_i}{a_i}\right)^p, \qquad \sum_{i=1}^{\tau}\frac{a_i}{A_i} \le \log\frac{A_\tau}{A_0} \le C_p\log(R\kappa/r),$$

Hölder's inequality yields

$$\sum_{i=1}^{\tau}\lambda_i A_i\|\widehat{x}_i - y_{i-1}\|_p^p \ge \frac{(3r/4)^p}{2^{p-1}}\frac{(\tau/2)^{p+1}}{(\sum_{i=1}^{\tau}a_i/A_i)^p} \ge \frac{c_p r^p \tau^{p+1}}{\log^p(R\kappa/r)}.$$

Increasing the iteration budget makes the displacement term in (48) exceed $4C_{0,p}R^p$ on $\{A_\tau < A_{\max}\}$. Markov's inequality therefore gives

$$\Pr(A_\tau \ge A_{\max}) \ge \frac{3}{4}.$$

The same potential bound gives

$$\mathbb{E}\left[\big(F(x_\tau) + L(\|x_\tau\|_p - R)_+ - F(x^\star)\big)\mathbf{1}_{\{A_\tau \ge A_{\max}\}}\right] \le \frac{C_{0,p}R^p}{A_{\max}} \le \frac{\epsilon_{\mathrm{opt}}}{64},$$

where $C_{\mathrm{ba}} \ge 64C_{0,p}$.

To enforce deterministic bounds on the additional calls, pre-sample independent geometric levels for all potential iterations, revealing each only when its iteration is reached. Level 0 is always used. For $t \ge 1$, a level-t call occurs only when the sampled level is t or $t+1$, with probability $O(2^{-t})$. A Chernoff bound and a union bound over all levels and executions give the per-execution bounds

$$C_p 2^{-t} K\log(R\kappa/r), \qquad 0 \le t \le \lceil \log_2 K\rceil,$$

except with total probability at most $1/(C_p\kappa)$. Abort before exceeding a bound and treat aborted executions as unsuccessful. Combining this event with the distance-test events, the implemented and analyzed executions agree except with probability at most $1/(C_p\kappa)$, after increasing constants.

Project each output onto $\mathbb{B}_p(R)$. Its excess loss lies in $[0, 2LR]$, so this comparison adds at most $2LR/(C_p\kappa)$ to the expected excess loss. Consequently, each implemented execution satisfies

$$\Pr(A_\tau \geq A_{\max}) \geq \frac{1}{2},$$
$$\mathbb{E}\left[\left(F\left(\frac{x_\tau}{\max\{1, \|x_\tau\|_p/R\}}\right) - F(x^\star)\right)\mathbf{1}_{\{A_\tau \geq A_{\max}\}}\right] \leq \frac{\epsilon_{\text{opt}}}{32}.$$

Independently restart from 0 until the first accepted execution and return its projected output. After $\lceil \log_2(128\kappa) \rceil$ unsuccessful executions, return 0. This selection uses no objective values, and

$$\mathbb{E}[F(x) - F(x^\star)] \leq \sum_{b=1}^{\lceil \log_2(128\kappa) \rceil} 2^{-(b-1)} \frac{\epsilon_{\text{opt}}}{32} + LR\, 2^{-\lceil \log_2(128\kappa) \rceil} < \epsilon_{\text{opt}}.$$

This proves the guarantee for every comparator in $\mathbb{B}_p(R)$. Finally, the total iteration budget is

$$\lceil \log_2(128\kappa) \rceil \lceil C_p K \log(R\kappa/r) \rceil = O(K \log\kappa \log(R\kappa/r)).$$

Index the primal and level-0 calls by $j = 1$, and level $t \geq 1$ by $j = t + 1$. The total number of iterations querying level j is

$$O\left(2^{-j} K \log\kappa \log(R\kappa/r)\right),$$

with objective accuracy

$$\frac{\lambda r^p}{C_p 2^j \log^{2p}(R\kappa/r)}.$$

We now prove the additional claim. Choose

$$x^\star \in \operatorname*{argmin}_{x \in \mathbb{R}^d} F(x) \cap \mathbb{B}_p(R).$$

Run the same construction directly on F, initialize A_0 using its Lipschitz bound L, and define

$$\widehat{x}_i = \operatorname*{argmin}_{x \in \mathbb{R}^d} \left\{ F(x) + \frac{\lambda_i}{p} \|x - y_{i-1}\|_p^p \right\}.$$

All local oracle calls are now for F itself. The oracle-call estimates and the conditional potential analysis remain unchanged up to constants depending only on p. Since $x^\star$ is a global minimizer, $F(x) - F(x^\star) \geq 0$ for every $x \in \mathbb{R}^d$. Let

$$N = \lceil C_p K \log(R\kappa/r) \rceil.$$

For the analyzed execution with exact distance tests, the same bounded-stopping argument gives

$$\mathbb{E}\left[A_\tau\big(F(x_\tau) - F(x^\star)\big) + \frac{1 - 2^{1-q}}{2q} \sum_{i=1}^{\tau} \lambda_i A_i \|\widehat{x}_i - y_{i-1}\|_p^p\right] \leq C_{0,p} R^p.$$

Both terms are nonnegative. The preceding acceptance argument therefore gives

$$\Pr(A_\tau \geq A_{\max}) \geq \frac{3}{4}$$

and, for sufficiently large C_{ba},

$$\mathbb{E}\left[\big(F(x_\tau)-F(x^\star)\big)\mathbf{1}_{\{A_\tau\geq A_{\max}\}}\right]\leq\frac{\epsilon_{\mathrm{opt}}}{64}.$$

For both the implemented and analyzed executions, ball-feasibility of the primal oracle outputs implies $\|x_i\|_p\leq R+ir$ for every completed iteration. Consequently, for every possible output with $\tau\leq N$,

$$0\leq F(x_\tau)-F(x^\star)\leq L\|x_\tau-x^\star\|_p\leq L(2R+Nr).$$

Strengthen the coupling failure probability over all planned executions to at most

$$\frac{1}{C_p\kappa(2+Nr/R)}.$$

This preserves the stated asymptotic oracle-call budgets. Indeed, $Kr\leq R$ implies

$$2+Nr/R=O(\log(R\kappa/r)),$$

so the additional logarithm in the amplification requirement is absorbed by the existing $O(\log(R\kappa/r))$ repetitions. The geometric-level caps are adjusted by constant factors in the same way.

The contribution of the coupling failures is now at most

$$L(2R+Nr)\frac{1}{C_p\kappa(2+Nr/R)}=\frac{LR}{C_p\kappa}=\frac{\epsilon_{\mathrm{opt}}}{C_p}.$$

Thus, after increasing constants, each implemented execution satisfies

$$\Pr(A_\tau\geq A_{\max})\geq\frac{1}{2}$$

and

$$\mathbb{E}\left[\big(F(x_\tau)-F(x^\star)\big)\mathbf{1}_{\{A_\tau\geq A_{\max}\}}\right]\leq\frac{\epsilon_{\mathrm{opt}}}{32}.$$

Independently restart until the first accepted execution, returning its unprojected output. After $\lceil\log_2(128\kappa)\rceil$ unsuccessful executions, return 0. Since $F(0)-F(x^\star)\leq LR$, the same restart calculation proves

$$\mathbb{E}[F(x)-F(x^\star)]<\epsilon_{\mathrm{opt}}.$$

The total iteration and oracle-call bounds are unchanged up to constants depending only on p. Since a global minimizer has objective value no larger than any comparator in $\mathbb{B}_p(R)$, the guarantee also holds for every such comparator. □

Proposition 12 (based on results in [GL12])**.** *Let f,g satisfy the assumptions of Problem 1. For $\bar{x}\in\mathcal{X}$, interpret local minimization as being over $\mathcal{X}\cap\mathbb{B}_p(\bar{x},r)$, as in Proposition 11. When $\rho=r$, Algorithm 10 is a (ϕ,λ)-ball optimization oracle for $\widehat{f}_\rho$ for $0<\phi\leq\lambda r^p/2$. It makes*

$$O\left(1+\log\frac{\lambda r^p}{\phi}+\left(\frac{\big(C_pL/\rho+(p-1)\lambda r^{p-2}\big)^p}{\lambda^2\phi^{p-2}}\right)^{\frac{1}{p+2}}+\frac{L^p}{\lambda\phi^{p-1}}\right)$$

total queries to g with constant query depth, and has an additional computational cost of

$$O\left(1+\log\frac{\lambda r^p}{\phi}+\left(\frac{\big(C_pL/\rho+(p-1)\lambda r^{p-2}\big)^p}{\lambda^2\phi^{p-2}}\right)^{\frac{1}{p+2}}\right)$$

sequential proximal-update depth and

$$O\left(1+\log\frac{\lambda r^p}{\phi}+\left(\frac{\left(C_pL/\rho+(p-1)\lambda r^{p-2}\right)^p}{\lambda^2\phi^{p-2}}\right)^{\frac{1}{p+2}}+\frac{L^p}{\lambda\phi^{p-1}}\right)$$

vector-arithmetic work.

Proof. We follow the proof of Proposition 4 in Appendix D of [CJJ+23]. Let

$$L_\Psi=\frac{C_pL}{\rho}+(p-1)\lambda r^{p-2},\qquad \Psi(x)=\widehat{f}_\rho(x)+\frac{\lambda}{p}\|x-\bar{x}\|_p^p.$$

The function Ψ is L_Ψ-smooth on $\mathcal{X}\cap\mathbb{B}_p(\bar{x},r)$ and λ-uniformly convex relative to

$$V(x,y)=D_{\|\cdot-\bar{x}\|_p^p/p}(y,x),$$

with $V(x,y)\geq\nu\|y-x\|_p^p/p$ and $\nu=2^{2-p}$. Let

$$x^\star\in\arg\min_{x\in\mathcal{X}\cap\mathbb{B}_p(\bar{x},r)}\Psi(x),\qquad \phi_k=\frac{\lambda r^p}{2^k}.$$

To justify the minibatch moment term below, condition on the past before a minibatch is used, and let x be its current iterate. Since $\rho=r$ and $x\in\mathcal{X}\cap\mathbb{B}_p(\bar{x},r)$, Lemma 1 implies that the N_k independent p-GREG estimates $\widetilde{\nabla}_{\bar{x}}^{g,j}\widehat{f}_\rho(x)$, $j=1,\dots,N_k$, are unbiased and have q-th moments at most $3L^q$. Since $q=p/(p-1)\in(1,2]$, applying the von Bahr–Esseen inequality coordinatewise and summing over coordinates gives

$$\mathbb{E}\left\|\frac{1}{N_k}\sum_{j=1}^{N_k}\left(\widetilde{\nabla}_{\bar{x}}^{g,j}\widehat{f}_{\rho,p}(x)-\nabla\widehat{f}_{\rho,p}(x)\right)\right\|_q^q\leq\frac{C_p}{N_k^q}\sum_{j=1}^{N_k}\mathbb{E}\left\|\widetilde{\nabla}_{\bar{x}}^{g,j}\widehat{f}_{\rho,p}(x)-\nabla\widehat{f}_{\rho,p}(x)\right\|_q^q\leq\frac{C_pL^q}{N_k^{q-1}}.$$

Applying Theorem 16 (Equation (72)) conditionally on the initial point of epoch k, with $\mu=\lambda$ and the parameters of Algorithm 10, gives

$$\mathbb{E}[\Psi(x_{T_k}^{\mathrm{ag}})-\Psi(x^\star)]\leq C_p\left[\frac{L_\Psi^{p-1}\mathbb{E}V(x_0,x^\star)}{\nu M_{f,k}^{p-2}T_k^p}+\frac{M_{f,k}^q+L^q/N_k^{q-1}}{(\nu\lambda)^{q-1}T_k^{q-1}}\right].\tag{49}$$

We prove by induction that

$$\mathbb{E}[\Psi(x_{T_k}^{\mathrm{ag}})-\Psi(x^\star)]\leq\phi_k.$$

For the first epoch, $x_0=\bar{x}$, and hence $V(x_0,x^\star)\leq r^p/p$. For subsequent epochs, first-order optimality at the constrained minimizer $x^\star$ and relative uniform convexity give

$$\Psi(x_0)-\Psi(x^\star)\geq\langle\nabla\Psi(x^\star),x_0-x^\star\rangle+\lambda V(x^\star,x_0)\geq\lambda V(x^\star,x_0).$$

By proof of Lemma 22 and Fact 3, we have, for all $a,b\in\mathbb{R}$,

$$D_{|\cdot|^p/p}(a,b)\leq C_pD_{|\cdot|^p/p}(b,a),$$

where C_p depends only on p. Translating by $\bar{x}$ and summing coordinatewise therefore gives

$$V(x_0,x^\star)=D_{\|\cdot-\bar{x}\|_p^p/p}(x^\star,x_0)\leq C_pD_{\|\cdot-\bar{x}\|_p^p/p}(x_0,x^\star)=C_pV(x^\star,x_0).$$

Consequently,

$$\mathbb{E}V(x_0, x^\star) \le \frac{C_p}{\lambda}\mathbb{E}[\Psi(x_0) - \Psi(x^\star)] \le \frac{C_p\phi_{k-1}}{\lambda}.$$

Substituting the choices of $M_{f,k}, T_k, N_k$ from Algorithm 10 into (49), and choosing the p-dependent constants sufficiently large, yields

$$\mathbb{E}[\Psi(x_{T_k}^{\mathrm{ag}}) - \Psi(x^\star)] \le \frac{\phi_{k-1}}{2} = \phi_k.$$

Since $K = \lceil \log_2(\lambda r^p/\phi) \rceil$, the final output satisfies

$$\mathbb{E}[\Psi(x_{T_K}^{\mathrm{ag}}) - \Psi(x^\star)] \le \frac{\lambda r^p}{2^K} \le \phi.$$

All iterates lie in $\mathcal{X} \cap \mathbb{B}_p(\bar{x}, r)$ by convexity of the feasible set. Finally, the geometric decrease of ϕ_k and $p > 2$ give

$$\sum_{k=1}^{K} T_k = O\left(1 + \log\frac{\lambda r^p}{\phi} + \left(\frac{L_\Psi^p}{\lambda^2\phi^{p-2}}\right)^{\frac{1}{p+2}}\right)$$

and

$$\sum_{k=1}^{K} T_k N_k = O\left(1 + \log\frac{\lambda r^p}{\phi} + \left(\frac{L_\Psi^p}{\lambda^2\phi^{p-2}}\right)^{\frac{1}{p+2}} + \frac{L^p}{\lambda\phi^{p-1}}\right).$$

All original stochastic-gradient queries are issued in parallel, since the anchor $\bar{x}$ and the batch sizes are fixed in advance. This proves the claimed oracle-query and proximal-update bounds. □

4.2 EpochDA Solver

Algorithm 9 ParallelEpochDA$(f, g, \bar{x}, r, \rho, \lambda, \bar{\phi})$

Input: $f : \mathbb{R}^d \to \mathbb{R}$ and $g : \mathbb{R}^d \to \mathbb{R}^d$ satisfying the assumptions of Problem 1, $\bar{x} \in \mathbb{R}^d$, $r = \rho > 0$, and $\lambda, \phi > 0$
$\eta_1 \leftarrow \frac{2^{2-p}}{4^{p-1}\lambda}$, $\quad T_1 \leftarrow \left\lceil \frac{4^p}{(2^{2-p})^2} \right\rceil$, $\quad T \leftarrow \left\lceil \frac{C_p L^p}{\lambda\phi^{p-1}} \right\rceil$ where $C_p \ge \frac{2\cdot 4^p\, 2^{p-1}\, 3^p}{(2^{2-p})^2(2^{p-1}-1)}$
Sample $\xi_i \sim \mathcal{GG}_p(0, \rho^2 I_d)$, $i \in [2T]$, independently
Query $g(\bar{x} + \xi_i)$ for all $i \in [2T]$ in parallel, using independent oracle randomness
$x_1^0 \leftarrow \bar{x}$, $\quad k \leftarrow 1$
while $\sum_{j=1}^{k} T_j \le T$ **do**
  $x_k^1 \leftarrow x_k^0$, $\quad \Phi_k(x) \leftarrow \frac{1}{p}\|x - x_k^0\|_p^p$
  for $t \in [T_k]$ **do**
    $i \leftarrow \sum_{j=1}^{k-1} T_j + t$
    $\widetilde{\nabla}_{\bar{x}}^{g} \widehat{f}_\rho(x_k^t) \leftarrow \frac{\gamma_{\rho,p}(x_k^t - \bar{x} - \xi_i)}{\gamma_{\rho,p}(\xi_i)} g(\bar{x} + \xi_i)$
    $x_k^{t+1} \leftarrow \operatorname{argmin}_{x \in \mathbb{B}_p(\bar{x}, r)} \left\{ \eta_k \left\langle \widetilde{\nabla}_{\bar{x}}^{g} \widehat{f}_\rho(x_k^t), x \right\rangle + \frac{\eta_k \lambda}{p}\|x - \bar{x}\|_p^p + D_{\Phi_k}(x, x_k^t) \right\}$
  end for
  $x_{k+1}^0 \leftarrow \frac{1}{T_k}\sum_{t=1}^{T_k} x_k^{t+1}$, $\quad T_{k+1} \leftarrow \lceil 2^{p-1} T_k \rceil$, $\quad \eta_{k+1} \leftarrow \frac{\eta_k}{2^{p-1}}$, $\quad k \leftarrow k+1$
end while
return x_k^0

Throughout, let f, g satisfy Problem 1, and let

$$x^\star = \operatorname*{argmin}_{x \in \mathbb{B}_p(\bar{x}, r)} \left\{ \widehat{f}_\rho(x) + \frac{\lambda}{p}\|x - \bar{x}\|_p^p \right\}.$$

Let $C_{\mathrm{res},p} \geq 1$ be the constant in the conditional q-moment bound for $\rho = r$.

Lemma 24. *In view of Problem 1, Each epoch k of Algorithm 9 satisfies*

$$\mathbb{E}\left[\widehat{f}_\rho(x_{k+1}^0) + \frac{\lambda}{p}\|x_{k+1}^0 - \bar{x}\|_p^p\right] - \widehat{f}_\rho(x^\star) - \frac{\lambda}{p}\|x^\star - \bar{x}\|_p^p \leq \frac{\mathbb{E}\|x^\star - x_k^0\|_p^p}{p\eta_k T_k} + \frac{(4L)^q}{q}\left(\frac{\eta_k}{2^{2-p}}\right)^{q-1}. \quad (50)$$

Proof. Problem 1 gives $\|\mathbb{E}g(x)\|_q \leq \mathbb{E}\|g(x)\|_q \leq L$; hence f and $\widehat{f}_\rho$ are convex and L-Lipschitz. Let $\mathcal{F}_{k,t-1}$ be the history of the samples already used, including x_k^t. By Lemma 1,

$$\mathbb{E}\left[\widetilde{\nabla}_{\bar{x}}^g \widehat{f}_\rho(x_k^t) \mid \mathcal{F}_{k,t-1}\right] = \nabla \widehat{f}_\rho(x_k^t) \quad , \quad \mathbb{E}\left[\left\|\widetilde{\nabla}_{\bar{x}}^g \widehat{f}_\rho(x_k^t)\right\|_q^q \mid \mathcal{F}_{k,t-1}\right] \leq 3L^q \quad (51)$$

The optimality condition of the update, convexity of the regularizer, and the Bregman three-point identity imply

$$\begin{aligned}
&\left\langle \widetilde{\nabla}_{\bar{x}}^g \widehat{f}_\rho(x_k^t), x_k^{t+1} - x^\star \right\rangle + \frac{\lambda}{p}\|x_k^{t+1} - \bar{x}\|_p^p - \frac{\lambda}{p}\|x^\star - \bar{x}\|_p^p \\
&\quad \leq \frac{1}{\eta_k}\left[D_{\Phi_k}(x^\star, x_k^t) - D_{\Phi_k}(x^\star, x_k^{t+1}) - D_{\Phi_k}(x_k^{t+1}, x_k^t)\right] \quad (52)
\end{aligned}$$

By Lipschitz continuity, convexity, and conditional unbiasedness,

$$\begin{aligned}
\mathbb{E}[\widehat{f}_\rho(x_k^{t+1}) - \widehat{f}_\rho(x^\star)] &\leq \mathbb{E}\left[\left\langle \widetilde{\nabla}_{\bar{x}}^g \widehat{f}_\rho(x_k^t), x_k^t - x^\star \right\rangle + L\|x_k^{t+1} - x_k^t\|_p\right] \\
&\leq \mathbb{E}\left\langle \widetilde{\nabla}_{\bar{x}}^g \widehat{f}_\rho(x_k^t), x_k^{t+1} - x^\star \right\rangle + \mathbb{E}\left[\left(\left\|\widetilde{\nabla}_{\bar{x}}^g \widehat{f}_\rho(x_k^t)\right\|_q + L\right)\|x_k^{t+1} - x_k^t\|_p\right]
\end{aligned}$$

Since $D_{\Phi_k}(u, v) \geq 2^{2-p}\|u - v\|_p^p/p$, Young's inequality gives

$$\left(\left\|\widetilde{\nabla}_{\bar{x}}^g \widehat{f}_\rho(x_k^t)\right\|_q + L\right)\|x_k^{t+1} - x_k^t\|_p - \frac{D_{\Phi_k}(x_k^{t+1}, x_k^t)}{\eta_k} \leq \frac{1}{q}\left(\frac{\eta_k}{2^{2-p}}\right)^{q-1}\left(\left\|\widetilde{\nabla}_{\bar{x}}^g \widehat{f}_\rho(x_k^t)\right\|_q + L\right)^q$$

Moreover, conditional Minkowski's inequality and (51) yield

$$\mathbb{E}\left[\left(\left\|\widetilde{\nabla}_{\bar{x}}^g \widehat{f}_\rho(x_k^t)\right\|_q + L\right)^q \mid \mathcal{F}_{k,t-1}\right] \leq (4L)^q.$$

Combining these estimates with (52) gives

$$\begin{aligned}
&\mathbb{E}\left[\widehat{f}_\rho(x_k^{t+1}) + \frac{\lambda}{p}\|x_k^{t+1} - \bar{x}\|_p^p\right] - \widehat{f}_\rho(x^\star) - \frac{\lambda}{p}\|x^\star - \bar{x}\|_p^p \\
&\quad \leq \frac{\mathbb{E}D_{\Phi_k}(x^\star, x_k^t) - \mathbb{E}D_{\Phi_k}(x^\star, x_k^{t+1})}{\eta_k} + \frac{(4L)^q}{q}\left(\frac{\eta_k}{2^{2-p}}\right)^{q-1}
\end{aligned}$$

Summing over $t = 1, \ldots, T_k$ telescopes the Bregman terms because Φ_k is fixed within the epoch. As $x_k^1 = x_k^0$,

$$D_{\Phi_k}(x^\star, x_k^1) = \frac{1}{p}\|x^\star - x_k^0\|_p^p.$$

Dropping the nonnegative final divergence, dividing by T_k, and using convexity at $x_{k+1}^0 = T_k^{-1}\sum_{t=1}^{T_k} x_k^{t+1}$ proves (50). □

Lemma 25. *After each completed epoch k of Algorithm 9,*

$$\mathbb{E}\left[\widehat{f}_\rho(x_{k+1}^0) + \frac{\lambda}{p}\|x_{k+1}^0 - \bar{x}\|_p^p\right] - \widehat{f}_\rho(x^\star) - \frac{\lambda}{p}\|x^\star - \bar{x}\|_p^p \le \frac{(4L)^q}{\lambda^{q-1}2^k}. \tag{53}$$

Proof. Convexity of $\widehat{f}_\rho$, p-uniform convexity of the regularizer, and constrained optimality of $x^\star$ imply, for every $w \in \mathbb{B}_p(\bar{x}, r)$,

$$\widehat{f}_\rho(w) + \frac{\lambda}{p}\|w - \bar{x}\|_p^p - \widehat{f}_\rho(x^\star) - \frac{\lambda}{p}\|x^\star - \bar{x}\|_p^p \ge \frac{2^{2-p}\lambda}{p}\|w - x^\star\|_p^p. \tag{54}$$

Since $x_1^0 = \bar{x}$ and $\widehat{f}_\rho$ is L-Lipschitz,

$$\widehat{f}_\rho(\bar{x}) - \widehat{f}_\rho(x^\star) - \frac{\lambda}{p}\|x^\star - \bar{x}\|_p^p \le \sup_{a\ge 0}\left\{La - \frac{\lambda}{p}a^p\right\} = \frac{L^q}{q\lambda^{q-1}} \le \frac{(4L)^q}{\lambda^{q-1}}.$$

The parameter choices in Algorithm 9 give

$$\eta_k = \frac{2^{2-p}}{4^{p-1}\lambda\, 2^{(p-1)(k-1)}}, \qquad T_k \ge \frac{4^p}{(2^{2-p})^2}\, 2^{(p-1)(k-1)},$$

and hence $\eta_k T_k \ge 4/(2^{2-p}\lambda)$. Suppose by induction that the expected objective error at the start of epoch k is at most $(4L)^q/(\lambda^{q-1}2^{k-1})$, the base case $k = 1$ is verified above. By (54),

$$\frac{\mathbb{E}\|x^\star - x_k^0\|_p^p}{p\eta_k T_k} \le \frac{1}{4}\frac{(4L)^q}{\lambda^{q-1}2^{k-1}}.$$

Using $(p-1)(q-1) = 1$, the other term in Lemma 24 is

$$\frac{(4L)^q}{q}\left(\frac{\eta_k}{2^{2-p}}\right)^{q-1} = \frac{(4L)^q}{4q\lambda^{q-1}2^{k-1}}$$

Applying Lemma 24 therefore gives

$$\mathbb{E}\left[\widehat{f}_\rho(x_{k+1}^0) + \frac{\lambda}{p}\|x_{k+1}^0 - \bar{x}\|_p^p\right] - \widehat{f}_\rho(x^\star) - \frac{\lambda}{p}\|x^\star - \bar{x}\|_p^p \le \left(\frac{1}{4} + \frac{1}{4q}\right)\frac{(4L)^q}{\lambda^{q-1}2^{k-1}} \le \frac{(4L)^q}{\lambda^{q-1}2^k}.$$

This is the required bound for epoch k and the induction hypothesis for the start of epoch $k+1$. □

Proposition 13 (Proposition 3, [CJJ+23])**.** *Let f, g satisfy the assumptions of Problem 1. When $\rho = r$, Algorithm 9 is a (ϕ, λ)-ball optimization oracle for $\widehat{f}_\rho$ which makes $O\left(\frac{L^p}{\lambda\phi^{p-1}}\right)$ total queries to g with constant query depth, and an additional computational cost of $O\left(\frac{L^p}{\lambda\phi^{p-1}}\right)$ depth and work.*

Proof. Let k be the epoch index when Algorithm 9 terminates, so that the returned point is x_k^0. The recurrence $T_{j+1} = \lceil 2^{p-1}T_j \rceil$ implies

$$T_j \le 2^{(p-1)(j-1)}\left(T_1 + \frac{1}{2^{p-1}-1}\right) \le \frac{2\cdot 4^p}{(2^{2-p})^2}\, 2^{(p-1)(j-1)}.$$

By the stopping condition,

$$T < \sum_{j=1}^{k} T_j \le \frac{2\cdot 4^p}{(2^{2-p})^2}\frac{2^{(p-1)k} - 1}{2^{p-1}-1} \le \frac{2\cdot 4^p\, 2^{p-1}}{(2^{2-p})^2(2^{p-1}-1)}\, 2^{(p-1)(k-1)}.$$

The initial-error bound and Lemma 25 therefore give

$$\mathbb{E}\left[\widehat{f}_\rho(x_k^0)+\frac{\lambda}{p}\|x_k^0-\bar{x}\|_p^p\right]-\min_{x\in\mathbb{B}_p(\bar{x},r)}\left\{\widehat{f}_\rho(x)+\frac{\lambda}{p}\|x-\bar{x}\|_p^p\right\}$$
$$\le\frac{(4L)^q}{\lambda^{q-1}2^{k-1}}\le\left[\frac{2\cdot 4^p\,2^{p-1}\,3^p}{(2^{2-p})^2(2^{p-1}-1)}\frac{L^p}{\lambda T}\right]^{1/(p-1)}\le\phi,$$

where the last inequality follows from $T=\lceil C_pL^p/(\lambda\phi^{p-1})\rceil$. The output lies in $\mathbb{B}_p(\bar{x},r)$ by construction and convexity, so the preceding error bound establishes the required guarantee. All $2T=O(1+L^p/(\lambda\phi^{p-1}))$ queries can be made in one parallel round. The algorithm performs at most T Bregman proximal updates and $O(T)$ vector operations, giving $O(T)$ additional computational depth and work if each update and vector operation has constant depth and work. □

Lemma 26 (Lemma 2.3, [CGJS23])**.** *Let f,g satisfy the assumptions of Problem 1, with $L>0$. For any $c\in\mathbb{R}^d$ and $R>0$, the function*

$$x\mapsto f(x)+2L(\|x-c\|_p-R)_+$$

is convex and $3L$-Lipschitz with respect to $\|\cdot\|_p$. For any y with $\|y-c\|_p\ge R$, define

$$y^{c,R}:=c+\frac{R}{\|y-c\|_p}(y-c).$$

Then $\|y^{c,R}-c\|_p=R$ and

$$f(y^{c,R})\le f(y)+2L(\|y-c\|_p-R)_+-L(\|y-c\|_p-R).$$

The penalized function attains a global minimum, and every global minimizer $x^\star$ satisfies

$$\|x^\star-c\|_p\le R,\qquad f(x^\star)+2L(\|x^\star-c\|_p-R)_+=f(x^\star)=\min_{x\in\mathbb{B}_p(c,R)}f(x).$$

Moreover, adding a deterministic subgradient of $2L(\|x-c\|_p-R)_+$ to $g(x)$ gives a stochastic subgradient oracle for the penalized function with q-th moment at most $(4L)^q$, using one query to g.

Proof. The proof follows the argument of Lemma 2.3 in [CGJS23]. We first show that $f(x)+2L(\|x-c\|_p-R)_+$ is convex and $3L$-Lipschitz. The assumptions of Problem 1 give

$$\|\mathbb{E}g(x)\|_q\le\left(\mathbb{E}\|g(x)\|_q^q\right)^{1/q}\le L.$$

Since $\mathbb{E}g(x)\in\partial f(x)$, the subgradient inequality implies that f is L-Lipschitz with respect to $\|\cdot\|_p$. The function $x\mapsto 2L(\|x-c\|_p-R)_+$ is convex and $2L$-Lipschitz. Adding f therefore gives a convex and $3L$-Lipschitz function. Next, consider $y\in\mathbb{R}^d$ with $\|y-c\|_p\ge R$. By the definition of $y^{c,R}$,

$$\|y^{c,R}-c\|_p=R,\qquad\|y^{c,R}-y\|_p=\|y-c\|_p-R.$$

In particular, $2L(\|y^{c,R}-c\|_p-R)_+=0$. Moreover,

$$\begin{aligned}f(y)+2L(\|y-c\|_p-R)_+&=f(y)+2L(\|y-c\|_p-R)\\&\ge f(y^{c,R})-L\|y^{c,R}-y\|_p+2L(\|y-c\|_p-R)\\&=f(y^{c,R})+L(\|y-c\|_p-R),\end{aligned}$$

where the inequality follows from the L-Lipschitz continuity of f. This yields the desired result, as it implies that the value of $f(x) + 2L(\|x-c\|_p - R)_+$ at any point outside the ball is larger than its value at some point satisfying $\|x-c\|_p \le R$, where $f(x) + 2L(\|x-c\|_p - R)_+ = f(x)$. Further, since the ball is compact and f is continuous, the minimum of f over the ball is attained. For the stochastic oracle, define

$$v(x) := \begin{cases} 0, & \|x-c\|_p \le R, \\ \dfrac{(|x_i - c_i|^{p-2}(x_i - c_i))_{i=1}^d}{\|x-c\|_p^{p-1}}, & \|x-c\|_p > R. \end{cases}$$

Then $v(x) \in \partial[(\|\cdot - c\|_p - R)_+](x)$ and $\|v(x)\|_q \le 1$, so

$$\mathbb{E}[g(x) + 2Lv(x)] \in \partial\big[f + 2L(\|\cdot - c\|_p - R)_+\big](x).$$

By Minkowski's inequality,

$$\big(\mathbb{E}\|g(x) + 2Lv(x)\|_q^q\big)^{1/q} \le \big(\mathbb{E}\|g(x)\|_q^q\big)^{1/q} + 2L\|v(x)\|_q \le 3L.$$

Computing $g(x) + 2Lv(x)$ requires only one query to g. □

Theorem 14 (Parallel EpochDA-based solver)**.** *BallAccel (Proposition 11) using parallel EpochDA (Algorithm 9) as a ball optimization oracle solves Problem 1 with expected error ϵ_{opt}, with*

$$O\Big(d^{\frac{p}{2(p+1)}}\kappa^{\frac{p}{p+1}}\log^4(d\kappa)\Big) \text{ query depth}$$

and

$$O\left(d^{\frac{p}{2(p+1)}}\kappa^{\frac{p}{p+1}}\log^4(d\kappa) + d^{\frac{p(p-2)}{2(p+1)}}\kappa^{\frac{p(2p-1)}{p+1}}\log^{2p(p-1)+2}(d\kappa)\right) \text{ total queries,}$$

and an additional computational cost of

$$O\left(d^{\frac{p}{2(p+1)}}\kappa^{\frac{p}{p+1}}\log^4(d\kappa) + d^{\frac{p(p-2)}{2(p+1)}}\kappa^{\frac{p(2p-1)}{p+1}}\log^{2p(p-1)+2}(d\kappa)\right) \text{ depth}$$

and

$$O\left(\left(d^{\frac{p}{2(p+1)}}\kappa^{\frac{p}{p+1}}\log^4(d\kappa) + d^{\frac{p(p-2)}{2(p+1)}}\kappa^{\frac{p(2p-1)}{p+1}}\log^{2p(p-1)+2}(d\kappa)\right)\cdot d\right) \text{ work.}$$

Proof. Let $x^\star \in \operatorname{argmin}_{x\in\mathbb{B}_p(R)} f(x)$, $\widehat{x}$ be the point returned by BallAccel and define

$$f_{\text{pen}}(x) = f(x) + 2L(\|x\|_p - R)_+.$$

We invoke Lemma 26 and apply Proposition 11 to $F = \widehat{f_{\text{pen}}}_\rho$, with

$$r = \rho = \frac{\epsilon_{\text{opt}}}{C_p L\sqrt{d}},$$

where C_p is a sufficiently large constant depending only on p. Since f_{pen} is $3L$-Lipschitz and $\mathbb{E}\|\xi_\rho\|_p \le \rho\sqrt{d}$, this choice gives

$$|F(x) - f_{\text{pen}}(x)| \le 3L\mathbb{E}\|\xi_\rho\|_p \le \frac{\epsilon_{\text{opt}}}{4}$$

for every $x \in \mathbb{R}^d$. We now show that F has a minimizer in $\mathbb{B}_p(3R)$. To see this, let $y \in \mathbb{R}^d$ satisfy $\|y\|_p > 3R$, and set $\widetilde{y} = \frac{R}{\|y\|_p} y$. By Lemma 26,

$$\begin{aligned} F(y) &\geq f_{\mathrm{pen}}(y) - \frac{\epsilon_{\mathrm{opt}}}{4} \geq f_{\mathrm{pen}}(\widetilde{y}) + L(\|y\|_p - R) - \frac{\epsilon_{\mathrm{opt}}}{4} \\ &\geq F(\widetilde{y}) + L(\|y\|_p - R) - \frac{\epsilon_{\mathrm{opt}}}{2} > F(\widetilde{y}), \end{aligned}$$

where the last inequality uses $\|y\|_p > 3R$ and $\epsilon_{\mathrm{opt}} \leq LR$. Since F is continuous, it attains its minimum over $\mathbb{B}_p(3R)$, and the inequality shows that this is also a global minimum. Thus Proposition 11 applies to F with radius $3R$, Lipschitz parameter $3L$, and target accuracy $\epsilon_{\mathrm{opt}}/2$. By Lemma 26, a stochastic subgradient oracle for f_{pen} is obtained from one query to g, and its dual q-th moment is at most $(4L)^q$. The estimator for F therefore has conditional q-th moment at most $(6L)^q$ on each local ball. The ball optimization calls required by Proposition 11 are for

$$x \mapsto F(x) + 3L(\|x\|_p - 3R)_+.$$

For these calls, we add $3Lv(x)$ to the estimator at the current iterate, where

$$v(x) \in \partial[(\|\cdot\|_p - 3R)_+](x), \qquad \|v(x)\|_q \leq 1,$$

is chosen deterministically. The resulting direction has conditional expectation in the subdifferential of this function and conditional q-th moment at most $(9L)^q$. The function is convex and $6L$-Lipschitz. Since $9L \leq 2(6L)$, the proofs of Lemmas 24 and 25 apply with L replaced by $6L$. Rescaling the Lipschitz parameter, radius, and target accuracy changes the corresponding BallAccel parameters only by constant factors depending on p. Hence, we retain the notation

$$\kappa = \frac{LR}{\epsilon_{\mathrm{opt}}}, \qquad K = \left(\frac{R}{r}\right)^{\frac{p}{p+1}} = \Theta\left(d^{\frac{p}{2(p+1)}} \kappa^{\frac{p}{p+1}}\right), \qquad \lambda^\star = \frac{\epsilon_{\mathrm{opt}} K^p}{R^p} \log^p \kappa$$

in the following complexity bounds, with constant-factor changes absorbed into the $O(\cdot)$ notation. For the query depth, each ball optimization oracle implemented by the above version of Algorithm 9 has constant query depth. Proposition 11 requires at most $O(K \log\kappa \log(R\kappa/r))$ iterations, with at most $O(\log^2(R\kappa/r))$ ball optimization oracle calls per iteration. Since $\log(R\kappa/r) = O(\log(d\kappa))$, the query depth is

$$O\left(K \log\kappa \log^3\left(\frac{R\kappa}{r}\right)\right) = O\left(d^{\frac{p}{2(p+1)}} \kappa^{\frac{p}{p+1}} \log^4(d\kappa)\right).$$

For the total oracle queries, we use Proposition 13 and the extension above, which bound the number of gradient queries for a (ϕ, λ)-BROO by $O\left(1 + \frac{L^p}{\lambda \phi^{p-1}}\right)$. The effective parameter $6L$ changes this bound only by a factor depending on p. We may upper bound every query cost using $\lambda = \Omega_p(\lambda^\star)$. Writing $\ell := \log(R\kappa/r)$ and $j_{\max} := \lceil \log_2 K + C_{\mathrm{ba}} \rceil$, and applying both requirements of Proposition 11, the total number of stochastic-gradient queries is at most

$$\begin{aligned} &O\left(K \log\kappa\, \ell^3 + \frac{K \log\kappa\, \ell^3 L^p}{(\lambda^\star)^p r^{p(p-1)}} + K \log\kappa\, \ell \sum_{j=1}^{j_{\max}} 2^{-j} \frac{L^p (2^j \ell^{2p})^{p-1}}{(\lambda^\star)^p r^{p(p-1)}}\right) \\ &\quad = O\left(K \log\kappa\, \ell^3 + \frac{K \log\kappa\, \ell^3 L^p}{(\lambda^\star)^p r^{p(p-1)}} + \frac{K \log\kappa\, \ell^{2p(p-1)+1} L^p}{(\lambda^\star)^p r^{p(p-1)}} \sum_{j=1}^{j_{\max}} 2^{j(p-2)}\right). \end{aligned}$$

For $p \geq 2$, the sum contains $O(\ell)$ terms, each bounded by $O(K^{p-2})$, and hence

$$\sum_{j=1}^{j_{\max}} 2^{j(p-2)} = O(K^{p-2}\ell).$$

Moreover, by $K^{p+1}r^p = R^p$,

$$\frac{K L^p}{(\lambda^\star)^p r^{p(p-1)}} = \frac{\kappa^p}{\log^{p^2}\kappa}.$$

Therefore, the total number of queries is bounded by

$$\begin{aligned}
&O\left(K\log\kappa\,\ell^3 + \frac{\kappa^p \log\kappa\,\ell^3}{\log^{p^2}\kappa} + \frac{\kappa^p K^{p-2}\log\kappa\,\ell^{2p(p-1)+2}}{\log^{p^2}\kappa}\right) \\
&\quad = O\left(K\log^4(d\kappa) + \kappa^p K^{p-2}\log^{2p(p-1)+2}(d\kappa)\right) \\
&\quad = O\left(d^{\frac{p}{2(p+1)}}\kappa^{\frac{p}{p+1}}\log^4(d\kappa) + d^{\frac{p(p-2)}{2(p+1)}}\kappa^{\frac{p(2p-1)}{p+1}}\log^{2p(p-1)+2}(d\kappa)\right),
\end{aligned}$$

where we used $\log\kappa/\log^{p^2}\kappa \leq 1$ and $\kappa^p K^{p-2} = \Theta\left(d^{\frac{p(p-2)}{2(p+1)}}\kappa^{\frac{p(2p-1)}{p+1}}\right)$. Finally, Proposition 11 gives

$$\mathbb{E}F(\widehat{x}) \leq \min_{x\in\mathbb{R}^d} F(x) + \frac{\epsilon_{\text{opt}}}{2} \leq F(x^\star) + \frac{\epsilon_{\text{opt}}}{2}.$$

Since $f(x) \leq f_{\text{pen}}(x)$ and f_{pen} is $3L$-Lipschitz,

$$f(x) \leq F(x) + 3L\mathbb{E}\|\xi_\rho\|_p$$

for all $x \in \mathbb{R}^d$. Therefore,

$$\begin{aligned}
\mathbb{E}f(\widehat{x}) &\leq \mathbb{E}F(\widehat{x}) + 3L\mathbb{E}\|\xi_\rho\|_p \\
&\leq F(x^\star) + \frac{\epsilon_{\text{opt}}}{2} + 3L\mathbb{E}\|\xi_\rho\|_p \\
&\leq f(x^\star) + \frac{\epsilon_{\text{opt}}}{2} + 6L\mathbb{E}\|\xi_\rho\|_p \\
&\leq f(x^\star) + \epsilon_{\text{opt}}.
\end{aligned}$$

□

4.3 AC-SA Solver

Algorithm 10 ℓ_p-AC-SA$(f, \bar{x}, r, \rho, \lambda, \bar{\phi})$

Require: $f : \mathbb{R}^d \to \mathbb{R}$ and $g : \mathbb{R}^d \to \mathbb{R}^d$ satisfying the assumptions of Problem 1, $\bar{x} \in \mathcal{X}$, $r, \rho, \lambda > 0$, $0 < \bar{\phi} \leq \lambda r^p/2$, $V(x, y) = D_{\|\cdot - \bar{x}\|_p^p/p}(y, x)$ and $\nu = 2^{2-p}$.

$K \leftarrow \lceil \log_2(\lambda r^p/\bar{\phi}) \rceil$.

For $k \in [K]$, set

$$M_{f,k} \leftarrow (\left(C_p L/\rho + (p-1)\lambda r^{p-2}\right) \nu\lambda \left(\lambda r^p/2^{k-1}\right)^p)^{1/(p+2)},$$

$$T_k \leftarrow \left\lceil C_p \left[1 + \left(\frac{\left(C_p L/\rho + (p-1)\lambda r^{p-2}\right)^p}{(\nu\lambda)^2 \left(\lambda r^p/2^{k-1}\right)^{p-2}} \right)^{1/(p+2)} \right] \right\rceil,$$

$$N_k \leftarrow \left\lceil C_p \max\left\{ 1, \frac{L^p}{\nu\lambda T_k \left(\lambda r^p/2^{k-1}\right)^{p-1}} \right\} \right\rceil.$$

Sample $\xi_i \sim \mathcal{GG}_p(0, \rho^2 I_d)$, $i \in [N]$ independently, for $N = \sum_{k\in[K]} T_k N_k$.

Query $g(\bar{x} + \xi_i)$ for all $i \in [N]$ in parallel

$x_0^{\mathrm{ag}} \leftarrow \bar{x}$, $x_0 \leftarrow \bar{x}$

for $k \in [K]$ **do**

  for $t \in [T_k]$ **do**

    $\alpha_t \leftarrow p/(t+p-1)$, $\Gamma_t \leftarrow \Gamma(p+1)\Gamma(t)/\Gamma(t+p)$.

    $\gamma_{t,k} \leftarrow \dfrac{2\left(\sup_{s\geq 1} \alpha_s^p/\Gamma_s\right)\left(C_p L/\rho + (p-1)\lambda r^{p-2}\right)^{p-1}}{\nu M_{f,k}^{p-2}} \Gamma_t$.

    $x_t^{\mathrm{md}} \leftarrow \frac{(1-\alpha_t)(\lambda+\gamma_{t,k})}{\gamma_{t,k}+(1-\alpha_t^2)\lambda} x_{t-1}^{\mathrm{ag}} + \frac{\alpha_t[(1-\alpha_t)\lambda+\gamma_{t,k}]}{\gamma_{t,k}+(1-\alpha_t^2)\lambda} x_{t-1}$.

    $N_{T,[k-1]} \leftarrow \sum_{k'\in[k-1]} T_{k'} N_{k'}$.

    $$\widehat{\nabla} f(x_t^{\mathrm{md}}) \leftarrow \frac{1}{N_k} \sum_{n\in[N_k]} \frac{\gamma_{\rho,p}(x_t^{\mathrm{md}} - \bar{x} - \xi_{N_{T,[k-1]}+(t-1)N_k+n})}{\gamma_{\rho,p}(\xi_{N_{T,[k-1]}+(t-1)N_k+n})} \cdot g(\bar{x} + \xi_{N_{T,[k-1]}+(t-1)N_k+n}).$$

    $$x_t \leftarrow \arg\min_{x\in\mathcal{X}\cap B_p(\bar{x},r)} \Big\{ \langle \alpha_t[\widehat{\nabla} f(x_t^{\mathrm{md}}) + \lambda J_p(x_t^{\mathrm{md}} - \bar{x})], x\rangle + \alpha_t\lambda V(x_t^{\mathrm{md}}, x) + [(1-\alpha_t)\lambda + \gamma_{t,k}] V(x_{t-1}, x) \Big\}.$$

    $x_t^{\mathrm{ag}} \leftarrow \alpha_t x_t + (1-\alpha_t) x_{t-1}^{\mathrm{ag}}$.

  end for

  $x_0^{\mathrm{ag}} \leftarrow x_{T_k}^{\mathrm{ag}}$, $x_0 \leftarrow x_{T_k}^{\mathrm{ag}}$

end for

return $x_{T_K}^{\mathrm{ag}}$

Definition 9. *Consider the class of stochastic convex optimization (SCO) problems*

$$\Psi^* := \min_{x\in X} \{\Psi(x) := f(x) + \mathcal{X}(x)\}, \tag{55}$$

where $X \subseteq \mathbb{R}^d$ is a nonempty closed convex set, $\mathcal{X} : X \to \mathbb{R}$ is a simple convex function, and $f : X \to \mathbb{R}$ is convex. Suppose $\omega : X \to \mathbb{R}$ is differentiable and its Bregman divergence

$$V(x, y) := \omega(y) - \omega(x) - \langle \nabla\omega(x), y - x\rangle$$

satisfies $V(x, y) \geq \nu\|y - x\|_p^p/p$ for some $\nu > 0$. Assume that $\lambda \geq 0$, $\mu \geq 0$, $L \geq 0$, and $M_f > 0$, and that, for every $x, y \in X$ and $f'(x) \in \partial f(x)$,

$$\mu V(x, y) \leq f(y) - f(x) - \langle f'(x), y - x\rangle \leq \frac{L}{2}\|y - x\|_p^2 + M_f\|y - x\|_p. \tag{56}$$

Define the lower model

$$l_\Psi(z, x) := f(z) + \langle f'(z), x - z\rangle + \mu V(z, x) + \mathcal{X}(x). \tag{57}$$

Lemma 27. *Let the convex function $p : X \to \mathbb{R}$, the points $\tilde{x}, \tilde{y} \in X$ and the scalars $\mu_1, \mu_2 \geq 0$ be given. Let $\omega : X \to \mathbb{R}$ be a differentiable convex function and $V(x, z)$ be defined in Definition 9 (Bregman divergence). If*

$$u^* \in \operatorname{Argmin}\left\{p(u) + \mu_1 V(\tilde{x}, u) + \mu_2 V(\tilde{y}, u) : u \in X\right\},$$

then for any $u \in X$, we have

$$p(u^*) + \mu_1 V(\tilde{x}, u^*) + \mu_2 V(\tilde{y}, u^*) \leq p(u) + \mu_1 V(\tilde{x}, u) + \mu_2 V(\tilde{y}, u) - (\mu_1 + \mu_2)V(u^*, u).$$

Proof. The proof is the same as in [GL12], it holds for any differentiable convex function ω. □

Lemma 28. *Let $x_t^{\text{ag}} := (1 - \alpha_t)x_{t-1}^{\text{ag}} + \alpha_t x_t$ for some $\alpha_t \in [0, 1]$ and $(x_{t-1}^{\text{ag}}, x_t) \in X \times X$. Suppose that f satisfies the upper-model bound in (56). Then, for any $z \in X$ and any $f'(z) \in \partial f(z)$, we have*

$$\Psi(x_t^{\text{ag}}) \leq (1 - \alpha_t)\Psi(x_{t-1}^{\text{ag}}) + \alpha_t\left[f(z) + \langle f'(z), x_t - z\rangle + \mathcal{X}(x_t)\right] + \frac{L}{2}\|x_t^{\text{ag}} - z\|_p^2 + M_f\|x_t^{\text{ag}} - z\|_p.$$

Proof. The proof is the same as in [GL12]. □

Let $\mathcal{F}_{t-1} := \sigma(\xi_1, \ldots, \xi_{t-1})$ denote the oracle history, with $\mathcal{F}_0$ trivial. Assume that x_t^{md} is $\mathcal{F}_{t-1}$-measurable and that the stochastic oracle has a selected subgradient $g(x) \in \partial f(x)$ satisfying

$$\mathbb{E}[G(x_t^{md}, \xi_t) \mid \mathcal{F}_{t-1}] = g(x_t^{md}), \qquad \mathbb{E}[\|G(x_t^{md}, \xi_t) - g(x_t^{md})\|_q^q \mid \mathcal{F}_{t-1}] \leq \sigma^q. \tag{58}$$

Throughout the proof use $f'(x_t^{md}) = g(x_t^{md})$ and write

$$G_t := G(x_t^{md}, \xi_t), \qquad \delta_t := G_t - g(x_t^{md}). \tag{59}$$

The following proposition establishes a basic recursion for the generic AC-SA algorithm.

Proposition 15. *Let $(x_{t-1}, x_{t-1}^{ag}) \in X \times X$ be given. Also let $(x_t^{md}, x_t, x_t^{ag}) \in X \times X \times X$ be computed according to (3.1), (3.2) and (3.3) in [GL12]. If $K_t := \nu(\mu + \gamma_t) - (L^{p-1}/M_f^{p-2})\alpha_t^p > 0$, then for any $x \in X$, we have*

$$\begin{aligned}\Psi(x_t^{ag}) + \mu V(x_t, x) \leq (1 - \alpha_t)\left[\Psi(x_{t-1}^{ag}) + \mu V(x_{t-1}, x)\right] + \alpha_t l_\Psi(x_t^{md}, x) \\ + \gamma_t\left[V(x_{t-1}, x) - V(x_t, x)\right] + \Delta_t(x),\end{aligned} \tag{60}$$

where $l_\Psi(z, x)$ is defined in (57),

$$\Delta_t(x) := \alpha_t\langle\delta_t, x - x_{t-1}^+\rangle + \frac{\alpha_t^q(2M_f + \|\delta_t\|_q)^q}{qK_t^{q-1}}, \tag{61}$$

$$x_{t-1}^+ := \frac{\alpha_t\mu}{\mu + \gamma_t}x_t^{md} + \frac{(1 - \alpha_t)\mu + \gamma_t}{\mu + \gamma_t}x_{t-1}. \tag{62}$$

Proof. We first establish some basic relations among the search points x_t^{ag}, x_t^{md} and x_t. Denote $d_t := x_t^{ag} - x_t^{md}$. It follows from (3.1) and (3.3) in [GL12] that

$$\begin{aligned} d_t &= \alpha_t x_t + (1-\alpha_t)x_{t-1}^{ag} - x_t^{md} \\ &= \alpha_t \left(x_t - \frac{\alpha_t \mu}{\mu+\gamma_t} x_t^{md} - \frac{(1-\alpha_t)\mu + \gamma_t}{\mu+\gamma_t} x_{t-1} \right) = \alpha_t (x_t - x_{t-1}^+), \end{aligned} \tag{63}$$

which, in view of the convexity of $\|\cdot\|_p^p$ and the uniform convexity of ω, implies that

$$\begin{aligned} \frac{\nu(\mu+\gamma_t)}{p\alpha_t^p}\|d_t\|_p^p &\le \frac{\nu(\mu+\gamma_t)}{p(\mu+\gamma_t)}\left[\alpha_t\mu\|x_t - x_t^{md}\|_p^p + ((1-\alpha_t)\mu+\gamma_t)\|x_t - x_{t-1}\|_p^p\right] \\ &\le \alpha_t \mu V(x_t^{md}, x_t) + ((1-\alpha_t)\mu+\gamma_t)V(x_{t-1}, x_t). \end{aligned} \tag{64}$$

Using the above result and Lemma 28 (with $z = x_t^{md}$), we have

$$\begin{aligned} \Psi(x_t^{ag}) &\le (1-\alpha_t)\Psi(x_{t-1}^{ag}) + \alpha_t\left[f(x_t^{md}) + \langle f'(x_t^{md}), x_t - x_t^{md}\rangle + \mathcal{X}(x_t)\right] + \frac{L}{2}\|d_t\|_p^2 + M_f\|d_t\|_p \\ &= (1-\alpha_t)\Psi(x_{t-1}^{ag}) + \alpha_t\left[f(x_t^{md}) + \langle f'(x_t^{md}), x_t - x_t^{md}\rangle + \mathcal{X}(x_t)\right] + \frac{\nu(\mu+\gamma_t)}{p\alpha_t^p}\|d_t\|_p^p \\ &\quad - \frac{\nu(\mu+\gamma_t)}{p\alpha_t^p}\|d_t\|_p^p + \frac{L}{2}\|d_t\|_p^2 + M_f\|d_t\|_p \\ &\le (1-\alpha_t)\Psi(x_{t-1}^{ag}) + \alpha_t\left[f(x_t^{md}) + \langle f'(x_t^{md}), x_t - x_t^{md}\rangle + \mathcal{X}(x_t) + \mu V(x_t^{md}, x_t)\right] \\ &\quad + ((1-\alpha_t)\mu+\gamma_t)V(x_{t-1}, x_t) - \frac{\nu(\mu+\gamma_t)}{p\alpha_t^p}\|d_t\|_p^p + \frac{L}{2}\|d_t\|_p^2 + M_f\|d_t\|_p. \end{aligned} \tag{65}$$

By using Lemma 27 with $p(u) = \alpha_t[\langle G_t, u\rangle + \mathcal{X}(u)]$, $\mu_1 = \alpha_t\mu$, $\mu_2 = (1-\alpha_t)\mu + \gamma_t$, $\tilde{x} = x_t^{md}$ and $\tilde{y} = x_{t-1}$, we have

$$\begin{aligned} &\alpha_t\left[f(x_t^{md}) + \langle G_t, x_t - x_t^{md}\rangle + \mathcal{X}(x_t) + \mu V(x_t^{md}, x_t)\right] + ((1-\alpha_t)\mu+\gamma_t)V(x_{t-1}, x_t) \\ &\le \alpha_t\left[f(x_t^{md}) + \langle G_t, x - x_t^{md}\rangle + \mathcal{X}(x) + \mu V(x_t^{md}, x)\right] + ((1-\alpha_t)\mu+\gamma_t)V(x_{t-1}, x) - (\mu+\gamma_t)V(x_t, x) \\ &= \alpha_t l_\Psi(x_t^{md}, x) + \alpha_t\langle\delta_t, x - x_t^{md}\rangle + ((1-\alpha_t)\mu+\gamma_t)V(x_{t-1}, x) - (\mu+\gamma_t)V(x_t, x), \end{aligned} \tag{66}$$

for any $x \in X$,where the last equality follows from (57) and (59). Combining (65) and (66), and using the fact that $f'(x_t^{md}) = G_t - \delta_t$ due to (59), we obtain

$$\begin{aligned} \Psi(x_t^{ag}) &\le (1-\alpha_t)\left[\Psi(x_{t-1}^{ag}) + \mu V(x_{t-1}, x)\right] + \alpha_t l_\Psi(x_t^{md}, x) + \gamma_t\left[V(x_{t-1}, x) - V(x_t, x)\right] \\ &\quad - \mu V(x_t, x) \underbrace{- \frac{\nu(\mu+\gamma_t)}{p\alpha_t^p}\|d_t\|_p^p + \frac{L}{2}\|d_t\|_p^2 + M_f\|d_t\|_p + \alpha_t\langle\delta_t, x - x_t\rangle}_{U_t}, \quad \forall x \in X. \end{aligned} \tag{67}$$

It remains to show that the term U_t defined above is bounded by $\Delta_t(x)$ as defined in the proposition.

Indeed, we have, by the last identity in (63),

$$\begin{aligned} U_t &= -\frac{\nu(\mu+\gamma_t)}{p\alpha_t^p}\|d_t\|_p^p + \frac{L}{2}\|d_t\|_p^2 + M_f\|d_t\|_p - \langle \delta_t, d_t\rangle + \langle \delta_t, d_t + \alpha_t(x - x_t)\rangle \\ &\le -\frac{\nu(\mu+\gamma_t)}{p\alpha_t^p}\|d_t\|_p^p + \frac{L^{p-1}}{pM_f^{p-2}}\|d_t\|_p^p + 2M_f\|d_t\|_p - \langle \delta_t, d_t\rangle + \alpha_t\langle \delta_t, x - x_{t-1}^+\rangle \\ &\le -\frac{\nu(\mu+\gamma_t) - (L^{p-1}/M_f^{p-2})\alpha_t^p}{p\alpha_t^p}\|d_t\|_p^p + (2M_f + \|\delta_t\|_q)\|d_t\|_p + \alpha_t\langle \delta_t, x - x_{t-1}^+\rangle \\ &\le \frac{\alpha_t^q(2M_f + \|\delta_t\|_q)^q}{qK_t^{q-1}} + \alpha_t\langle \delta_t, x - x_{t-1}^+\rangle = \Delta_t(x), \end{aligned} \tag{68}$$

where the last inequality follows by the Young's inequality w.r.t. $\|d_t\|$. □

Lemma 29 below follows from Proposition 15 by taking summation of the relations in (60).

Lemma 29. *Let $\{x_t^{ag}\}_{t\ge1}$ be computed by the generic AC-SA algorithm for solving $F_{X,X}(L, M_f, \sigma, \mu)$. Also assume that $\{\alpha_t\}_{t\ge1}$ and $\{\gamma_t\}_{t\ge1}$ are chosen so that $K_t > 0$ for all $t \ge 1$. We have*

$$\Psi(x_t^{ag}) + \mu V(x_t, x) - \Gamma_t\sum_{\tau=1}^t\left[\frac{\alpha_\tau}{\Gamma_\tau}l_\Psi(x_\tau^{md}, x)\right] \le \Gamma_t\sum_{\tau=1}^t\frac{\gamma_\tau}{\Gamma_\tau}\left[V(x_{\tau-1}, x) - V(x_\tau, x)\right] + \Gamma_t\sum_{\tau=1}^t\frac{\Delta_\tau(x)}{\Gamma_\tau}, \tag{69}$$

for any $x \in X$ and any $t \ge 1$, where $l_\Psi(z, x)$ and $\Delta_\tau(x)$ are defined in (57) and (61), respectively.

Proof. Dividing both sides of relation (60) by Γ_t, and using the definition of Γ_t in (72) and the fact that $\alpha_1 = 1$, we have

$$\begin{aligned} \frac{1}{\Gamma_t}\left[\Psi(x_t^{ag}) + \mu V(x_t, x)\right] \le{}& \frac{1}{\Gamma_{t-1}}\left[\Psi(x_{t-1}^{ag}) + \mu V(x_{t-1}, x)\right] + \frac{\alpha_t}{\Gamma_t}l_\Psi(x_t^{md}, x) \\ &+ \frac{\gamma_t}{\Gamma_t}\left[V(x_{t-1}, x) - V(x_t, x)\right] + \frac{\Delta_t(x)}{\Gamma_t}, \quad \forall t \ge 2, \end{aligned}$$

and

$$\frac{1}{\Gamma_1}\left[\Psi(x_1^{ag}) + \mu V(x_1, x)\right] \le \frac{\alpha_1}{\Gamma_1}l_\Psi(x_1^{md}, x) + \frac{\gamma_1}{\Gamma_1}\left[V(x_0, x) - V(x_1, x)\right] + \frac{\Delta_1(x)}{\Gamma_1}.$$

Summing up the above inequalities, we have

$$\frac{1}{\Gamma_t}\left[\Psi(x_t^{ag}) + \mu V(x_t, x)\right] \le \sum_{\tau=1}^t\frac{\alpha_\tau}{\Gamma_\tau}l_\Psi(x_\tau^{md}, x) + \sum_{\tau=1}^t\frac{\gamma_\tau}{\Gamma_\tau}\left[V(x_{\tau-1}, x) - V(x_\tau, x)\right] + \sum_{\tau=1}^t\frac{\Delta_\tau(x)}{\Gamma_\tau},$$

which clearly implies (69). □

Theorem 16. *Consider the generic AC-SA algorithm for $F_{X,X}(L, M_f, \sigma, \mu)$. Assume $\alpha_1 = 1$, $0 < \alpha_t < 1$ for $t \ge 2$, and that $\{\alpha_t\}_{t\ge1}$ and $\{\gamma_t\}_{t\ge1}$ are chosen such that*

$$K_t := \nu(\mu + \gamma_t) - \frac{L^{p-1}}{M_f^{p-2}}\alpha_t^p > 0, \tag{70}$$

$$\gamma_1/\Gamma_1 = \gamma_2/\Gamma_2 = \dots, \tag{71}$$

where

$$\Gamma_t := \begin{cases} 1, & t = 1, \\ (1-\alpha_t)\Gamma_{t-1}, & t \geq 2. \end{cases} \tag{72}$$

Then,
a) under the stochastic-oracle conditions (58), we have

$$\mathbb{E}[\Psi(x_t^{ag}) - \Psi^*] \leq B_e(t) := \Gamma_t \gamma_1 V(x_0, x^*) + \Gamma_t \sum_{\tau=1}^{t} \frac{2^{q-1}\alpha_\tau^q[(2M_f)^q + \sigma^q]}{q\Gamma_\tau K_\tau^{q-1}} \tag{73}$$

for any $t \geq 1$, where x^* is an arbitrary optimal solution of (55).

Proof. We show this following the original proof of Theorem 1 in [GL12]. Observe that by the definition of Γ_t and the fact that $\alpha_1 = 1$, we have

$$\sum_{\tau=1}^{t} \frac{\alpha_\tau}{\Gamma_\tau} = \frac{\alpha_1}{\Gamma_1} + \sum_{\tau=2}^{t} \frac{1}{\Gamma_\tau}\left(1 - \frac{\Gamma_\tau}{\Gamma_{\tau-1}}\right) = \frac{1}{\Gamma_1} + \sum_{\tau=2}^{t}\left(\frac{1}{\Gamma_\tau} - \frac{1}{\Gamma_{\tau-1}}\right) = \frac{1}{\Gamma_t}. \tag{74}$$

The lower-model inequality follows directly from (56): for every $x, z \in X$,

$$l_\Psi(z, x) = f(z) + \langle f'(z), x - z\rangle + \mu V(z, x) + \mathcal{X}(x) \leq f(x) + \mathcal{X}(x) = \Psi(x).$$

Using this inequality and the previous observation, we obtain

$$\Gamma_t \sum_{\tau=1}^{t}\left[\frac{\alpha_\tau}{\Gamma_\tau} l_\Psi(x_\tau^{md}, x)\right] \leq \Gamma_t \sum_{\tau=1}^{t}\left[\frac{\alpha_\tau}{\Gamma_\tau}\Psi(x)\right] = \Psi(x), \qquad \forall x \in X. \tag{75}$$

Moreover, by the condition $\gamma_1/\Gamma_1 = \gamma_2/\Gamma_2 = \cdots$, we have

$$\Gamma_t \sum_{\tau=1}^{t} \frac{\gamma_\tau}{\Gamma_\tau}\left[V(x_{\tau-1}, x) - V(x_\tau, x)\right] = \Gamma_t \frac{\gamma_1}{\Gamma_1}\left[V(x_0, x) - V(x_t, x)\right] \leq \Gamma_t \gamma_1 V(x_0, x), \tag{76}$$

where the last inequality follows from $\Gamma_1 = 1$ and $V(x_t, x) \geq 0$. Substituting (75) and (76) into the cumulative recursion (69), and using $\mu V(x_t, x) \geq 0$, gives

$$\Psi(x_t^{ag}) - \Psi(x) \leq \Psi(x_t^{ag}) + \mu V(x_t, x) - \Psi(x) \leq \Gamma_t \gamma_1 V(x_0, x) + \Gamma_t \sum_{\tau=1}^{t} \frac{\Delta_\tau(x)}{\Gamma_\tau}, \qquad \forall x \in X. \tag{77}$$

Note that $(x_\tau^{md}, x_{\tau-1}, x_{\tau-1}^{ag})$ is determined by the history $\mathcal{F}_{\tau-1} = \sigma(\xi_1, \dots, \xi_{\tau-1})$. Consequently, $x_{\tau-1}^+$ is $\mathcal{F}_{\tau-1}$-measurable. Taking $x = x^\star$ and using conditional unbiasedness of the stochastic oracle, we obtain

$$\mathbb{E}\left[\left\langle \delta_\tau, x^\star - x_{\tau-1}^+\right\rangle \middle| \mathcal{F}_{\tau-1}\right] = \left\langle \mathbb{E}[\delta_\tau \mid \mathcal{F}_{\tau-1}], x^\star - x_{\tau-1}^+\right\rangle = 0. \tag{78}$$

By the conditional q-moment assumption in (58),

$$\mathbb{E}[\|\delta_\tau\|_q^q \mid \mathcal{F}_{\tau-1}] \leq \sigma^q.$$

Furthermore, the elementary inequality $(a+b)^q \le 2^{q-1}(a^q+b^q)$, valid for $a, b \ge 0$, implies

$$\begin{aligned}\mathbb{E}[(2M_f + \|\delta_\tau\|_q)^q] &\le 2^{q-1}\big((2M_f)^q + \mathbb{E}\|\delta_\tau\|_q^q\big) \\ &\le 2^{q-1}\left((2M_f)^q + \sigma^q\right).\end{aligned}$$

Recalling the definition of $\Delta_\tau(x)$ in the preceding proposition and observing that $q - 1 = 1/(p-1)$, we conclude that

$$\begin{aligned}\mathbb{E}[\Delta_\tau(x^\star)] &= \alpha_\tau \mathbb{E}\big[\langle \delta_\tau, x^\star - x_{\tau-1}^+\rangle\big] + \frac{\alpha_\tau^q}{qK_\tau^{q-1}}\mathbb{E}[(2M_f + \|\delta_\tau\|_q)^q] \\ &\le \frac{2^{q-1}\alpha_\tau^q\big((2M_f)^q + \sigma^q\big)}{qK_\tau^{q-1}}.\end{aligned}$$

Finally, taking expectations on both sides of (77) with $x = x^\star$, and applying the previous estimate, we have

$$\begin{aligned}\mathbb{E}[\Psi(x_t^{ag}) - \Psi^\star] &\le \Gamma_t\gamma_1 V(x_0, x^\star) + \Gamma_t \sum_{\tau=1}^t \frac{\mathbb{E}[\Delta_\tau(x^\star)]}{\Gamma_\tau} \\ &\le \Gamma_t\gamma_1 V(x_0, x^\star) + \frac{2^{q-1}\Gamma_t}{q}\sum_{\tau=1}^t \frac{\alpha_\tau^q\big((2M_f)^q + \sigma^q\big)}{\Gamma_\tau K_\tau^{q-1}},\end{aligned}$$

as claimed. □

Theorem 17. *Assume that the conditions of Propositions 11 and 12 hold, then* BallAccel *(Proposition 11), using parallel ℓ_p-AC-SA (Algorithm 10) as a ball optimization oracle, solves Problem 1 with expected error ϵ_{opt}, with*

$$O\Big(d^{\frac{p}{2(p+1)}}\kappa^{\frac{p}{p+1}}\log^4(d\kappa)\Big) \quad \textit{query depth}$$

and

$$O\left(\left(d^{\frac{p}{2(p+1)}}\kappa^{\frac{p}{p+1}} + d^{\frac{p}{2(p+2)}}\kappa^{\frac{2p}{p+2}}\right)\log^{2p}(d\kappa) + d^{\frac{p(p-2)}{2(p+1)}}\kappa^{\frac{p(2p-1)}{p+1}}\log^{2p(p-1)+2}(d\kappa)\right) \quad \textit{total queries,}$$

and an additional computational cost of

$$O\Big(\Big(d^{\frac{p}{2(p+1)}}\kappa^{\frac{p}{p+1}} + d^{\frac{p}{2(p+2)}}\kappa^{\frac{2p}{p+2}}\Big)\log^{2p}(d\kappa)\Big) \quad \textit{depth}$$

and

$$O\left(\left[\left(d^{\frac{p}{2(p+1)}}\kappa^{\frac{p}{p+1}} + d^{\frac{p}{2(p+2)}}\kappa^{\frac{2p}{p+2}}\right)\log^{2p}(d\kappa) + d^{\frac{p(p-2)}{2(p+1)}}\kappa^{\frac{p(2p-1)}{p+1}}\log^{2p(p-1)+2}(d\kappa)\right]d\right) \quad \textit{work}$$

The computational bounds use the vector-operation cost convention of Problem 1 and Proposition 12.

Proof. Let $x^\star \in \operatorname{argmin}_{x\in\mathbb{B}_p(R)} f(x)$, and define

$$f_{\text{pen}}(x) = f(x) + 2L(\|x\|_p - R)_+, \qquad F = \widehat{f_{\text{pen}}}_\rho, \qquad r = \rho = \frac{\epsilon_{\text{opt}}}{C_p' L\sqrt{d}}.$$

By Lemma 26, f_{pen} is convex and $3L$-Lipschitz, and its stochastic subgradient oracle has dual q-th moment at most $(4L)^q$. Convexity and smoothing give $F(x) \geq f_{\mathrm{pen}}(x)$. Choosing C_p' sufficiently large, the smoothing estimate yields

$$0 \leq F(x) - f_{\mathrm{pen}}(x) \leq 3L\mathbb{E}\|\xi_\rho\|_p \leq \frac{\epsilon_{\mathrm{opt}}}{4}, \qquad x \in \mathbb{R}^d.$$

By the Lipschitz continuity of f and projection onto $\mathbb{B}_p(R)$,

$$f_{\mathrm{pen}}(x) \geq f(x^\star) + L(\|x\|_p - R)_+, \qquad x \in \mathbb{R}^d.$$

Consequently, $F \geq f_{\mathrm{pen}}$ is coercive and attains its global minimum. For any global minimizer z of F,

$$L(\|z\|_p - R)_+ \leq F(z) - f(x^\star) \leq F(x^\star) - f(x^\star) \leq \frac{\epsilon_{\mathrm{opt}}}{4}.$$

Since $\epsilon_{\mathrm{opt}} \leq LR$, this implies

$$\|z\|_p \leq R + \frac{\epsilon_{\mathrm{opt}}}{4L} \leq 2R.$$

Thus F has a global minimizer in $\mathbb{B}_p(2R)$. Apply the implementation in Proposition 11 to F, with comparison point $x^\star$, comparison radius $2R$, Lipschitz parameter $3L$, and target accuracy $\epsilon_{\mathrm{opt}}/2$. All ball optimization oracles are applied directly to F. Each required regularized subproblem therefore has objective

$$x \mapsto F(x) + \frac{\lambda}{p}\|x - \bar{x}\|_p^p.$$

Proposition 12, applied to f_{pen} and its stochastic subgradient oracle, implements these calls using Algorithm 10, with L replaced by $3L$. The BallAccel parameters corresponding to this invocation differ from the original κ, K, and $\lambda^\star$ only by constant factors depending on p. In the complexity estimates below, we retain their definitions from Proposition 10. In particular,

$$K = \Theta\left(d^{\frac{p}{2(p+1)}}\kappa^{\frac{p}{p+1}}\right), \qquad \log(R\kappa/r) = O(\log(d\kappa)).$$

Each AC-SA ball optimization call has constant query depth. Thus Proposition 11 gives total query depth

$$O\left(K\log\kappa\log^3(R\kappa/r)\right) = O\left(d^{\frac{p}{2(p+1)}}\kappa^{\frac{p}{p+1}}\log^4(d\kappa)\right).$$

For the computational depth, all calls use $\lambda = \Omega_p(\lambda^\star)$, and hence

$$\frac{L}{\lambda r^{p-1}} \leq \frac{C_p\kappa}{K^{1/p}\log^p\kappa} \leq \frac{C_p\kappa}{K^{1/p}}.$$

By Proposition 12, each call in the first requirement of Proposition 11 requires

$$O\left(1 + \left(\frac{\kappa}{K^{1/p}}\right)^{\frac{p}{p+2}}\right)$$

sequential updates. A level-j call in the second requirement requires

$$O\left(1 + j + \log\log(R\kappa/r) + \left(1 + \frac{\kappa}{K^{1/p}}\right)^{\frac{p}{p+2}} 2^{\frac{j(p-2)}{p+2}}\log^{\frac{2p(p-2)}{p+2}}(R\kappa/r)\right)$$

updates. Using the call counts in the two requirements and

$$\sum_{j\geq 1} 2^{-j}(1+j) = O(1), \qquad \sum_{j\geq 1} 2^{-\frac{4j}{p+2}} = O(1),$$

the combined computational depth is

$$O\left(\left[K+\kappa^{\frac{p}{p+2}}K^{\frac{p+1}{p+2}}\right]\log^{2p}(d\kappa)\right).$$

Here we used

$$4\le 2p, \qquad 2+\frac{2p(p-2)}{p+2}\le 2p.$$

Substituting the expression for K gives the stated computational-depth bound. The contribution of the $L^p/(\lambda\phi^{p-1})$ term in Proposition 12 is summed exactly as in the proof of Theorem 14, and is at most

$$O\left(\kappa^p K^{p-2}\log^{2p(p-1)+2}(d\kappa)\right).$$

Adding the update bound above gives total queries

$$O\left(\left[K+\kappa^{\frac{p}{p+2}}K^{\frac{p+1}{p+2}}\right]\log^{2p}(d\kappa)+\kappa^p K^{p-2}\log^{2p(p-1)+2}(d\kappa)\right).$$

The identities

$$\kappa^{\frac{p}{p+2}}K^{\frac{p+1}{p+2}}=\Theta\left(d^{\frac{p}{2(p+2)}}\kappa^{\frac{2p}{p+2}}\right)$$

and

$$\kappa^p K^{p-2}=\Theta\left(d^{\frac{p(p-2)}{2(p+1)}}\kappa^{\frac{p(2p-1)}{p+1}}\right)$$

give the stated query bound. Under the stated computational cost convention, multiplication by d gives the work bound. Finally, the unpenalized implementation of Proposition 11 returns $\widehat{x}$ satisfying

$$\mathbb{E}F(\widehat{x})\le F(x^\star)+\frac{\epsilon_{\mathrm{opt}}}{2}.$$

Since $f\le f_{\mathrm{pen}}\le F$ and $f_{\mathrm{pen}}(x^\star)=f(x^\star)$,

$$\mathbb{E}f(\widehat{x})\le\mathbb{E}F(\widehat{x})\le F(x^\star)+\frac{\epsilon_{\mathrm{opt}}}{2}\le f(x^\star)+\frac{3\epsilon_{\mathrm{opt}}}{4}\le f(x^\star)+\epsilon_{\mathrm{opt}}.$$

This completes the proof. □

5 ℓ_p Parallel Stochastic Convex Optimization: Lower Bounds

We consider minimizing a convex function $f:\mathbb{R}^d\to\mathbb{R}$ that is 1-Lipschitz with respect to $\|\cdot\|_p$. To fix notation, let

$$B_p^d=\{x\in\mathbb{R}^d:\|x\|_p\le 1\}, \qquad B_q^d=\{s\in\mathbb{R}^d:\|s\|_q\le 1\}.$$

Then f is 1-Lipschitz with respect to $\|\cdot\|_p$ if and only if all its subgradients lie in B_q^d. An algorithm accesses f only through a *local oracle*, meaning that the answer at a point x depends only on the restriction of f to an arbitrarily small neighborhood of x. This class contains the usual (sub)gradient oracle as well as its noisy variants, and our lower bound holds for any fixed local oracle.

We consider K-*parallel* algorithms, whereby in each round the algorithm makes up to K queries in parallel, chosen based on the answers from previous rounds but not on the answers to other queries in the same round. Following [BS18], we measure complexity through *adaptivity*, i.e., the number of sequential rounds M. Throughout the section, we focus on $K\le\mathrm{poly}(d)$. Otherwise, as noted by [DG19], with exponentially many queries per round, one can query an ε-net of the feasible set and return the best point found.

Our hard instance is constructed on a restricted region of the unit ball B_p^d in which every coordinate is of order $d^{-1/p}$. Specifically, we define the feasible region

$$X := \left\{ x \in \mathbb{R}^d : \|x\|_p \le 1, \quad \|x\|_\infty \le 4d^{-1/p} \right\},$$

and require that all queries and the final output of the algorithm belong to X.

Theorem 18. *There are universal constants $c, C > 0$ with the following property. Let $M \ge 2$, $K \ge 1$, $\rho \in (0, 1/4)$, and suppose $d \ge 16M$. Further, assume that*

$$M^{-1/p} \ge C\varepsilon, \tag{79}$$

$$M^{\frac{3}{2}-\frac{1}{p}} \sqrt{\frac{\log(16KM^2/\rho)}{d}} \le c. \tag{80}$$

Then for every randomized K-parallel algorithm whose queries and output belong to X, and every fixed local oracle, there exists a convex function $f : \mathbb{R}^d \to \mathbb{R}$, 1-Lipschitz with respect to ℓ_p, such that the algorithm returns an ε-minimizer of f over X in fewer than M rounds with probability at most ρ. Consequently, up to logarithmic factors,

$$M = \widetilde{\Omega}\left(\min\left\{ \varepsilon^{-p}, d^{\frac{p}{3p-2}} \right\} \right).$$

5.1 Random block directions

The directions below follow the disjoint-support construction of [DG19] for $p \ge 2$, with two key differences: the supports are drawn at random and each z_i comes with a direction h_i that is used to construct the wall function. For the rest of the section let $L = \lfloor \frac{d}{8M} \rfloor$ and choose pairwise disjoint sets $J_1, \dots, J_M \subset [d]$, each of size L, uniformly among all ordered disjoint collections. Independently choose Rademacher signs $r_{ij} \in \{-1, +1\}$ and set

$$z_i = L^{-1/q} \sum_{j \in J_i} r_{ij} e_j, \qquad h_i = L^{-1/p} \sum_{j \in J_i} r_{ij} e_j, \qquad x^* = -M^{-1/p} \sum_{i=1}^{M} h_i.$$

Since the blocks are disjoint, $\|z_i\|_q = \|h_i\|_p = \|x^*\|_p = 1$ and $\langle z_i, x^* \rangle = -M^{-1/p}$ for every $i \in [M]$. Moreover, $x^* \in X$: $d \ge 16M$ implies $L \ge \frac{d}{16M}$, and therefore

$$\|x^*\|_\infty = (ML)^{-1/p} \le \left(\frac{16}{d} \right)^{1/p} \le 4d^{-1/p}.$$

5.2 The wall function

As in [BJL+19], who introduced a wall in the ℓ_2 setting, our objective also contains a wall W. For $t \in [M]$, let $R_t := [d] \setminus \bigcup_{i<t} J_i$ denote the coordinates outside the first $t-1$ blocks. The wall is a convex function that is small at x^* and, at round t, dominates all future affine pieces at every query $x \in X$ with $\|x_{R_t}\|_p > 4M^{-1/p}$ (Lemmas 30 and 32). At such queries the objective carries no information about the future directions, so concentration is needed only for the remaining queries. Fix a parameter $\theta > 0$, to be chosen in Lemma 34. For $t \in [M]$ let

$$\mathcal{C}_t := \left\{ s \in B_q^d : \|s\|_\infty \le L^{-1/q}, \quad |\langle h_i, s \rangle| \le \theta \text{ for all } i < t \right\}, \tag{81}$$

and let

$$\mathcal{C} := \left\{ s \in B_q^d : \|s\|_\infty \le L^{-1/q}, \quad |\langle h_i, s\rangle| \le \theta \text{ for all } i \le M \right\}. \tag{82}$$

The caps $|\langle h_i, s\rangle| \le \theta$ keep the slope s nearly orthogonal to the block directions, which makes the wall small at x^*. We define the *truncated wall*

$$W_t(x) := \max_{s \in \mathcal{C}_t} \left(\langle x, s\rangle - \frac{M^{-1/p}}{32} \|s\|_2^2 \right) - \frac{33}{32} M^{-1/p}, \tag{83}$$

which depends only on $h_1, \dots, h_{t-1}$, and the *full wall*

$$W(x) := \max_{s \in \mathcal{C}} \left(\langle x, s\rangle - \frac{M^{-1/p}}{32} \|s\|_2^2 \right) - \frac{33}{32} M^{-1/p}. \tag{84}$$

The quadratic term is used only to make the maximizing slope unique and continuous, so that $W_t = W$ on a neighborhood of each query (Lemma 31).

Now, we establish the two properties of the wall that we need: it is large on queries with large residual norm on R_t, and small at x^*.

Lemma 30. *Let $R_t := [d] \setminus \bigcup_{i<t} J_i$. For every $x \in X$ satisfying $\|x_{R_t}\|_p > 4M^{-1/p}$, we have*

$$W_t(x) \ge \|x_{R_t}\|_p - \frac{17}{16} M^{-1/p}. \tag{85}$$

Moreover, if $M\theta \le 1$, then

$$W(x^*) \le -\frac{1}{32} M^{-1/p}.$$

Proof. Fix $x \in X$ with $\|x_{R_t}\|_p > 4M^{-1/p}$ and define the ℓ_p-norming functional of x_{R_t} by

$$s_j := \frac{\operatorname{sgn}(x_j)|x_j|^{p-1}}{\|x_{R_t}\|_p^{p-1}} \mathbf{1}_{\{j \in R_t\}}.$$

Then $\|s\|_q = 1$, $\langle x, s\rangle = \|x_{R_t}\|_p$. We claim that $s \in \mathcal{C}_t$. Since $x \in X$, $\|x\|_\infty \le 4d^{-1/p}$. Hence

$$\|s\|_\infty = \left(\frac{\|x_{R_t}\|_\infty}{\|x_{R_t}\|_p} \right)^{p-1} < \left(\frac{4d^{-1/p}}{4M^{-1/p}} \right)^{p-1} = \left(\frac{M}{d} \right)^{1/q}.$$

Since $L \le d/(8M)$,

$$L^{-1/q} \ge \left(\frac{8M}{d} \right)^{1/q} > \left(\frac{M}{d} \right)^{1/q},$$

so $\|s\|_\infty \le L^{-1/q}$. Moreover, s is supported on R_t, whereas every h_i with $i < t$ is supported on J_i. Therefore, $\langle h_i, s\rangle = 0$ for $i < t$. Thus $s \in \mathcal{C}_t$. Since $q \le 2$ and $\|s\|_q = 1$, $\|s\|_2 \le 1$. Using s as a feasible candidate in (83),

$$W_t(x) \ge \langle x, s\rangle - \frac{M^{-1/p}}{32}\|s\|_2^2 - \frac{33}{32} M^{-1/p} \ge \|x_{R_t}\|_p - \frac{34}{32} M^{-1/p} = \|x_{R_t}\|_p - \frac{17}{16} M^{-1/p},$$

which proves (85). We next bound the full wall at x^*. Let $s \in \mathcal{C}$. Since

$$x^* = -M^{-1/p} \sum_{i=1}^{M} h_i,$$

the cap constraints give

$$\langle x^*, s\rangle = -M^{-1/p}\sum_{i=1}^{M}\langle h_i, s\rangle \le M^{-1/p}\sum_{i=1}^{M}|\langle h_i, s\rangle| \le M^{1-1/p}\theta.$$

If $M\theta \le 1$, then $M^{1-1/p}\theta = M^{-1/p}(M\theta) \le M^{-1/p}$. The quadratic term is nonpositive, so for all $s \in \mathcal{C}$, it satisfies

$$\langle x^*, s\rangle - \frac{M^{-1/p}}{32}\|s\|_2^2 \le M^{-1/p}.$$

Taking the maximum over $s \in \mathcal{C}$ and subtracting $33M^{-1/p}/32$ yields

$$W(x^*) \le -\frac{1}{32}M^{-1/p}.$$

□

Lemma 31. *For every $t \in [M]$, the truncated wall W_t and the full wall W are convex and 1-Lipschitz with respect to ℓ_p. Let*

$$s_t(x) := \arg\max_{s\in\mathcal{C}_t}\left\{\langle x, s\rangle - \frac{M^{-1/p}}{32}\|s\|_2^2\right\}.$$

The maximizer is unique. Moreover, if

$$|\langle h_i, s_t(x)\rangle| \le \frac{\theta}{2} \qquad \text{for every } i = t, \dots, M, \tag{86}$$

then there exists an open neighborhood U of x such that

$$W_t(y) = W(y) \qquad \text{for every } y \in U.$$

Proof. For every fixed $s \in \mathcal{C}_t$, the map

$$x \mapsto \langle x, s\rangle - \frac{M^{-1/p}}{32}\|s\|_2^2$$

is affine with slope $s \in B_q^d$. Hence the maximum over $s \in \mathcal{C}_t$ is convex and 1-Lipschitz with respect to ℓ_p. The same argument applies to W. Subtracting a constant would not affect both properties. Completing the square gives $s_t(x) = \Pi_{\mathcal{C}_t}(16M^{1/p}x)$, so the maximizer is unique and, as the Euclidean projection is 1-Lipschitz,

$$\|s_t(y) - s_t(x)\|_2 \le \left\|16M^{1/p}y - 16M^{1/p}x\right\|_2 = 16M^{1/p}\|y - x\|_2. \tag{87}$$

Suppose (86) holds. For every $i = t, \dots, M$,

$$\begin{aligned}|\langle h_i, s_t(y)\rangle| &\le |\langle h_i, s_t(x)\rangle| + |\langle h_i, s_t(y) - s_t(x)\rangle| \\ &\le \frac{\theta}{2} + \|h_i\|_2\|s_t(y) - s_t(x)\|_2 \\ &\le \frac{\theta}{2} + 16M^{1/p}\|h_i\|_2\|y - x\|_2.\end{aligned}$$

Since there are finitely many indices, there exists $\delta > 0$ such that, whenever $\|y - x\|_2 < \delta$,

$$|\langle h_i, s_t(y)\rangle| < \theta \qquad \text{for every } i = t, \dots, M.$$

The vector $s_t(y)$ already satisfies the caps corresponding to $i < t$ because $s_t(y) \in \mathcal{C}_t$. Thus $s_t(y) \in \mathcal{C}$. Since $\mathcal{C} \subseteq \mathcal{C}_t$, the full and truncated optimization problems have the same optimal value at such y. Therefore $W_t(y) = W(y)$ throughout a neighborhood of x. □

5.3 Hard objective and hiding future pieces

For $\tau > 0$, define

$$f(x) := \max\left\{\max_{1\le i\le M}\left(\frac{1}{32}\langle z_i, x\rangle - \frac{i\tau}{8}\right), W(x)\right\}, \tag{88}$$

$$f_t(x) := \max\left\{\max_{1\le i\le t}\left(\frac{1}{32}\langle z_i, x\rangle - \frac{i\tau}{8}\right), W_t(x)\right\}. \tag{89}$$

Lemma 32. *Fix a round $t < M$ and a query $x \in X$. Assume (86). If $\|x_{R_t}\|_p \le 4M^{-1/p}$, also assume*

$$|\langle z_i, x\rangle| \le \tau \qquad \text{for every } i \ge t. \tag{90}$$

Then $f = f_t$ on a neighborhood of x.

Proof. Lemma 31 gives $W = W_t$ on a neighborhood of x. First suppose $\|x_{R_t}\|_p > 4M^{-1/p}$. Every future z_i, $i \ge t$, is supported on R_t, and $\|z_i\|_q = 1$, so

$$\langle z_i, x\rangle = \langle z_i, x_{R_t}\rangle \le \|x_{R_t}\|_p.$$

By Lemma 30,

$$W_t(x) \ge \|x_{R_t}\|_p - \frac{17}{16}M^{-1/p}.$$

Therefore, for every $i \ge t$,

$$W_t(x) - \frac{1}{32}\langle z_i, x\rangle + \frac{i\tau}{8} \ge \frac{31}{32}\|x_{R_t}\|_p - \frac{17}{16}M^{-1/p} > \frac{45}{16}M^{-1/p}.$$

Thus the wall strictly dominates every future affine piece, and the strict inequality holds locally. Now suppose $\|x_{R_t}\|_p \le 4M^{-1/p}$. For every $i > t$, (90) gives

$$\left(\frac{1}{32}\langle z_t, x\rangle - \frac{t\tau}{8}\right) - \left(\frac{1}{32}\langle z_i, x\rangle - \frac{i\tau}{8}\right) \ge -\frac{\tau}{32} - \frac{t\tau}{8} - \frac{\tau}{32} + \frac{(t+1)\tau}{8} = \frac{\tau}{16} > 0.$$

Hence every future affine piece lies strictly below the t-th piece. Combining the affine and wall parts proves the claim. □

5.4 Random-block concentration

Lemma 33. *Let $R \subseteq [d]$ have cardinality $n \ge 7d/8$, and let J be a uniformly random L-subset of R, where $L = \lfloor \frac{d}{8M} \rfloor$. Let $(r_j)_{j\in J}$ be independent Rademacher signs and define*

$$z = L^{-1/q}\sum_{j\in J} r_j e_j, \qquad h = L^{-1/p}\sum_{j\in J} r_j e_j.$$

Then there exists a universal constant $C > 0$ such that, for every $u \ge 1$, the following hold.

(i) For every deterministic $x \in X$ satisfying $\|x_R\|_p \le 4M^{-1/p}$, with probability at least $1 - 2e^{-u}$,

$$|\langle z, x\rangle| \le CM^{-1/p}\left[M^{\frac{1}{2}-\frac{1}{p}}\sqrt{\frac{u}{d}} + \frac{Mu}{d}\right]. \tag{91}$$

(ii) For every deterministic $s \in B_q^d$ satisfying $\|s\|_\infty \le L^{-1/q}$, with probability at least $1 - 2e^{-u}$,

$$|\langle h, s\rangle| \le C\left[\sqrt{\frac{u}{d}} + \frac{Mu}{d}\right]. \tag{92}$$

Proof. Fix a deterministic vector $a \in \mathbb{R}^R$ and write

$$Q_J(a) := \sum_{j\in J} a_j^2.$$

Since J is a uniform sample without replacement from R, Hoeffding's comparison theorem allows us to compare the moment generating function of $Q_J(a)$ with that of a with-replacement sample. Let $Y_1, \dots, Y_L$ be independent and uniformly distributed over the finite population $\{a_j^2 : j \in R\}$. Then

$$\mathbb{E}Y_\ell = \frac{\|a\|_2^2}{n}, \qquad 0 \le Y_\ell \le \|a\|_\infty^2,$$

and

$$\mathrm{Var}(Y_\ell) \le \frac{\|a\|_\infty^2 \|a\|_2^2}{n}.$$

The standard Bernstein bound for the independent sample gives: for every $t > 0$, with probability at least $1 - e^{-t}$,

$$Q_J(a) \le 2\left[\frac{L}{n}\|a\|_2^2 + \|a\|_\infty^2 t\right]. \tag{93}$$

Conditioning on J, since the signs are independent Rademachers, we have

$$\Pr\left(\left|\sum_{j\in J} r_j a_j\right| \ge r \,\middle|\, J\right) \le 2\exp\left(-\frac{r^2}{2Q_J(a)}\right)$$

Applying (93) with $t = 2u$ and then the Rademacher tail bound, we obtain, with probability at least $1 - 2e^{-u}$,

$$\left|\sum_{j\in J} r_j a_j\right| \le C\left[\sqrt{\frac{L}{n}}\|a\|_2\sqrt{u} + \|a\|_\infty u\right]. \tag{94}$$

For part (i), apply (94) with $a = x_R$ and multiply by $L^{-1/q}$. Since $p \ge 2$,

$$\|x_R\|_2 \le n^{\frac{1}{2}-\frac{1}{p}}\|x_R\|_p \le 4n^{\frac{1}{2}-\frac{1}{p}}M^{-1/p}.$$

Hence the first term satisfies

$$L^{-1/q}\sqrt{\frac{L}{n}}\|x_R\|_2\sqrt{u} \le CM^{-1/p}L^{\frac{1}{p}-\frac{1}{2}}n^{-1/p}\sqrt{u} \le CM^{-1/p}M^{\frac{1}{2}-\frac{1}{p}}\sqrt{\frac{u}{d}},$$

where we used

$$\frac{d}{16M} \le L \le \frac{d}{8M}, \qquad n \ge \frac{7d}{8}.$$

For the second term, because $x \in X$, $\|x_R\|_\infty \le 4d^{-1/p}$, and therefore

$$L^{-1/q}\|x_R\|_\infty u \le C\left(\frac{M}{d}\right)^{1/q} d^{-1/p}u = C\frac{M^{1/q}u}{d} = CM^{-1/p}\frac{Mu}{d}.$$

This proves (91). For part (ii), apply (94) with $a = s_R$ and multiply by $L^{-1/p}$. Since $q \le 2$,

$$\|s_R\|_2^2 = \sum_{j \in R} |s_j|^{2-q} |s_j|^q \le \|s\|_\infty^{2-q} \|s\|_q^q \le L^{-(2-q)/q},$$

so $\|s_R\|_2 \le L^{\frac{1}{p}-\frac{1}{2}}$. Thus $L^{-1/p}\sqrt{\frac{L}{n}}\|s_R\|_2\sqrt{u} \le C\sqrt{\frac{u}{d}}$, while

$$L^{-1/p}\|s_R\|_\infty u \le L^{-1/p}L^{-1/q}u = \frac{u}{L} \le C\frac{Mu}{d}.$$

which proves (92). □

5.5 The high-probability event

Lemma 34. *With the choices*

$$\tau = CM^{-1/p}\left[M^{\frac{1}{2}-\frac{1}{p}}\sqrt{\frac{\log(16KM^2/\rho)}{d}} + \frac{M\log(16KM^2/\rho)}{d}\right] \tag{95}$$

and

$$\theta = C\left[\sqrt{\frac{\log(16KM^2/\rho)}{d}} + \frac{M\log(16KM^2/\rho)}{d}\right], \tag{96}$$

and with C sufficiently large, the following statements hold simultaneously with probability at least $1-\rho$:

(i) for every round $t < M$, every query x in that round satisfying $\|x_{R_t}\|_p \le 4M^{-1/p}$, and every $i \ge t$,

$$|\langle z_i, x\rangle| \le \tau;$$

(ii) for every round $t < M$, every query x in that round, and every $i \ge t$,

$$|\langle h_i, s_t(x)\rangle| \le \theta/2;$$

(iii) after round $M-1$, the final output satisfies the analogous two bounds with the fresh last block J_M.

Proof. For each $t \in [M]$, let $\mathcal{F}_{t-1}$ denote the σ-field generated by all randomness revealed before round t. In particular, $J_1, \dots, J_{t-1}$ and their signs are $\mathcal{F}_{t-1}$-measurable, and therefore

$$R_t = [d] \setminus \bigcup_{j<t} J_j$$

is fixed under this conditioning. Since

$$|R_t| = d - (t-1)L \ge d - ML \ge \frac{7d}{8},$$

Lemma 33 applies on R_t. Let

$$x_{t,1}, \dots, x_{t,K_t}, \qquad K_t \le K,$$

be the queries in round t. Conditioned on $\mathcal{F}_{t-1}$, these queries are fixed. Moreover, every query belongs to X. For every $i \geq t$, the conditional marginal distribution of J_i is uniform over the L-subsets of R_t, and its signs are fresh. Thus Lemma 33 applies to each fixed triple (t, k, i). Set

$$u := \log\left(\frac{16KM^2}{\rho}\right).$$

If $\|(x_{t,k})_{R_t}\|_p \leq 4M^{-1/p}$, part (i) of Lemma 33, together with the choice of τ, gives

$$\Pr\left(|\langle z_i, x_{t,k}\rangle| > \tau \,|\, \mathcal{F}_{t-1}\right) \leq 4e^{-u} = \frac{\rho}{4KM^2}.$$

If the residual norm is larger, the corresponding hard-direction event is irrelevant. For the wall bounds, conditioned on $\mathcal{F}_{t-1}$, the set $\mathcal{C}_t$ and the query $x_{t,k}$ are fixed. By Lemma 31, $s_t(x_{t,k})$ is uniquely determined, and

$$\|s_t(x_{t,k})\|_q \leq 1, \qquad \|s_t(x_{t,k})\|_\infty \leq L^{-1/q}.$$

Part (ii) of Lemma 33, together with the choice of θ, gives

$$\Pr\left(|\langle h_i, s_t(x_{t,k})\rangle| > \theta/2 \,|\, \mathcal{F}_{t-1}\right) \leq \frac{\rho}{4KM^2}.$$

The same argument applies to the final output $\widehat{x}$ after round $M-1$. By assumption $\widehat{x} \in X$, and conditioned on $\mathcal{F}_{M-1}$ the block J_M and its signs are still fresh. Therefore the analogous hard-direction and wall bounds each fail with probability at most $\rho/(4KM^2)$. There are fewer than KM^2 hard-direction events and fewer than KM^2 wall events, together with two final-output events. Since $M \geq 2$ and $K \geq 1$,

$$2KM^2 + 2 \leq 4KM^2.$$

A union bound therefore gives total failure probability at most ρ. □

5.6 Proof of Theorem 18

Proof of Theorem 18. First consider a deterministic algorithm. The function f is convex and 1-Lipschitz with respect to ℓ_p: every affine hard piece has dual norm $1/32$, while Lemma 31 gives that the wall is 1-Lipschitz. On the event in Lemma 34, Lemma 32 shows that f and f_t agree on a neighborhood of every round-t query. Induction over the rounds therefore shows that the complete transcripts coincide through round $M-1$. Let

$$u = \log\left(\frac{16KM^2}{\rho}\right).$$

The assumption (80), with c sufficiently small relative to the constant in (95), implies

$$M\tau \leq \frac{1}{64}M^{-1/p}. \tag{97}$$

Thus (97) follows by choosing c sufficiently small. The same assumption also implies $M\theta \leq 1$. We next bound the optimum. By Lemma 30,

$$W(x^*) \leq -\frac{1}{32}M^{-1/p}.$$

Also, for every i,

$$\frac{1}{32}\langle z_i, x^*\rangle - \frac{i\tau}{8} = -\frac{1}{32}M^{-1/p} - \frac{i\tau}{8} \le -\frac{1}{32}M^{-1/p}.$$

Since $x^* \in X$,

$$f^* := \min_{x\in X} f(x) \le f(x^*) \le -\frac{1}{32}M^{-1/p}. \tag{98}$$

Consider any query x made in round $t < M$. If $\|x_{R_t}\|_p > 4M^{-1/p}$, then Lemmas 32 and 30 give

$$f(x) = f_t(x) \ge W_t(x) \ge \|x_{R_t}\|_p - \frac{17}{16}M^{-1/p} > \frac{47}{16}M^{-1/p}.$$

If instead $\|x_{R_t}\|_p \le 4M^{-1/p}$, Lemma 34 gives $|\langle z_t, x\rangle| \le \tau$, and therefore

$$f(x) = f_t(x) \ge \frac{1}{32}\langle z_t, x\rangle - \frac{t\tau}{8} \ge -\frac{\tau}{32} - \frac{M\tau}{8}.$$

Combining this with (97) and (98), in either case,

$$f(x) - f^* \ge \frac{1}{64}M^{-1/p}. \tag{99}$$

It remains to treat the final output $\widehat{x}$ after round $M-1$. Since $\widehat{x}$ is determined by the first $M-1$ blocks, J_M is still uniform over the L-subsets of R_M, with independent signs. We now apply part (iii) of Lemma 34. If $\|\widehat{x}_{R_M}\|_p > 4M^{-1/p}$, then part (iii) of Lemma 34 gives

$$|\langle h_M, s_M(\widehat{x})\rangle| \le \theta/2.$$

Hence Lemma 31 gives $W(\widehat{x}) = W_M(\widehat{x})$, and Lemma 30 yields

$$f(\widehat{x}) \ge W(\widehat{x}) = W_M(\widehat{x}) > \frac{47}{16}M^{-1/p}.$$

If instead $\|\widehat{x}_{R_M}\|_p \le 4M^{-1/p}$, part (iii) gives $|\langle z_M, \widehat{x}\rangle| \le \tau$, so the M-th affine piece gives

$$f(\widehat{x}) \ge -\frac{\tau}{32} - \frac{M\tau}{8}.$$

Thus (99) also holds for the final output. Choose the constant C in (79) sufficiently large that

$$\frac{1}{64}M^{-1/p} > \varepsilon.$$

Then, on the event of Lemma 34, no output produced in fewer than M rounds is an ε-minimizer. This event has probability at least $1-\rho$. Finally, let the algorithm be randomized, and let ω denote its random choices, which are independent of the blocks and signs. For every fixed ω the algorithm is deterministic, so the argument above shows that it returns an ε-minimizer in fewer than M rounds with probability at most ρ over the blocks and signs. Averaging over ω, the same bound holds over ω, the blocks and the signs jointly. Hence, for some fixed choice of blocks and signs, the probability over ω alone is at most ρ, and the corresponding f is the function in the theorem.The condition (79) gives

$$M \lesssim \varepsilon^{-p},$$

while (80) gives, up to logarithmic factors,

$$M \lesssim d^{\frac{p}{3p-2}}.$$

This proves the theorem. □

6 Conclusion

In this work, we studied parallel SCO in ℓ_p geometries for $p \in [2, \infty)$. We first introduced a class of ODC_p algorithms, and presented various applications of these tools. We then showed how to use these algorithms to develop two efficient ℓ_p parallel algorithms, which in favorable geometries, improves upon the corresponding sequential stochastic query depth complexity. We finally present a dimension-dependent lower bound on the number of adaptive rounds required for parallel ℓ_p nonsmooth convex optimization, improving upon that of [DG19]. One interesting extension to this work would be to consider how these tools could be used in a private SCO setting, where the goal is to produce an iterate that has at most $\varepsilon_{\mathrm{opt}}$ expected population risk more than the optimal risk. One work that has studied private SCO in ℓ_p geometries is [BGN21], and it would be interesting to improve upon the upper and lower bounds presented here.

6.1 AI Methodology Statement.

We used ChatGPT 5.6 and Claude to assist with checking and aiding some steps of our proofs. In particular, the addition of the quadratic term in the wall function was suggested by ChatGPT. Generative AI tools were not used for devising the main ideas and conclusions of the paper. The authors have reviewed all AI-assisted work and take full responsibility for the contents of the paper.

References

[ABJS26] Deeksha Adil, Brian Bullins, Arun Jambulapati, and Aaron Sidford. Convex optimization with p-norm oracles. In *37th International Conference on Algorithmic Learning Theory*, 2026. (Cited on pages 1, 2, 3, and 22.)

[ACJ+21] Hilal Asi, Yair Carmon, Arun Jambulapati, Yujia Jin, and Aaron Sidford. Stochastic bias-reduced gradient methods. *Advances in Neural Information Processing Systems*, 34:10810–10822, 2021. (Cited on pages 1, 2, 3, 11, 16, 21, 22, 25, 32, and 40.)

[AFKT21] Hilal Asi, Vitaly Feldman, Tomer Koren, and Kunal Talwar. Private stochastic convex optimization: Optimal rates in ℓ_1 geometry. In *International Conference on Machine Learning*, pages 393–403. PMLR, 2021. (Cited on page 2.)

[APS19] Deeksha Adil, Richard Peng, and Sushant Sachdeva. Fast, provably convergent IRLS algorithm for p-norm linear regression. *Advances in Neural Information Processing Systems*, 32, 2019. (Cited on page 2.)

[AS20] Deeksha Adil and Sushant Sachdeva. Faster p-norm minimizing flows, via smoothed q-norm problems. In *Proceedings of the Fourteenth Annual ACM-SIAM Symposium on Discrete Algorithms*, pages 892–910. SIAM, 2020. (Cited on page 1.)

[Bec17] Amir Beck. *First-order methods in optimization*. SIAM, 2017. (Cited on page 8.)

[BG15] Jose H Blanchet and Peter W Glynn. Unbiased monte carlo for optimization and functions of expectations via multi-level randomization. In *2015 Winter Simulation Conference (WSC)*, pages 3656–3667. IEEE, 2015. (Cited on pages 2 and 11.)

[BGN21] Raef Bassily, Cristóbal Guzmán, and Anupama Nandi. Non-Euclidean differentially private stochastic convex optimization. In *Conference on Learning Theory*, pages 474–499. PMLR, 2021. (Cited on pages 4, 7, and 84.)

[BJL+19] Sébastien Bubeck, Qijia Jiang, Yin-Tat Lee, Yuanzhi Li, and Aaron Sidford. Complexity of highly parallel non-smooth convex optimization. *Advances in neural information processing systems*, 32, 2019. (Cited on pages 1, 4, 7, and 76.)

[BS18] Eric Balkanski and Yaron Singer. Parallelization does not accelerate convex optimization: Adaptivity lower bounds for non-smooth convex minimization. *arXiv preprint arXiv:1808.03880*, 2018. (Cited on pages 4 and 75.)

[Bul20] Brian Bullins. Highly smooth minimization of non-smooth problems. In *Conference on Learning Theory*, pages 988–1030. PMLR, 2020. (Cited on page 2.)

[CGJS23] Deeparnab Chakrabarty, Andrei Graur, Haotian Jiang, and Aaron Sidford. Parallel submodular function minimization. *Advances in Neural Information Processing Systems*, 36:68639–68654, 2023. (Cited on pages 1 and 64.)

[CGMR25] Juan Pablo Contreras, Cristóbal Guzmán, and David Martínez-Rubio. Non-Euclidean high-order smooth convex optimization extended abstract. In *The Thirty Eighth Annual Conference on Learning Theory*, pages 1330–1330. PMLR, 2025. (Cited on pages 1 and 2.)

[CJJ+20] Yair Carmon, Arun Jambulapati, Qijia Jiang, Yujia Jin, Yin Tat Lee, Aaron Sidford, and Kevin Tian. Acceleration with a ball optimization oracle. *Advances in Neural Information Processing Systems*, 33:19052–19063, 2020. (Cited on page 43.)

[CJJ+23] Yair Carmon, Arun Jambulapati, Yujia Jin, Yin Tat Lee, Daogao Liu, Aaron Sidford, and Kevin Tian. ReSQueing parallel and private stochastic convex optimization. In *2023 IEEE 64th Annual Symposium on Foundations of Computer Science (FOCS)*, pages 2031–2058. IEEE, 2023. (Cited on pages 1, 4, 5, 6, 7, 47, 55, 60, and 63.)

[CJJS21] Yair Carmon, Arun Jambulapati, Yujia Jin, and Aaron Sidford. Thinking inside the ball: Near-optimal minimization of the maximal loss. In *Conference on Learning Theory*, pages 866–882. PMLR, 2021. (Cited on pages 2 and 32.)

[DBW12] John C Duchi, Peter L Bartlett, and Martin J Wainwright. Randomized smoothing for stochastic optimization. *SIAM Journal on Optimization*, 22(2):674–701, 2012. (Cited on pages 1 and 91.)

[DG19] Jelena Diakonikolas and Cristóbal Guzmán. Lower bounds for parallel and randomized convex optimization. In *Conference on Learning Theory*, pages 1132–1157. PMLR, 2019. (Cited on pages 1, 2, 4, 75, 76, and 84.)

[DG24] Jelena Diakonikolas and Cristóbal Guzmán. Complementary composite minimization, small gradients in general norms, and applications. *Mathematical Programming*, 208(1):319–363, 2024. (Cited on pages 3, 25, and 26.)

[FHL25] Ilyas Fatkhullin, Florian Hübler, and Guanghui Lan. Can sgd handle heavy-tailed noise? *arXiv preprint arXiv:2508.04860*, 2025. (Cited on page 1.)

[FWY20] Kimon Fountoulakis, Di Wang, and Shenghao Yang. p-norm flow diffusion for local graph clustering. In *International Conference on Machine Learning*, pages 3222–3232. PMLR, 2020. (Cited on page 1.)

[Gil15] Michael B Giles. Multilevel Monte Carlo methods. *Acta numerica*, 24:259–328, 2015. (Cited on page 2.)

[GL12] Saeed Ghadimi and Guanghui Lan. Optimal stochastic approximation algorithms for strongly convex stochastic composite optimization I: A generic algorithmic framework. *SIAM Journal on Optimization*, 22(4):1469–1492, 2012. (Cited on pages 59, 69, 70, and 72.)

[HK11] Elad Hazan and Satyen Kale. Beyond the regret minimization barrier: an optimal algorithm for stochastic strongly-convex optimization. In *Proceedings of the 24th Annual Conference on Learning Theory*, pages 421–436. JMLR Workshop and Conference Proceedings, 2011. (Cited on page 2.)

[IN14] Anatoli Iouditski and Yuri Nesterov. Primal-dual subgradient methods for minimizing uniformly convex functions. *arXiv preprint arXiv:1401.1792*, 2014. (Cited on page 17.)

[JLS22] Arun Jambulapati, Yang P Liu, and Aaron Sidford. Improved iteration complexities for overconstrained p-norm regression. In *Proceedings of the 54th Annual ACM SIGACT Symposium on Theory of Computing*, pages 529–542, 2022. (Cited on page 2.)

[JST24] Arun Jambulapati, Aaron Sidford, and Kevin Tian. Closing the computational-query depth gap in parallel stochastic convex optimization. In *The Thirty Seventh Annual Conference on Learning Theory*, pages 2608–2643. PMLR, 2024. (Cited on pages 5 and 6.)

[KLL+23] Jonathan Kelner, Jerry Li, Allen X Liu, Aaron Sidford, and Kevin Tian. Semi-random sparse recovery in nearly-linear time. In *The Thirty Sixth Annual Conference on Learning Theory*, pages 2352–2398. PMLR, 2023. (Cited on page 93.)

[KLOS14] Jonathan A Kelner, Yin Tat Lee, Lorenzo Orecchia, and Aaron Sidford. An almost-linear-time algorithm for approximate max flow in undirected graphs, and its multicommodity generalizations. In *Proceedings of the twenty-fifth annual ACM-SIAM symposium on Discrete algorithms*, pages 217–226. SIAM, 2014. (Cited on page 1.)

[Lan16] Guanghui Lan. Gradient sliding for composite optimization. *Mathematical Programming*, 159(1):201–235, 2016. (Cited on page 2.)

[LBL+25] Xinyu Luo, Cedar Site Bai, Bolian Li, Petros Drineas, Ruqi Zhang, and Brian Bullins. Stacey: Promoting stochastic steepest descent via accelerated ℓ_p-smooth nonconvex optimization. *arXiv preprint arXiv:2506.06606*, 2025. (Cited on page 1.)

[Lin17] Peter Lindqvist. *Notes on the p-Laplace equation*. Number 161. University of Jyväskylä, 2017. (Cited on page 22.)

[LS20] Yang P Liu and Aaron Sidford. Faster energy maximization for faster maximum flow. In *Proceedings of the 52nd Annual ACM SIGACT Symposium on Theory of Computing*, pages 803–814, 2020. (Cited on page 1.)

[LZ20] Yunwen Lei and Ding-Xuan Zhou. Convergence of online mirror descent. *Applied and Computational Harmonic Analysis*, 48(1):343–373, 2020. (Cited on page 2.)

[MFS26] Thomas Massena, Corentin Friedrich, and Mathieu Serrurier. From SGD to Muon: Adaptive optimization via Schatten-p norms. *arXiv preprint arXiv:2605.19781*, 2026. (Cited on page 1.)

[MS13] Renato DC Monteiro and Benar Fux Svaiter. An accelerated hybrid proximal extra-gradient method for convex optimization and its implications to second-order methods. *SIAM Journal on Optimization*, 23(2):1092–1125, 2013. (Cited on page 2.)

[Nem94] Arkadi Nemirovski. On parallel complexity of nonsmooth convex optimization. *Journal of Complexity*, 10(4):451–463, 1994. (Cited on page 1.)

[Nes09] Yurii Nesterov. Primal-dual subgradient methods for convex problems. *Mathematical programming*, 120(1):221–259, 2009. (Cited on pages 2, 16, and 17.)

[NY83] Arkadi Nemirovski and David Borisovich Yudin. *Problem complexity and method efficiency in optimization*. Wiley-Interscience, 1983. (Cited on page 1.)

[PB14] Neal Parikh and Stephen Boyd. Proximal algorithms. *Foundations and Trends in optimization*, 1(3):127–239, 2014. (Cited on page 2.)

[Pet26] Thomas Pethick. When to use what Schatten-p norm in deep learning? *arXiv preprint arXiv:2606.15268*, 2026. (Cited on page 1.)

[RSS11] Alexander Rakhlin, Ohad Shamir, and Karthik Sridharan. Making gradient descent optimal for strongly convex stochastic optimization. *arXiv preprint arXiv:1109.5647*, 2011. (Cited on page 2.)

[SBZ+26] Zitao Song, Cedar Site Bai, Zhe Zhang, Brian Bullins, and David F Gleich. Can entry-wise clipping give spectral control of stochastic gradients? *Advances in Neural Information Processing Systems*, 2026. (Cited on page 1.)

[Ver22] Roman Vershynin. *High-dimensional probability*. Cambridge University Press, 2022. (Cited on page 10.)

[VYB+22] Nuri Mert Vural, Lu Yu, Krishna Balasubramanian, Stanislav Volgushev, and Murat A Erdogdu. Mirror descent strikes again: Optimal stochastic convex optimization under infinite noise variance. In Po-Ling Loh and Maxim Raginsky, editors, *Proceedings of Thirty Fifth Conference on Learning Theory*, volume 178 of *Proceedings of Machine Learning Research*, pages 65–102. PMLR, 02–05 Jul 2022. (Cited on pages 1, 5, and 6.)

[Zăl83] C Zălinescu. On uniformly convex functions. *Journal of Mathematical Analysis and Applications*, 95(2):344–374, 1983. (Cited on page 21.)

A Helper Facts

We present a few useful properties and facts about Bregman divergences, which we use in our analysis.

Property 1. *(Three Point Property). For all $a, b, c \in \mathbb{R}^d$ and convex Φ, it holds that*

$$\langle \nabla\Phi(a) - \nabla\Phi(b), c - a\rangle = D_\Phi(c, b) - D_\Phi(c, a) - D_\Phi(a, b).$$

Fact 2. *Let $\Phi(x) = \frac{1}{p}||x||_p^p$ for $p \geq 2$. For all $x, y \in \mathbb{R}^d$, it holds that*

$$D_\Phi(x, y) \geq \frac{2^{2-p}}{p}||x - y||_p^p.$$

Proof. Fix $x, y \in \mathbb{R}^d$. It holds that

$$D_\Phi(x, y) = \int_0^1 \langle \nabla\Phi(y + t(x - y)) - \nabla\Phi(y), x - y\rangle dt$$

Since $\Phi(x) = \frac{1}{p}\sum_{i=1}^d |x_i|^p$, we analyze (slightly overloading notation here) for $\psi(t) = |t|^{p-2}t$:

$$I_\psi(x, y) = \int_0^1 (\psi(y + t(x - y)) - \psi(y))(x - y)dt$$

We now show that for all $a, b \in \mathbb{R}$, it holds that $(\psi(a) - \psi(b))(a - b) \geq 2^{2-p}|a - b|^p$. If $a = b$, the result trivially holds, so we only consider the setting in which $a \neq b$. Since $\psi'(t) = (p - 1)|t|^{p-2}$, it holds that

$$(\psi(a) - \psi(b))(a - b) = (a - b)^2(p - 1)\int_0^1 |b + s(a - b)|^{p-2}ds$$

Letting $u = b/(a - b)$ so $b + (a - b)s = (a - b)(u + s)$, we write

$$\int_0^1 |b + s(a - b)|^{p-2}ds = |a - b|^{p-2}\int_0^1 |u + s|^{p-2}ds$$

Letting $F(u) = \int_0^1 |u + s|^{p-2}ds$, observe that the minimum of F occurs at $u = -\frac{1}{2}$, since $|x|^{p-2}$ is convex for $p \geq 2$. Therefore,

$$|a - b|^{p-2}\int_0^1 |u + s|^{p-2} \geq |a - b|^{p-2}\int_0^1 |s - \frac{1}{2}|^{p-2} = |a - b|^{p-2}\int_{-1/2}^{1/2} |u|^{p-2}du = |a - b|^{p-2} \cdot \frac{2^{2-p}}{p - 1}$$

which implies that $(\psi(a) - \psi(b))(a - b) \geq 2^{2-p}|a - b|^p$. Further, it holds that

$$(\psi(y + t(x - y)) - \psi(y))(x - y) = \frac{1}{t} \cdot (\psi(y + t(x - y)) - \psi(y))(tx - ty) \geq 2^{2-p}|x - y|^p t^{p-1}$$

which means that $I_\psi(x, y) \geq 2^{2-p}|x - y|^p \int_0^1 t^{p-1}dt = \frac{2^{2-p}}{p}|x - y|^p$. As $D_\Phi = \sum_{i=1}^d I_\psi(x_i, y_i)$, the claim follows. □

Fact 3. *Let $\Phi(x) = \frac{1}{p}||x||_p^p$ for some $p \geq 2$. For all $x, y \in \mathbb{R}^d$, it holds that*

$$D_\Phi(x, y) \leq \frac{p - 1}{2}(||x||_p + 2||y||_p)^{p-2}||x - y||_p^2.$$

Proof. We first prove a smoothness-like guarantee for Φ: $||\nabla\Phi(x) - \nabla\Phi(y)||_q \leq (p-1)(||x||_p + ||y||_p)^{p-2}||x-y||_p$. Since $\nabla\Phi(x)_i = \psi(x_i), \psi(t) = |t|^{p-2}t$, we have that for all $a, b \in \mathbb{R}$:

$$\begin{aligned}
|\psi(a) - \psi(b)| &= (p-1)|a-b| \int_0^1 |b + s(a-b)|^{p-2} ds \\
&\leq (p-1)|a-b| \int_0^1 ((1-s)|b| + s|a|)^{p-2} ds \\
&\leq (p-1)|a-b| \int_0^1 (|a| + |b|)^{p-2} ds = (p-1)|a-b|(|a| + |b|)^{p-2}
\end{aligned}$$

Thus, for each coordinate: $|\nabla\Phi(x_i) - \nabla\Phi(y_i)| \leq (p-1)(|x_i| + |y_i|)^{p-2}|x_i - y_i|$ so we have that

$$||\nabla\Phi(x) - \nabla\Phi(y)||_q = (\sum_i |\nabla\Phi(x_i) - \nabla\Phi(y_i)|^q)^{1/q} \leq (p-1)(\sum_i (|x_i| + |y_i|)^{(p-2)q}|x_i - y_i|^q)^{1/q}$$

Applying Hölder's inequality, we have that

$$\begin{aligned}
\sum_i (|x_i| + |y_i|)^{(p-2)q}|x_i - y_i|^q &\leq (\sum_i (|x_i| + |y_i|)^{\frac{(p-2)q\cdot(p-1)}{p-2}})^{\frac{p-2}{p-1}} \cdot (\sum_i |x_i - y_i|^{q(p-1)})^{\frac{1}{p-1}} \\
&\leq (\sum_i (|x_i| + |y_i|)^p)^{\frac{p-2}{p-1}} \cdot (\sum_i |x_i - y_i|^p)^{\frac{1}{p-1}}
\end{aligned}$$

which implies that

$$\begin{aligned}
(\sum_i (|x_i| + |y_i|)^{(p-2)q}|x_i - y_i|^q)^{1/q} &\leq (\sum_i (|x_i| + |y_i|)^p)^{\frac{p-2}{q(p-1)}} \cdot (\sum_i |x_i - y_i|^p)^{\frac{1}{q(p-1)}} \\
&= (\sum_i (|x_i| + |y_i|)^p)^{\frac{p-2}{p}} \cdot (\sum_i |x_i - y_i|^p)^{\frac{1}{p}} \\
&= ((\sum_i (|x_i| + |y_i|)^p)^{1/p})^{p-2} \cdot ||x-y||_p \\
&\leq ((\sum_i |x_i|^p)^{1/p} + (\sum_i |y_i|^p)^{1/p})^{p-2} \cdot ||x-y||_p \\
&= (||x||_p + ||y||_p)^{p-2} \cdot ||x-y||_p
\end{aligned}$$

where we used Minkowski's inequality in the last inequality. Therefore, it holds that

$$\begin{aligned}
D_\Phi(x,y) &= \int_0^1 \langle \nabla\Phi(y + t(x-y)) - \nabla\Phi(y), x - y\rangle dt \\
&\leq \int_0^1 ||\nabla\Phi(y + t(x-y)) - \nabla\Phi(y)||_q \cdot ||x-y||_p dt \\
&\leq \int_0^1 (p-1) \cdot (||y + t(x-y)||_p + ||y||_p)^{p-2} \cdot ||t(x-y)||_p \cdot ||x-y||_p dt \\
&= (p-1)||x-y||_p^2 \int_0^1 t(||y + t(x-y)||_p + ||y||_p)^{p-2} dt \\
&\leq (p-1)||x-y||_p^2 \cdot (||x||_p + 2||y||_p)^{p-2} \int_0^1 t dt \\
&= \frac{p-1}{2}(||x||_p + 2||y||_p)^{p-2} \cdot ||x-y||_p^2
\end{aligned}$$

which finishes the proof. □

Now, we present a few useful facts regarding our Generalized Gaussian ($\mathcal{GG}_p$) distribution.

Fact 4. *Suppose that $f : \mathbb{R}^d \to \mathbb{R}$ is convex and L-Lipschitz with respect to $\|\cdot\|_p$. Then its generalized Gaussian convolution $\hat{f}_{\rho,p}$ (Definition 1) satisfies the following:*

1. *$\hat{f}_{\rho,p}$ is $(C_p' L/\rho)$-smooth with respect to $\|\cdot\|_p$, for some constant $C_p' > 0$ depending only on p.*
2. *For every $x \in \mathbb{R}^d$,*
$$f(x) \le \hat{f}_{\rho,p}(x) \le f(x) + L\rho\sqrt{d}.$$
The dependence on d in this bound is asymptotically tight.

Proof. First consider $x, x' \in \mathbb{R}^d$ with $\|x - x'\|_p \le \rho/2$. Use the same center $\overline{x} = x$ in both estimators from Definition 2. By the unbiasedness assertion in Lemma 1, for $\xi \sim \mathcal{GG}_p(0, \rho^2)$,

$$\nabla \hat{f}_{\rho,p}(x) - \nabla \hat{f}_{\rho,p}(x') = \mathbb{E}_\xi \left[\frac{\gamma_{\rho,p}(-\xi) - \gamma_{\rho,p}(x' - x - \xi)}{\gamma_{\rho,p}(\xi)} g(x + \xi) \right].$$

The triangle inequality, the bound $\|g(x+\xi)\|_q \le L$, and Cauchy–Schwarz give

$$\begin{aligned}
\|\nabla \hat{f}_{\rho,p}(x) - \nabla \hat{f}_{\rho,p}(x')\|_q &\le L\, \mathbb{E}_\xi \left| \frac{\gamma_{\rho,p}(-\xi) - \gamma_{\rho,p}(x' - x - \xi)}{\gamma_{\rho,p}(\xi)} \right| \\
&\le L \left(\mathbb{E}_\xi \left| \frac{\gamma_{\rho,p}(-\xi) - \gamma_{\rho,p}(x' - x - \xi)}{\gamma_{\rho,p}(\xi)} \right|^2 \right)^{1/2} \\
&\le \frac{\sqrt{2}\, e^{(p-1)/2} C_p L}{\rho} \|x - x'\|_p.
\end{aligned}$$

The last inequality follows from the second bound in Lemma 2 with $m = 2$ and $\overline{x} = x$. For arbitrary $x, x' \in \mathbb{R}^d$, choose an integer $N \ge 1$ such that $\|x - x'\|_p / N \le \rho/2$, and define

$$x_i = x + \frac{i}{N}(x' - x), \qquad i = 0, \dots, N.$$

Then $x_0 = x$, $x_N = x'$, and $\|x_i - x_{i-1}\|_p = \|x - x'\|_p / N \le \rho/2$. Applying the local bound to each consecutive pair yields

$$\begin{aligned}
\|\nabla \hat{f}_{\rho,p}(x) - \nabla \hat{f}_{\rho,p}(x')\|_q &\le \sum_{i=1}^{N} \|\nabla \hat{f}_{\rho,p}(x_i) - \nabla \hat{f}_{\rho,p}(x_{i-1})\|_q \\
&\le \frac{\sqrt{2}\, e^{(p-1)/2} C_p L}{\rho} \sum_{i=1}^{N} \|x_i - x_{i-1}\|_p \\
&= \frac{\sqrt{2}\, e^{(p-1)/2} C_p L}{\rho} \|x - x'\|_p.
\end{aligned}$$

Thus the first assertion holds with $C_p' = \sqrt{2}\, e^{(p-1)/2} C_p$.

For the second property, we use the randomized smoothing argument in [DBW12]. Let $\xi \sim \mathcal{GG}_p(0, \rho^2)$. By symmetry, $\mathbb{E}[\xi] = 0$, so convexity and Jensen's inequality imply

$$f(x) = f\big(x + \mathbb{E}[\xi]\big) \le \mathbb{E}[f(x + \xi)] = \hat{f}_{\rho,p}(x).$$

On the other hand, Lipschitz continuity and Cauchy–Schwarz give

$$\hat{f}_{\rho,p}(x) \le f(x) + L\,\mathbb{E}\|\xi\|_p \le f(x) + L\big(\mathbb{E}\|\xi\|_p^2\big)^{1/2}$$

Since $\gamma_{\rho,p}(z)$ is proportional to $\exp(-\|z\|_p^2/(2\rho^2))$, radial integration gives

$$\mathbb{E}\|\xi\|_p^2 = \frac{\int_0^\infty r^{d+1}\exp\left(-\frac{r^2}{2\rho^2}\right)\,dr}{\int_0^\infty r^{d-1}\exp\left(-\frac{r^2}{2\rho^2}\right)\,dr} = 2\rho^2\frac{\Gamma(d/2+1)}{\Gamma(d/2)} = d\rho^2.$$

Consequently, $\hat{f}_{\rho,p}(x) \le f(x) + L\rho\sqrt{d}$. To see this bound is tight, we take $f(x) = L\|x\|_p$. Notice that this yields

$$\hat{f}_{\rho,p}(0) - f(0) = L\,\mathbb{E}\|\xi\|_p = \sqrt{2}L\rho\,\frac{\Gamma((d+1)/2)}{\Gamma(d/2)} \sim L\rho\sqrt{d} \qquad \text{as } d\to\infty.$$

□

Fact 5. *Let $p \in [2,\infty)$ and let $\Phi(x) = \frac{1}{2}||x||_p^2$. Furthermore, let $\gamma_{\rho,p}(x) = Z_{\rho,p}^{-1}\exp(-\frac{\Phi(x)}{\rho^2})$ where $Z_{\rho,p}^{-1}$ is the appropriate normalization constant. Then, for every $u \in \mathbb{R}^d$,*

$$\mathbb{E}_{\xi\sim\gamma_{\rho,p}}\left[\exp\left(\frac{\langle\nabla\Phi(\xi),u\rangle}{\rho^2}\right)\right] \le \exp\left(\frac{p-1}{2\rho^2}\|u\|_p^2\right)$$

Proof. Since $\Phi(x) = \frac{1}{2}\|x\|_p^2$ is $(p-1)$-smooth with respect to $\|\cdot\|_p$, for every $x, h \in \mathbb{R}^d$,

$$\Phi(x+h) \le \Phi(x) + \langle\nabla\Phi(x),h\rangle + \frac{p-1}{2}\|h\|_p^2$$

Taking $x = \xi$ and $h = -u$ gives

$$\Phi(\xi-u) \le \Phi(\xi) - \langle\nabla\Phi(\xi),u\rangle + \frac{p-1}{2}\|u\|_p^2,$$

and therefore

$$\langle\nabla\Phi(\xi),u\rangle \le \Phi(\xi) - \Phi(\xi-u) + \frac{p-1}{2}\|u\|_p^2$$

Hence

$$\begin{aligned}\mathbb{E}_{\xi\sim\gamma_{\rho,p}}\left[\exp\left(\frac{\langle\nabla\Phi(\xi),u\rangle}{\rho^2}\right)\right] &= \frac{1}{Z_{\rho,p}}\int_{\mathbb{R}^d}\exp\left(\frac{\langle\nabla\Phi(\xi),u\rangle - \Phi(\xi)}{\rho^2}\right)d\xi \\ &\le \exp\left(\frac{p-1}{2\rho^2}\|u\|_p^2\right)\frac{1}{Z_{\rho,p}}\int_{\mathbb{R}^d}\exp\left(-\frac{\Phi(\xi-u)}{\rho^2}\right)d\xi.\end{aligned}$$

By translation invariance of Lebesgue measure,

$$\int_{\mathbb{R}^d}\exp\left(-\frac{\Phi(\xi-u)}{\rho^2}\right)d\xi = \int_{\mathbb{R}^d}\exp\left(-\frac{\Phi(\xi)}{\rho^2}\right)d\xi = Z_{\rho,p}$$

Therefore,

$$\mathbb{E}_{\xi\sim\gamma_{\rho,p}}\left[\exp\left(\frac{\langle\nabla\Phi(\xi),u\rangle}{\rho^2}\right)\right] \le \exp\left(\frac{p-1}{2\rho^2}\|u\|_p^2\right)$$

□

Algorithm 11 Aggregate

Input: $\mathcal{S} = \{y_i\}_{i\in[k]} \subset \mathbb{R}^d$, $R \geq 0$ such that for some $z \in \mathbb{R}^d$ at least $0.51k$ points $y_i \in \mathcal{S}$ have $||y_i - z||_p \leq R/3$
for $i \in [k]$ **do**
    if at least 0.51k points $y_j \in \mathcal{S}$ satisfy $||y_i - y_j||_p \leq 2R/3$ **then return** $\tilde{z} \leftarrow y_i$
    end if
end for

Fact 6. *Let $\mathcal{S} = (y_1, \ldots, y_k)$ with $y_i \in \mathbb{R}^d$, and let $R > 0$. Suppose that there exists $z \in \mathbb{R}^d$ such that*

$$|\{i \in [k] : \|y_i - z\|_p \leq R/3\}| \geq 0.51k.$$

Then Aggregate$(\mathcal{S}, R)$ *runs in $O(k^2 d)$ time and returns $\tilde{z} \in \mathcal{S}$ satisfying $\|\tilde{z} - z\|_p \leq R$.*

Proof. Following [KLL+23], let

$$T = \{i \in [k] : \|y_i - z\|_p \leq R/3\}.$$

For every $i, j \in T$, the triangle inequality gives $\|y_i - y_j\|_p \leq 2R/3$. Thus every $i \in T$ has at least $0.51k$ points within distance $2R/3$, so the algorithm has a valid point to return.

Suppose that the algorithm returns $\tilde{z} = y_i$, and let

$$\mathcal{Q} = \{j \in [k] : \|y_i - y_j\|_p \leq 2R/3\}.$$

By the selection rule, $|\mathcal{Q}| \geq 0.51k$. Since $|T| \geq 0.51k$,

$$|\mathcal{Q} \cap T| \geq |\mathcal{Q}| + |T| - k \geq 0.02k > 0.$$

Choose $j^\star \in \mathcal{Q} \cap T$. Then

$$\|\tilde{z} - z\|_p \leq \|y_i - y_{j^\star}\|_p + \|y_{j^\star} - z\|_p \leq \frac{2R}{3} + \frac{R}{3} = R.$$

The running time is $O(k^2 d)$ because the algorithm performs at most $O(k^2)$ pairwise distance computations, each costing $O(d)$ time. □